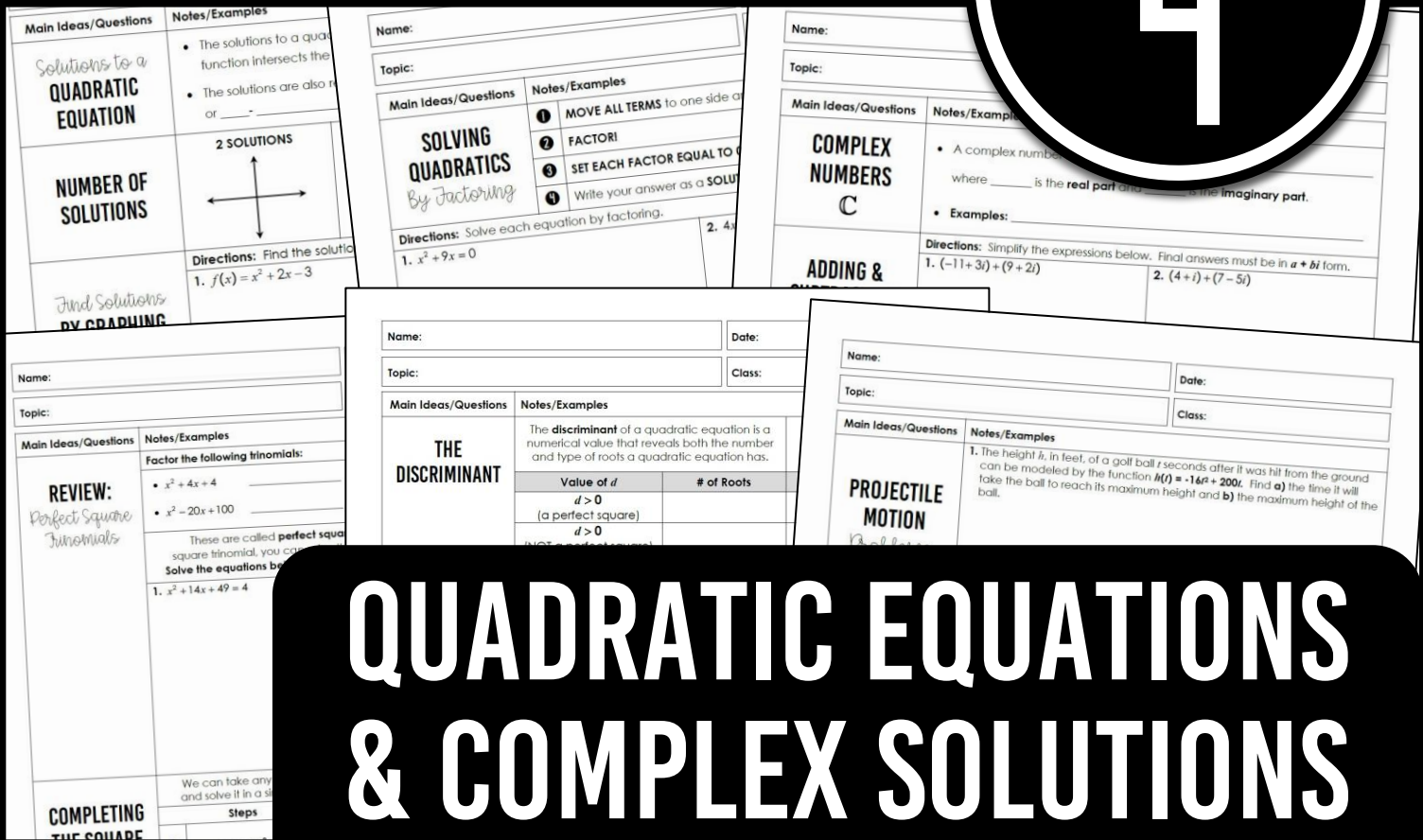


ALGEBRA 2

Unit 4



QUADRATIC EQUATIONS & COMPLEX SOLUTIONS

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 4 - Solving Quadratics & Complex Solutions: Sample Unit Outline

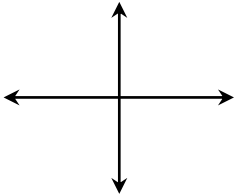
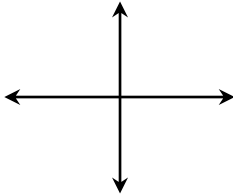
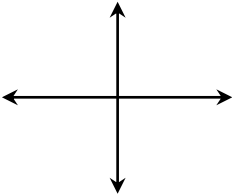
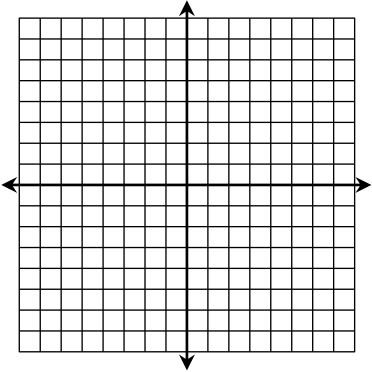
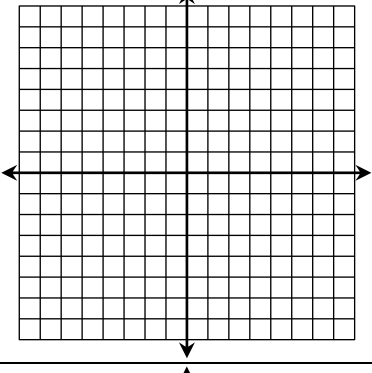
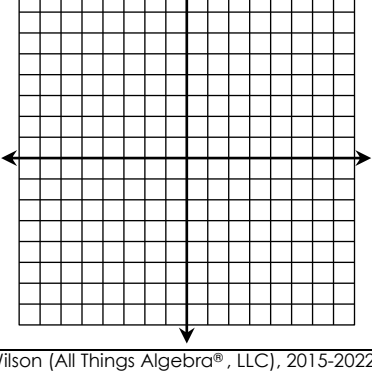
	TOPIC	HOMEWORK
DAY 1	Solving Quadratics by Graphing (Includes a review of factoring for next Day 2 lesson)	HW #1
DAY 2	Solving Quadratics by Factoring (Includes Standard Form/Factored Form/Vertex Form Reinforcement)	HW #2
DAY 3	Solving Quadratics by Square Roots Method (Includes review of simplifying radicals)	HW #3
DAY 4	Quiz 4-1	None
DAY 5	Pure Imaginary Numbers, Powers of i , Solving Quadratics with Imaginary Solutions (Square Roots Method)	HW #4
DAY 6	Adding, Subtracting, and Multiplying Complex Numbers; Solving Quadratics with Complex Solutions (Square Roots Method)	HW #5
DAY 7	Dividing Complex Numbers; Classifying & Properties of Complex Numbers	HW #6
DAY 8	Quiz 4-2	None
DAY 9	Solving Quadratics by Completing the Square (Rational, Irrational, and Complex Solutions)	HW #7
DAY 10	Solving Quadratics by the Quadratic Formula (Rational, Irrational, and Complex Solutions)	HW #8
DAY 11	Discriminant; Review all Methods to Solving Quadratics	HW #9
DAY 12	Quiz 4-3	None
DAY 13	Applications: Geometric & Consecutive Integers	HW #10
DAY 14	Applications: Projectile Motion & Quadratic Regression	HW #11
DAY 15	Unit 4 Review	Study for Test
DAY 16	UNIT 4 TEST	None
DAY 17	Nonlinear Systems: Linear-Quadratic & Quadratic-Quadratic (Day 1 – Solve Graphically)	HW #12
DAY 18	Nonlinear Systems: Linear-Quadratic & Quadratic-Quadratic (Day 2 – Solve Algebraically)	HW #13
DAY 19	Quiz 4-4	None

Note 1: Graphing quadratic functions is included in the previous unit:

[Introduction to Parent Functions & Transformations](#)

Note 2: Linear-Quadratic & Quadratic-Quadratic Systems were added in April 2022. They are not on the unit test as it would make the test too long. This topic is also in [Unit 9 \(Conics\)](#), however, there are also circles, ellipses, and hyperbolas included in the nonlinear systems portions of that unit. Nonlinear systems in this unit focuses on lines and parabolas only.

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
<i>Solutions to a</i> QUADRATIC EQUATION	- The solutions to a quadratic equation are the points at which the function intersects the ____ - ____. - The solutions are also referred to as _____, _____ or ____ - _____		
NUMBER OF SOLUTIONS	2 SOLUTIONS 	1 SOLUTION 	NO REAL SOLUTIONS 
<i>Find Solutions</i> BY GRAPHING SOLUTIONS: 1. _____ 2. _____ 3. _____	Directions: Find the solutions to each quadratic equation by graphing. 1. $f(x) = x^2 + 2x - 3$  2. $f(x) = -x^2 - 10x - 25$  3. $f(x) = x^2 - 6x + 8$ 		

SOLUTIONS:

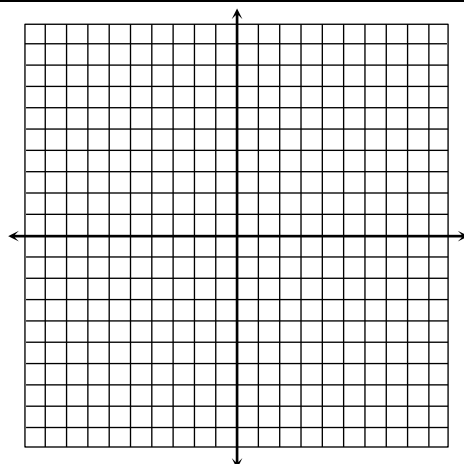
4. _____

5. _____

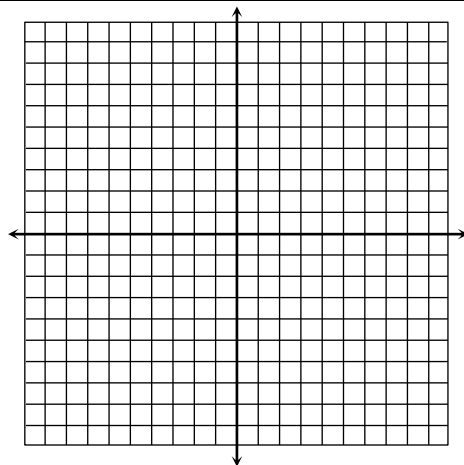
6. _____

7. _____

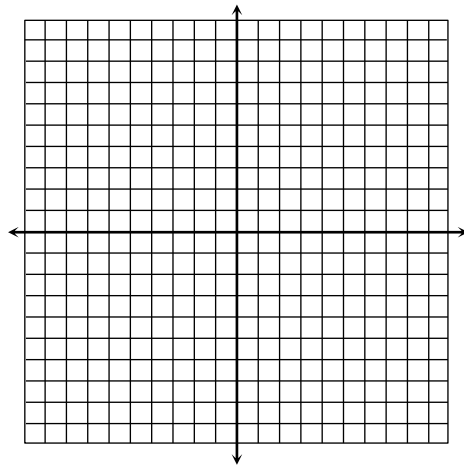
4. $f(x) = -x^2 - 11x - 28$



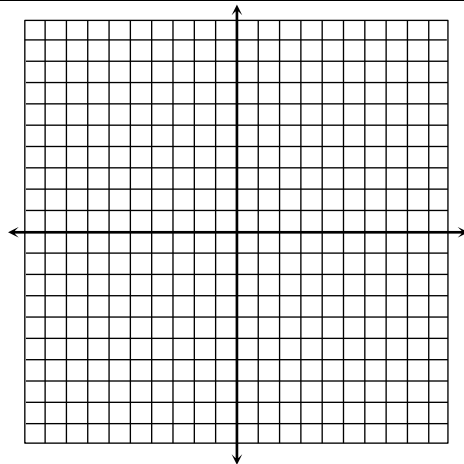
5. $f(x) = x^2 - 9$



6. $f(x) = 2x^2 + 6x$



7. $f(x) = -x^2 + 8x - 17$



Factoring Refresher

Quadratic equations can be solved by factoring as well.
Use the problems below to refresh your factoring skills ☺

SET 1 Greatest Common Factor	
1. $12k - 18$	2. $40x^8y + 64x^4y$
3. $14m^6 - 35m^3 - 7m^2$	4. $3ab^2 - 5a^2b + 8ab$

**** When factoring, always look for a _____ first! ****

SET 2 Trinomials ($ax^2 + bx + c$, where $a = 1$)	
5. $x^2 + 14x + 45$	6. $w^2 - 15w + 26$
7. $c^2 + 2c - 48$	8. $n^2 - n - 72$
9. $a^2 + 12a + 36$	10. $n^2 - 2n + 1$
11. $2k^2 - 16k - 40$	12. $5z^2 - 25z + 30$

SET 3 Trinomials ($ax^2 + bx + c$, where $a > 1$)	
13. $2x^2 - 15x + 18$	14. $5p^2 + 22p - 48$

15. $3h^2 - 19h - 40$	16. $12x^2 + 5x - 2$
17. $6y^2 - 5y - 21$	18. $16m^2 + 60m - 54$
19. $9x^2 - 12x + 4$	20. $8y^2 + 56y + 96$

SET 4 Difference of Squares	
21. $x^2 - 4$	22. $y^2 - 81$
23. $9x^2 - 25y^4$	24. $m^2n^2 - 49$
25. $72r^6 - 2s^2$	26. $2a^2b - 32a^4b$

Name: _____

Unit 4: Solving Quadratic Equations

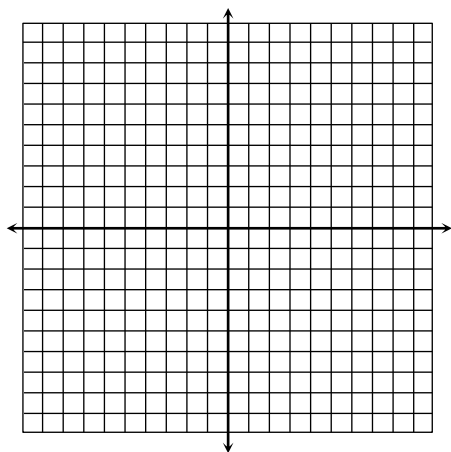
Date: _____ Per: _____

Homework 1: Solving Quadratics by Graphing
& Factoring Review

**** This is a 2-page document! ****

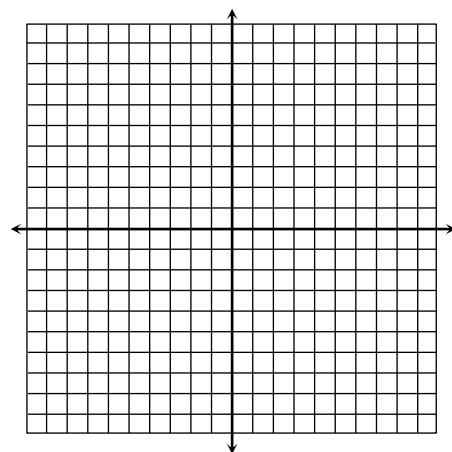
Directions: Find the solutions of each quadratic equation by graphing.

1. $f(x) = x^2 - 4x - 5$



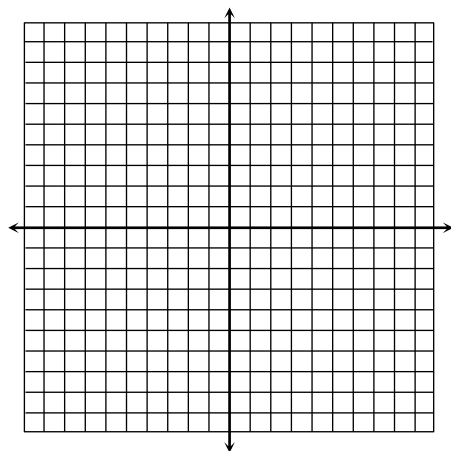
Solutions:

2. $f(x) = -x^2 + 15x - 56$



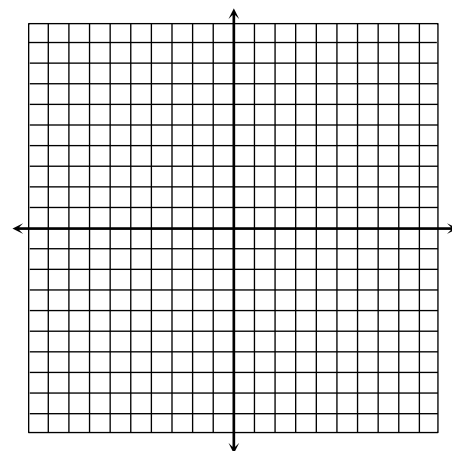
Solutions:

3. $f(x) = x^2 - 8x + 16$



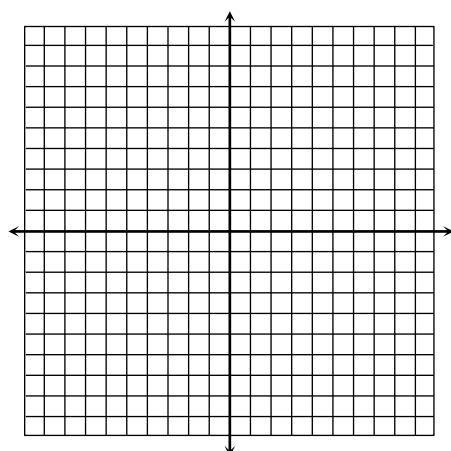
Solutions:

4. $f(x) = -3x^2 + 9x$



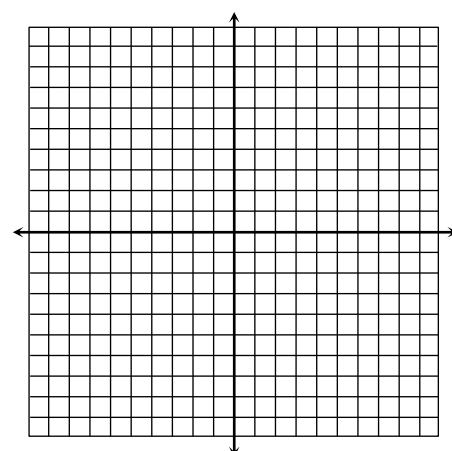
Solutions:

5. $f(x) = 2x^2 - 8$



Solutions:

6. $f(x) = \frac{1}{2}x^2 + 3x + 7$



Solutions:

Directions: Factor each expression. Be sure to check for a GCF first.

7. $x^2 + 10x + 24$

8. $n^2 + 12n - 85$

9. $6p^2 - 24p + 24$

10. $3a^2 - 3a - 36$

11. $2x^2 - 15x - 8$

12. $9m^2 + 24m + 7$

13. $16w^2 + 8w + 1$

14. $20y^2 - 34y + 6$

15. $k^2 - 25$

16. $9m^2 - 49$

17. $64x^2y - y$

18. $20c^2 - 45d^2$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
SOLVING QUADRATICS <i>By Factoring</i>	❶	MOVE ALL TERMS to one side and set the equation EQUAL TO 0 .
	❷	FACTOR!
	❸	SET EACH FACTOR EQUAL TO 0 and SOLVE for each x-value.
	❹	Write your answer as a SOLUTION SET . $x = \{_, _ \}$

Directions: Solve each equation by factoring.

1. $x^2 + 9x = 0$	2. $4x^2 - 8x = 0$
3. $x^2 - 4 = 0$	4. $9x^2 - 25 = 0$
5. $x^2 - 10x + 21 = 0$	6. $2x^2 - 2x - 24 = 0$
7. $3x^2 - 13x - 30 = 0$	8. $6x^2 + 5x - 4 = 0$

9. $2x^2 = 5x$

10. $7x^2 + 8x = x^2$

11. $2x^2 - 50 = x^2 - 1$

12. $3x^2 = 81 - x^2$

13. $x^2 + 2x = 80$

14. $x^2 + 11x + 29 = x + 4$

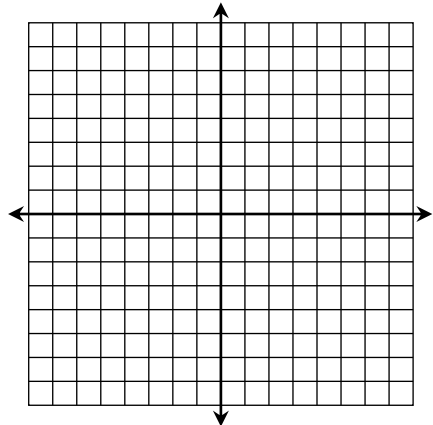
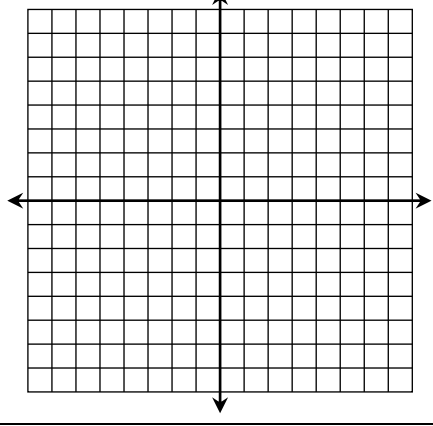
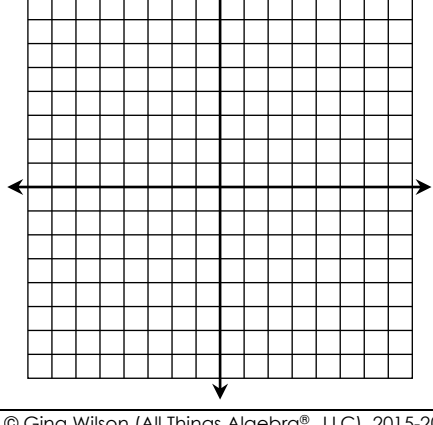
15. $2x^2 - 9x - 65 = x^2 - x$

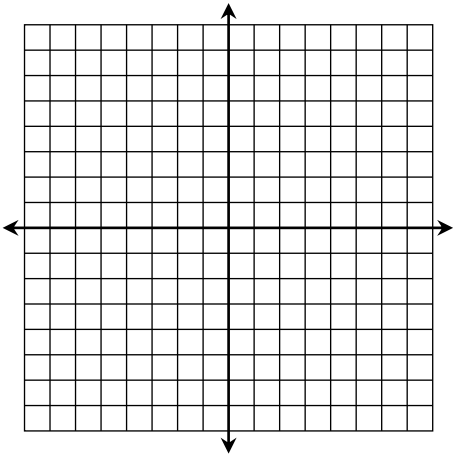
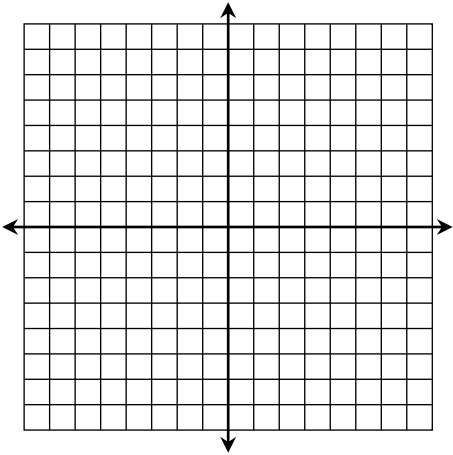
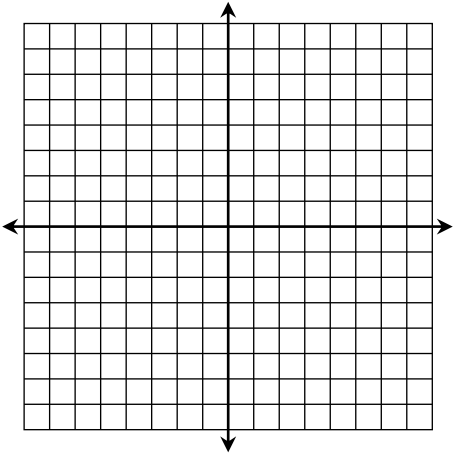
16. $4x^2 + 41x + 98 = 2 - 3x$

17. $2x^2 = x + 21$

18. $4x^2 - 11x = x - 9$

REPRESENTING QUADRATIC FUNCTIONS

STANDARD FORM	VERTEX FORM	FACTORED FORM	GRAPH
$f(x) = x^2 - 8x + 15$			
$f(x) = -2x^2 - 4x + 6$			
	$f(x) = -(x - 7)^2 + 1$		

STANDARD FORM	VERTEX FORM	FACTORED FORM	GRAPH
	$f(x) = -3(x - 1)^2 + 3$		
		$f(x) = (x + 3)(x + 7)$	
		$f(x) = -(x + 4)^2$	

Name: _____

Unit 4: Solving Quadratic Equations



Date: _____ Per: _____

Homework 2: Solving Quadratics by Factoring

**** This is a 2-page document! ****

Directions: Find the solutions of each quadratic equation factoring.

1. $x^2 - 4x = 0$

2. $10x^2 = 6x$

3. $x^2 - 36 = 0$

4. $16x^2 = 81$

5. $x^2 + 12x + 27 = 0$

6. $x^2 - 4x = 77$

7. $x^2 = x + 30$

8. $x^2 + 8x - 65 = x - 5$

9. $2x^2 + 6x - 80 = 0$

10. $5x^2 + 11x + 2 = 0$

11. $8x^2 - 2x = 15$

12. $12x^2 - 26x - 7 = 3x + 1$

Directions: Identify the roots of each quadratic function below.

13. $f(x) = (x + 5)(x + 6)$

14. $f(x) = -2(x + 9)(x - 1)$

15. $f(x) = x^2 - 5x + 6$

16. $f(x) = x^2 - 16x + 64$

17. $f(x) = (x - 1)^2 - 25$

18. $f(x) = -(x - 2)^2 + 4$

19. Give the vertex form and factored form of $f(x) = x^2 - 12x + 35$. What specific information does each form provide?

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
RADICALS Review	- A radical is any expression with a square root, cube root, etc. - Radical Symbol: _____ - List the first 20 perfect squares: _____	
SIMPLIFYING Radicals	Case 1: Perfect Squares – Take the square root of the number. Case 2: Non-Perfect Squares – Break the radical down using the largest perfect square. Take the square root of the perfect square, and leave the “leftover” under the radical symbol. For both cases: If there is a coefficient outside the radical, multiply this with any number you take out.	
Directions: Write each expression below in simplest radical form.		
1. $\sqrt{49}$	2. $\sqrt{225}$	3. $2\sqrt{144}$
4. $\sqrt{\frac{4}{9}}$	5. $\sqrt{\frac{1}{81}}$	6. $\sqrt{\frac{36}{121}}$
7. $\sqrt{112}$	8. $\sqrt{245}$	9. $5\sqrt{72}$
10. $9\sqrt{45}$	11. $\sqrt{250}$	12. $3\sqrt{578}$

SOLVING QUADRATICS by Square Roots	If a quadratic equation can be written in the form $x^2 = n$, then the quadratic can be solved using the Square Root Property :	
	If $x^2 = n$, then _____	
	❶	REWRITE the equation as $x^2 = n$
	❷	Take the SQUARE ROOT of both sides.
❸ SIMPLIFY the radical. Place " $\pm$ " to indicate both answers.		
Directions: Solve each equation using the square roots method.		
13. $x^2 - 64 = 0$	14. $7x^2 + 8 = 15$	15. $81x^2 + 5 = 21$
16. $8x^2 + 1 = 17$	17. $2x^2 - 9 = 55$	18. $9x^2 + 3 = 111$
19. $4 - 3x^2 = -77$	20. $5x^2 + 10 = 310$	21. $-\frac{1}{2}x^2 + 1 = -39$
22. $(x + 2)^2 = 49$	23. $3(x - 1)^2 = 18$	24. $-2(x + 7)^2 = -40$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 3: Solving Quadratics by
Square Roots

**** This is a 2-page document! ****

Directions: Solve each equation by square roots. Give all solutions in simplest radical form.

1. $x^2 - 100 = 0$

2. $25x^2 = 4$

3. $5x^2 - 11 = 234$

4. $\frac{1}{3}x^2 + 14 = 26$

5. $2x^2 - 3 = -3$

6. $9x^2 - 1 = 63$

7. $x^2 - 1 = 39$

8. $5x^2 = 90$

9. $4x^2 - 30 = 22$

10. $3x^2 + 4 = 166$

11. $10 - 2x^2 = -198$	12. $\frac{1}{4}x^2 - 7 = 25$
13. $(x - 4)^2 = 9$	14. $(x + 1)^2 = 75$
15. $2(x + 3)^2 = 56$	16. $\frac{3}{2}(x - 2)^2 = 72$
17. Which of the equations below can be solved by square roots? How would you solve the other equation? Explain your reasoning. Give the solutions to both equations. Equation A: $x^2 - 9x = 0$ Equation B: $x^2 - 9 = 0$	

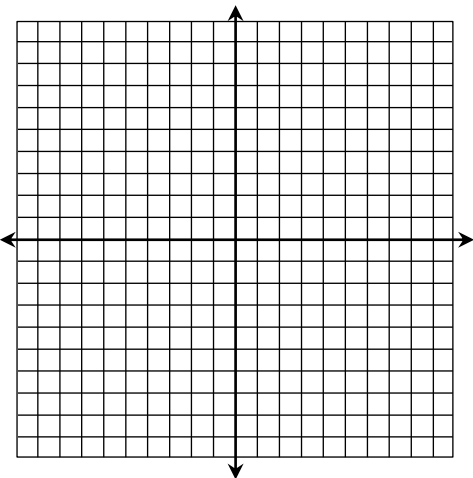
Name: _____ Algebra 2

Date: _____ Per: _____ Unit 4: Solving Quadratic Equations

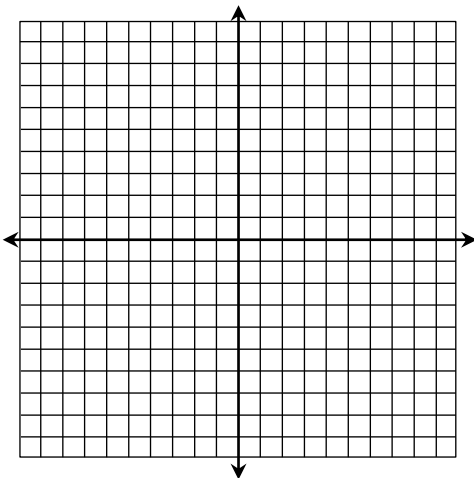
Quiz 4-1: Graphing, Factoring, & Square Roots Methods

For questions 1-2, find the roots of the function by graphing.

1. $f(x) = x^2 - 8x + 12$



2. $f(x) = -2x^2 - 4x - 2$



1. _____
2. _____

For questions 3-12, solve by factoring.

3. $x^2 + 5x - 36 = 0$

4. $x^2 + 61 = 1 - 17x$

3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

5. $3x^2 + 112 = 2x^2 + 22x$

6. $2x^2 + 7x = 42 - x$

7. $12x^2 + 17x = 5$

8. $25x^2 = 20x - 4$

9. $x^2 + 5x = 0$

10. $8x^2 + 30x = 2x$

9. _____

10. _____

11. _____

12. _____

11. $x^2 = 100$

12. $9x^2 - 6 = 10$

For questions 13-16, solve using the square roots method. Write answers in simplest radical form.

13. $x^2 - 121 = 0$

14. $4x^2 - 64 = 17$

13. _____

14. _____

15. _____

16. _____

15. $(x - 7)^2 = 60$

16. $\frac{1}{4}(x + 2)^2 = 27$

17. Write the function below in factored form and identify the axis of symmetry, vertex, and roots.

$f(x) = -(x + 3)^2 + 16$

Factored Form:	
Axis of Symmetry:	Vertex:
Roots:	

18. Write the function below in vertex form and identify the axis of symmetry, vertex, and roots.

$f(x) = 2(x + 9)(x + 5)$

Vertex Form:	
Axis of Symmetry:	Vertex:
Roots:	

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
IMAGINARY NUMBERS	- Equations such as $x^2 + 1 = 0$ have no real solution, so mathematicians defined the imaginary numbers to represent their solutions. - The imaginary unit, _____, is defined as _____. This is useful when working with square roots of negative numbers. - A pure imaginary number is written in the form _____, where _____ is the real number and _____ is the imaginary part.		
<i>Simplifying</i> NEGATIVE SQUARE ROOTS	Step 1: Rewrite $\sqrt{-a}$ as $\sqrt{-1 \cdot a}$ Step 2: Break a down if it is not a perfect square. Step 3: Simplify the radical, recalling that $\sqrt{-1} = i$.		
	1. $\sqrt{-9}$	2. $\sqrt{-196}$	3. $\sqrt{-5}$
	4. $\sqrt{-80}$	5. $\sqrt{-32}$	6. $\sqrt{-192}$
QUADRATIC EQUATIONS <i>with Imaginary Solutions</i>	Solve using the square roots method. Write all answers in simplest form.		
	7. $x^2 + 81 = 0$	8. $2x^2 + 9 = 1$	
	9. $4x^2 + 15 = -9$	10. $x^2 + 13 = 1$	

	11. $3x^2 - 5 = -446$	12. $-\frac{2}{3}x^2 - 1 = 17$
POWERS OF i TIP: Break apart into a power of i^2 to simplify!	$i^1 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^5 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^2 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^6 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^3 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^7 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^4 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$ $i^8 = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$	
	13. i^{15}	14. i^{62}
	15. i^{76}	16. i^{43}
	Simplify each expression.	
FINDING PRODUCTS WATCH OUT: Rewrite negative square roots as imaginary numbers BEFORE performing operations!	17. $4i \cdot 7i$	18. $(-4i)(2i)(-9i)$
	19. $(2i)^3 \cdot (5i)$	20. $(i\sqrt{3})^2 \cdot (-8i)^2$
	21. $\sqrt{-18} \cdot \sqrt{-10}$	22. $\sqrt{-24} \cdot \sqrt{-3} \cdot \sqrt{-2}$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 4: Pure Imaginary Numbers

**** This is a 2-page document! ******Directions:** Simplify the expressions below.

1. $\sqrt{-25}$

2. $\sqrt{-324}$

3. $\sqrt{-15}$

4. $\sqrt{-44}$

5. $\sqrt{-252}$

6. $\sqrt{-288}$

7. $\sqrt{-176}$

8. $\sqrt{-8} \cdot \sqrt{24}$

9. $\sqrt{-6} \cdot \sqrt{-12} \cdot \sqrt{-5}$

10. i^{28}

11. i^{49}

12. i^{86}

13. $8i \cdot -9i \cdot 6i$

14. $(-4i)^3 \cdot 2i$

15. $(2i)^5 \cdot (i\sqrt{6})^2$

Directions: Solve using the square roots method. Write all answers in simplest form.

16. $x^2 + 4 = 0$

17. $-3x^2 = 150$

18. $2x^2 + 71 = -163$

19. $18 - x^2 = 39$

20. $-4x^2 - 3 = 481$

21. $-\frac{4}{3}x^2 - 15 = 201$

Directions: The solutions of a quadratic equation are given below. Give an example of an equation with the given solutions.

22. $\pm 4i$

23. $\pm i\sqrt{7}$

24. $\pm 3i\sqrt{2}$

25. $\pm 8i\sqrt{3}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
COMPLEX NUMBERS $\mathbb{C}$	- A complex number is a number written in the form _____ where _____ is the real part and _____ is the imaginary part. Examples: _____	
ADDING & SUBTRACTING <i>Complex Numbers</i>	Directions: Simplify the expressions below. Final answers must be in $a + bi$ form.	
	1. $(-11 + 3i) + (9 + 2i)$	2. $(4 + i) + (7 - 5i)$
	3. $(7 - 2i) - (2 + 6i)$	4. $6i - (14 - i) + (5 - 3i)$
MULTIPLYING <i>Complex Numbers</i> **Recall**	Directions: Simplify the expressions below. Final answers must be in $a + bi$ form.	
	5. $2i(8 - 3i)$	6. $-i(-2 + 10i)$
	7. $(7 + i)(4 - i)$	8. $(2 - 4i)(-5 - 3i)$
	9. $(6 - 2i)^2$	10. $(1 + 7i)(9 + 3i) - (4 + 2i)$

COMPLEX CONJUGATES	Two complex numbers in the form $a + bi$ and $a - bi$ are called complex conjugates. The product of two conjugates is always a real number.	
	11. $(8 + i)(8 - i)$	12. $(3 - i)(3 + i)$
	13. $(5 - 4i)(5 + 4i)$	14. $(-7 + 2i)(-7 - 2i)$
QUADRATIC EQUATIONS <i>with Complex Solutions</i>	Solve using the square roots method. Write all answers in simplest form.	
	15. $(x + 2)^2 = -81$	16. $(x - 6)^2 = -32$
	17. $-(x + 4)^2 = 75$	18. $2(x - 1)^2 = -56$
	19. $-\frac{1}{2}(x - 7)^2 = 27$	20. $\frac{3}{4}(x + 3)^2 = -96$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 5: Complex Numbers

**** This is a 2-page document! ******Simplify each expression below.**

1. $(11 - 7i) + (2 - 5i)$

2. $(-6 + 10i) - (1 - 2i)$

3. $(7 + 3i) + (-1 - 14i)$

4. $(-3 + 8i) - (1 + 8i)$

5. $-5i(-2 - i)$

6. $8i(3 - 6i)$

7. $(7 + i)(4 + 3i)$

8. $(-3 - 8i)(1 - 2i)$

9. $(-7 + i)^2$

10. $(5 - 2i)(2 + 7i) - 2i(3i)^2$

11. $(3 + 5i)(3 - 5i)$

12. $(-2 - 9i)(-2 + 9i)$

Solve using the square roots method. Write all answers in simplest form.

13. $(x - 1)^2 = -16$

14. $(x + 4)^2 = -90$

15. $(x + 9)^2 = -24$

16. $-(x - 5)^2 = 108$

17. $-3(x + 2)^2 = 24$

18. $5(x - 3)^2 = -225$

19. $\frac{1}{3}(x - 7)^2 = -28$

20. $-\frac{5}{2}(x + 1)^2 = 30$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
DIVIDING Complex Numbers WATCH OUT! "i" can not be in the denominator of a complex number.	CASE 1	THE DENOMINATOR IS A MONOMIAL: Multiply both the numerator and denominator by "i"
	1. $\frac{10}{2i}$	2. $\frac{4}{-9i}$
	3. $\frac{-7}{5i}$	4. $\frac{28-8i}{4i}$
	5. $\frac{1+4i}{2i}$	6. $\frac{-8-2i}{6i}$
BINOMIAL Denominators	CASE 2	THE DENOMINATOR IS A BINOMIAL: Multiply both the numerator and denominator by the conjugate of the denominator.
	7. $\frac{-2}{5-i}$	8. $\frac{5i}{6+2i}$

$$9. \frac{3i}{4-3i}$$

$$10. \frac{2}{-3-5i}$$

$$11. \frac{7i}{-4+8i}$$

$$12. \frac{7+3i}{2+i}$$

$$13. \frac{1+8i}{2-4i}$$

$$14. \frac{-3+7i}{-9+i}$$

$$15. \frac{9+6i}{5+5i}$$

$$16. \frac{-8-4i}{-4-3i}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ORGANIZING <i>Number Sets</i>	Pure imaginary numbers and real numbers make up the complex numbers. The complex numbers and its subsets are organized as:	
CLASSIFYING <i>Numbers</i>	Directions: Simplify, then name all sets to which each value belongs.	
	1. $\sqrt{-64}$	2. $10i \cdot -2i$
	3. $\sqrt{\frac{1}{9}}$	4. $(15 - 8i) - (9 - 3i)$
	5. $(4 - i)(9 + 5i)$	6. i^{41}
	7. $(-2i)^3 \cdot (i\sqrt{7})^2$	8. $(3 - 5i)(3 + 5i)$
	9. i^{16}	10. $\sqrt{150}$

PROPERTIES OF COMPLEX NUMBERS

The properties of real numbers (Commutative, Associative, Identity, Inverse, and Distributive) can also be applied to complex numbers.

Directions: For questions 11 – 15, write an example of the property using the complex numbers.

11. Commutative Property of Addition: _____
12. Associative Property of Multiplication: _____
13. Distributive Property: _____
14. Identity Property of Multiplication: _____
15. Inverse Property of Addition: _____

Directions: For questions 16 – 21, name the property that justifies the statement.

16. $2i(7 - 5i) = 2i \cdot 7 - 2i \cdot 5i$

17. $\frac{1}{3i} \cdot 3i = 1$

18. $(-5 + 4i) + 9i = -5 + (4i + 9i)$

19. $(-12 + i) + 0 = (-12 + i)$

20. $(-6i \cdot 2i) \cdot 8i = 8i \cdot (-6i \cdot 2i)$

21. $(7 - 2i) + -(7 - 2i) = 0$

22. Name the additive and multiplicative identity of $6i$.

23. Name the additive and multiplicative inverse of $(-8 - 3i)$.

Directions: For questions 24 – 28 answer true or false. If false, provide a counterexample.

24. The set of complex numbers is closed under addition. _____
25. The set of complex numbers is closed under division. _____
26. All complex numbers are pure imaginary numbers. _____
27. A number cannot be both irrational and complex. _____
28. The product of a real number and a complex number is a real number. _____

Name: _____

Unit 4: Solving Quadratic Equations



Date: _____ Per: _____

Homework 6: Dividing Complex Numbers;
Classifying & Properties

**** This is a 2-page document! ****

Simplify each expression below.

1. $\frac{6}{i}$

2. $\frac{-4}{7i}$

3. $\frac{18}{9i}$

4. $\frac{8+4i}{2i}$

5. $\frac{-35-15i}{5i}$

6. $\frac{2+i}{3i}$

7. $\frac{6i}{1-i}$

8. $\frac{10}{6+2i}$

9. $\frac{2i}{-9+7i}$

10. $\frac{4+10i}{3+i}$

11. $\frac{4-i}{4-5i}$	12. $\frac{5-8i}{-1-4i}$
Simplify, then name all sets to which each value belongs.	
13. i^{67}	14. $(-8i^9)^2$
15. $(11-7i) - (-4-2i)$	16. $\sqrt{-63}$
17. $(-9+4i)(-9-4i)$	18. $\sqrt{\frac{49}{25}}$
For questions 19-20, write an example of each property using the complex numbers.	
19. Commutative Property of Multiplication	20. Distributive Property
21. Name the additive inverse of $5+i$.	22. Name the multiplicative identity of $-12i$.
For questions 23-24, answer true/false. If false, give a counterexample.	
23. The set of complex numbers is closed under subtraction.	24. The set of irrational numbers is closed under multiplication.

Name: _____

Algebra 2

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Unit 4: Solving Quadratic Equations

Quiz 4-2: Complex Numbers, Equations with Complex Solutions

Simplify each expression.

1. $\sqrt{-64}$

2. $2\sqrt{-108}$

3. $(-5i)(-3i)$

4. $(i\sqrt{7})^2 \cdot 2i$

5. $(8 + 3i) + (-6 - 12i)$

6. $(5 - 9i) - (1 + 4i)$

7. $(9 + 5i)(4 - 2i)$

8. $(-1 - 3i)^2$

9. $\frac{20 + 8i}{4i}$

10. $\frac{-4 - 4i}{10 - 2i}$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

Simplify, then name all sets to which each value belongs.

11. i^{49}

12. $(6 + i)(6 - i)$

11. _____

Sets: _____

12. _____

Sets: _____

Solve using the square root property. Give all solutions in simplest form.

13. $x^2 + 100 = 0$

14. $2x^2 + 27 = -13$

13. _____

14. _____

15. _____

16. _____

17. _____

18. _____

19. _____

20. _____

15. $11 - 4x^2 = 187$

16. $\frac{4}{3}x^2 - 6 = -70$

17. $(x - 5)^2 = -16$

18. $-(x + 2)^2 = 54$

19. $3(x + 1)^2 = -240$

20. $-\frac{1}{2}(x - 7)^2 = 96$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
REVIEW: <i>Perfect Square Trinomials</i>	Factor the following trinomials:	
	$x^2 + 4x + 4$ _____ $x^2 - 20x + 100$ _____	$x^2 - 14x + 49$ _____ $x^2 + 16x + 64$ _____
	These are called perfect square trinomials . If you have a perfect square trinomial, you can solve the quadratic equation by square roots! Solve the equations below by factoring and using the square root property.	
	1. $x^2 + 14x + 49 = 4$	2. $x^2 - 6x + 9 = 20$
COMPLETING THE SQUARE <i>Method</i>	We can take any quadratic equation, create a perfect square trinomial, and solve it in a similar way. This method is called completing the square .	
	Steps	Example
	① Rewrite as $ax^2 + bx = c$.	$x^2 + 8x - 34 = 0$
	② Divide both sides by "a" to create $x^2 + bx = c$.	
	③ Take half of b , square it, then add this to both sides of the equation.	
	④ Factor the perfect square trinomial.	
	⑤ Take the square root of both sides and solve for both cases.	
	⑥ Simplify all irrational or complex solutions.	

RATIONAL <i>Solutions</i>	Solve by completing the square. Give all solutions in simplest form.	
	3. $x^2 - 18x + 56 = 0$	4. $2x^2 - 16x = -30$
	5. $4x^2 - 8x = -3$	6. $3x^2 + 10x + 8 = 0$
	7. $x^2 + 16x - 21 = -5$	8. $3x^2 - 30x = 69$
IRRATIONAL <i>Solutions</i>		

	9. $x^2 - 6x - 31 = 0$	10. $2x^2 + 4x - 57 = 87$
COMPLEX <i>Solutions</i>	11. $x^2 + 12x + 43 = 0$	12. $x^2 - 18x + 97 = 0$
	13. $4x^2 + 76 = 16x$	14. $3x^2 - 6x + 80 = -4$

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 7: Solving Quadratics by
Completing the Square



**** This is a 2-page document! ****

Directions: Solve each equation by completing the square. Give all solutions in simplest form.

1. $x^2 + 4x + 22 = 67$

2. $x^2 = 16x - 63$

3. $2x^2 - x - 3 = 0$

4. $4x^2 + 24x = -35$

5. $x^2 - 12x - 44 = 0$

6. $2x^2 - 16x + 8 = 0$

7. $x^2 + 10x + 1 = 24$

8. $3x^2 = 3 - 42x$

9. $x^2 + 2x + 37 = 0$

10. $5x^2 - 30x + 85 = 0$

11. $x^2 - 14x + 62 = -77$

12. $2x^2 - 20x + 89 = -87$

Name:	Date:
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Topic:	Class:
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Main Ideas/Questions	Notes/Examples	
THE QUADRATIC FORMULA	①	Write the equation in STANDARD FORM , $ax^2 + bx + c = 0$
	②	Identify a , b , and c . SUBSTITUTE them into the formula.
	③	SIMPLIFY!
$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$		

Directions: Solve each quadratic equation below using the quadratic formula.

1. $x^2 + 2x - 63 = 0$	2. $6x^2 - 11x - 10 = 0$
3. $-x^2 + 2x = 1$	4. $-2x^2 + 3x = 4x - 15$
5. $9x^2 - 1 = 24$	6. $8x^2 = 10x$

7. $x^2 - 9 = 4x$

8. $-2x^2 + 17 = 8x - 1$

9. $-8x^2 + 7 = 4$

10. $-x^2 + 10x = 8$

11. $-3x^2 + 8x - 7 = 0$

12. $5x^2 + 9 = 4x + 5$

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 8: The Quadratic Formula



**** This is a 2-page document! ****

Directions: Solve each equation by the quadratic formula.

1. $x^2 + 12x - 8 = 0$

2. $-2x^2 + 7x = 3$

3. $-x^2 + 5x - 6 = 2$

4. $13x^2 - 16x = x^2 - x$

5. $8x^2 + 15x = 7x - 4$

6. $-x^2 + 18 = 93$

7. $3x^2 + 5x = 8$

8. $5x^2 + 1 = 4x - 7$

9. $7x^2 - 10x + 9 = 5 - 2x$

10. $-12x^2 + 21 = 0$

11. $-2x^2 + 24 = 8x$

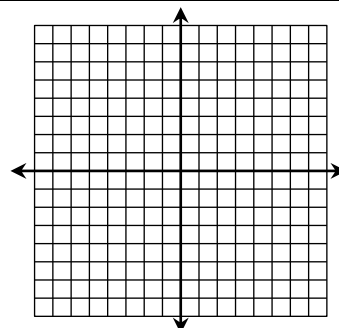
12. $3x^2 + x = -2x - 7$

METHODS FOR SOLVING QUADRATIC EQUATIONS

Directions: Solve the equation below using each method.
If a method can't be used, explain why.

$$x^2 + 6x + 5 = 0$$

GRAPHING



FACTORING

SQUARE ROOTS

**COMPLETING
THE SQUARE**

**QUADRATIC
FORMULA**

Name:	Date:
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Main Ideas/Questions	Notes/Examples		
THE DISCRIMINANT	The discriminant of a quadratic equation is a numerical value that reveals both the number and type of roots a quadratic equation has.		Formula:
	Value of d	# of Roots	Type of Roots
	$d > 0$ (a perfect square)		
	$d > 0$ (NOT a perfect square)		
	$d = 0$		
	$d < 0$		
EXAMPLES	Find the value of the discriminant, then determine the number and type of roots.		
	1. $2x^2 + 12x - 13 = 0$	2. $-x^2 + 16x - 64 = 0$	
	3. $-4x^2 + 84 = 10x$	4. $3x^2 + 24 = 0$	
<i>Choosing the</i> BEST METHOD	Method	Can be used...	When?
	GRAPHING		
	FACTORING		
	SQUARE ROOTS		
	COMPLETING THE SQUARE		
	QUADRATIC FORMULA		

Choosing the Best Method to Solve a Quadratic Equation

Step 1: Find the discriminant of each equation and give the number and type of roots.

EQUATION A

$$x^2 - 9x + 21 = 0$$

EQUATION B

$$-2x^2 + 144 = 0$$

EQUATION C

$$3x^2 - 22x + 7 = 0$$

EQUATION D

$$x^2 + 6x - 15 = 0$$

Step 2: Solve each equation using the methods below. You can only use each method once.

FACTORING

☐

SQUARE ROOTS

☐

COMPLETE THE SQUARE

☐

THE QUADRATIC FORMULA

☐

Be prepared to justify your choices!

Group Members:

Per: _____



REVIEW: *Solving Quadratic Equations*

Directions: Solve using the indicated method. Work together to complete each problem. Do not divide up the work! Each person should be participating.

FACTORING

1. $x^2 - 7 = 6x$

2. $4x^2 + 35x + 24 = 0$

SQUARE ROOTS

3. $10x^2 + 2 = 282$

4. $2x^2 + 50 = -166$

COMPLETING THE SQUARE

5. $x^2 + 12x + 3 = 7$

6. $6x^2 - 12x - 38 = 10$

QUADRATIC FORMULA	7. $5x^2 + 4x + 1 = 0$	8. $-2x^2 + 2x = 5$

CHOOSE EACH METHOD ONCE!	9. $2x^2 = 3x + 10$ ☐ F ☐ SR ☐ CS ☐ QF	10. $x^2 + 10x = 2x$ ☐ F ☐ SR ☐ CS ☐ QF
	11. $4x^2 - 39 = 10$ ☐ F ☐ SR ☐ CS ☐ QF	12. $-x^2 + 2x = 7$ ☐ F ☐ SR ☐ CS ☐ QF

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 9: Solving Quadratics Review

**** This is a 2-page document! ****

Directions: Solve each equation using the indicated method.

1. $x^2 = 6x + 91$

(Factoring)

2. $-2x^2 + 146 = -34$

(Square Roots)

3. $x^2 - 14x + 53 = 9$

(Completing the Square)

4. $3x^2 + 6x - 62 = 7$

(Completing the Square)

5. $-x^2 + 7x = 1$

(Quadratic Formula)

6. $5x^2 = x - 2$

(Quadratic Formula)

7. Step 1: Find the discriminant of each equation, then determine the number and type of roots.

EQUATION A

$$-x^2 + 4x - 11 = 0$$

EQUATION B

$$4x^2 - 20x + 25 = 0$$

EQUATION C

$$-3x^2 - 11x + 2 = 0$$

EQUATION D

$$\frac{1}{2}x^2 - 8 = 0$$

Step 2: Solve each equation using the methods below. You can only use each method once.

FACTORING

☐

SQUARE ROOTS

☐

COMPLETE THE SQUARE

☐

THE QUADRATIC FORMULA

☐

Name: _____

Algebra 2

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Unit 4: Solving Quadratic Equations

Quiz 4-3: Solving Quadratic Equations (All Methods)

Solve by factoring:

1. $3x^2 - 3x - 60 = 0$

2. $4x^2 + 31x = 3x$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

Solve by square roots:

3. $9x^2 - 18 = 7$

4. $\frac{1}{3}x^2 - 7 = 43$

Solve by completing the square:

5. $2x^2 + 8x + 134 = 0$

Solve by the quadratic formula:

6. $x^2 + 12x = 7x + 1$

7. Step 1: Find the discriminant of each equation, then determine the number and type of solutions.

A. $-2x^2 - 196 = 0$	B. $2x^2 - 9x + 8 = 0$	C. $4x^2 - 21x - 18 = 0$	D. $x^2 + 18x + 97 = 0$
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Step 2: Solve each equation using the methods below. Each method may only be used once.

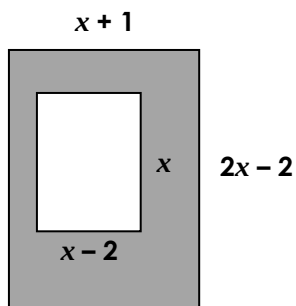
FACTORING <input data-bbox="688 732 782 827" type="checkbox"/>	SQUARE ROOTS <input data-bbox="1388 732 1482 827" type="checkbox"/>
COMPLETE THE SQUARE <input data-bbox="688 1369 782 1463" type="checkbox"/>	THE QUADRATIC FORMULA <input data-bbox="1388 1369 1482 1463" type="checkbox"/>

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
GEOMETRIC <i>Problems</i>	1. The width of a rectangle is six meters less than its length. If the area of the rectangle is 112 m^2 , find the dimensions of the rectangle.	2. The length of a rectangle is one foot more than twice its width. If the area of the rectangle is 300 ft^2 , find the dimensions of the rectangle.
	3. The dimensions of a rectangle can be expressed as $x + 6$ and $x - 2$. If the area of the rectangle is 65 in^2 , find the dimensions of the rectangle.	4. The base of a triangle is seven less than twice its height. If the area of the triangle is 102 cm^2 , find the length of the base.
	5. The hypotenuse of a triangle is one foot more than twice the length of the shorter leg. The longer leg is seven feet longer than the shorter leg. Find the dimensions of the triangle.	6. The height of a rectangular prism is 3 feet. The length and width can be expressed as $x - 1$ and $x + 4$. If the volume of the prism is 198 ft^3 , find the value of x .

7. Ken has a rectangular shaped vegetable garden that is 15 feet by 9 feet. He would like to increase the length and width by the same amount, x , so the new garden is twice the current area. Find the value of x to the nearest tenth of a foot.

8. Given the diagram to the left, if the area of the shaded region is 78 square inches, what are the dimensions of the outside rectangle?



CONSECUTIVE INTEGER

Problems

9. Find two consecutive positive odd integers such that the square of the smaller integer subtracted from 13 times the larger integer is 68.

10. Find three consecutive positive integers such that the product of the median and largest integer is 8 times the sum of the smallest and largest integer.

Name: _____

Unit 4: Solving Quadratic Equations



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Homework 10: Quadratic Equation Applications

**** This is a 2-page document! ****

1. The length of a rectangle is 3 meters less than twice its width. The area of the rectangle is 104 m^2 . Find the dimensions of the rectangle.

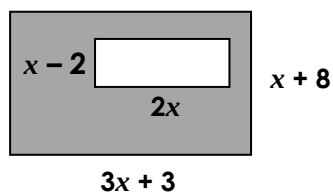
2. The dimensions of a square are altered so that one dimension is increased by 5 feet, while the other is decreased by 3 feet. The area of the resulting rectangle is 84 ft^2 . What was the area of the original square?

3. The shorter leg of a right triangle is 8 less than the hypotenuse. The longer leg is 1 less than the hypotenuse. Find the perimeter of the triangle.

4. The height of a trapezoid can be expressed as $x - 4$, while the bases can be expressed as $x + 4$ and $x + 9$. If the area of the trapezoid is 99 cm^2 , find the length of the larger base.

5. An inground rectangular pool has a concrete pathway surrounding the pool. If the pool is 16 feet by 32 feet and the entire area of the pool including the walkway is 924 ft^2 , find the width of the walkway.

6. Given the diagram below, if the area of the shaded region is 290 ft^2 , what are the dimensions of the inside rectangle?



7. Find two positive even consecutive integers such that one less than the square of the smaller integer is three more than the four times the larger integer.

8. Find three positive consecutive integers such that the product of the smallest integer and largest integer is 47 more than 8 times the median integer.

Name:

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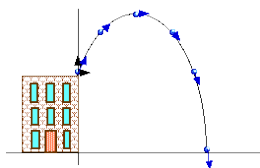
Class:

Main Ideas/Questions

Notes/Examples

PROJECTILE MOTION

Problems



1. The height h , in feet, of a golf ball t seconds after it was hit from the ground can be modeled by the function $h(t) = -16t^2 + 200t$. Find **a)** the time it will take the ball to reach its maximum height and **b)** the maximum height of the ball.

2. A rock is thrown off a bridge 40 feet above a river at an initial velocity of 96 feet per second. The height h , in feet, of the rock t seconds after it was thrown can be modeled by the function $h(t) = -16t^2 + 96t + 40$. Find the height of the rock 4 seconds after it was thrown.

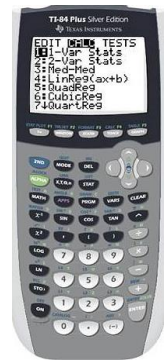
3. A cannon launched a cannonball from a height of 4 feet with an upward velocity of 125 feet per second. The height h , in feet, of the cannonball t seconds after it was launched can be modeled by the function $h(t) = -16t^2 + 125t + 4$. Find **a)** the maximum height of the cannonball and **b)** the time, in seconds, it will take the cannonball to reach the ground.

4. A dirt bike launches from a 9-foot-tall ramp at an initial velocity of 30 feet per second. The height h , in feet, of the bike t seconds after it launched from the ramp can be modeled by the function $h(t) = -16t^2 + 30t + 9$. After how many seconds will the bike reach the ground?

5. The height h , in feet, of a kangaroo t seconds after it jumps from the ground can be modeled by the function $h(t) = -16t^2 + 28t$. What is the maximum height of the kangaroo?
6. An arrow is shot from a bow from a height of 4 feet with an initial velocity of 150 feet per second. **a)** Using the formula $h(t) = -16t^2 + vt + s$, if v is the initial velocity, and s is the initial height, write an equation to model the height h of the arrow t seconds after the arrow was shot. Then **b)** find the height of the arrow 7 seconds after it was shot.
7. A diver jumps from a diving board 32 feet above the water with an initial velocity of 12 feet per second. The height h , in feet, of a the diver t seconds after he jumped can modeled by the function $h(t) = -16t^2 + 12t + 32$. How many seconds will it take the diver to reach the surface of the water?
8. The height h , in feet, of a a soccer ball t seconds after it was hit by a player with his head can be modeled by the function $h(t) = -16t^2 + 20t + 6.5$. What is the domain and range of this function?
9. The height h , in feet, of a baseball t seconds after it is thrown can be modeled by the function $h(t) = -16t^2 + 86t + 5$. If the ball is caught when it is two feet from the ground, how long was the ball in the air?

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
QUADRATIC Regression	In a previous unit, you learned how to take data following a linear pattern and write a line of best fit. Sometimes, data follows a curve, and it's more appropriate to write a curve of best fit using quadratic regression.
	Enter your data. Find the curve of best fit.
	- Hit STAT, ENTER - Enter x-values in L₁ - Enter y-values in L₂ - Hit STAT - Arrow over to CALC - Choose #5-QuadReg - Hit ENTER twice



EXAMPLES													
1. An amusement park ride takes riders to the top of a tower and drops them. The table below shows the height, h, of the tower at time t seconds after drop.	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="padding: 5px;">t</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">0.5</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">1.5</td> <td style="padding: 5px;">2</td> </tr> <tr> <td style="padding: 5px;">h</td> <td style="padding: 5px;">200</td> <td style="padding: 5px;">164</td> <td style="padding: 5px;">120</td> <td style="padding: 5px;">68</td> <td style="padding: 5px;">8</td> </tr> </table>	t	0	0.5	1	1.5	2	h	200	164	120	68	8
t	0	0.5	1	1.5	2								
h	200	164	120	68	8								
a. Write an equation for the curve of best fit.	b. How many seconds will it take the tower to reach the ground?												
2. Braden jumped off a cliff and into the ocean while vacationing in Hawaii. His height h after time t seconds is given in the table below.	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="padding: 5px;">t</td> <td style="padding: 5px;">0</td> <td style="padding: 5px;">1</td> <td style="padding: 5px;">2</td> <td style="padding: 5px;">3</td> <td style="padding: 5px;">4</td> </tr> <tr> <td style="padding: 5px;">h</td> <td style="padding: 5px;">480</td> <td style="padding: 5px;">482</td> <td style="padding: 5px;">452</td> <td style="padding: 5px;">390</td> <td style="padding: 5px;">296</td> </tr> </table>	t	0	1	2	3	4	h	480	482	452	390	296
t	0	1	2	3	4								
h	480	482	452	390	296								
a. Write an equation for the curve of best fit.	b. How long will it take Braden to reach a height of 150 feet?												

3. Sarah is standing on a bridge that stands 50 feet above the water. She tosses a coin into the water. The height, h , of the coin at time t seconds is given in the table below.

t	0	0.5	1	1.5	2
h	50	64	70	68	58

- a. Write an equation for the curve of best fit. b. What is the height of the coin after 3 seconds?

4. The table below shows how wind affects a runner's performance in the 200 meter dash, where s represents the speed of the wind in meters per second and t represents the change in finishing time. Negative wind speed means the runner is running against the wind while positive wind speed means the runner is running with the wind.

s	-4	-2	0	2	4
t	1.48	0.6	0	-0.5	-1.52

- a. Write an equation for the curve of best fit. b. Predict the change in finishing time when the wind is -8 meters per second.

5. A study compared the speed in miles per hour, x , and the average fuel economy in miles per gallon, y , for cars. The results are shown in the table below.

x	20	30	40	50	60	70
y	25.2	28.5	30.1	30.4	27.8	25.0

- a. Write an equation for the curve of best fit. b. Predict the fuel economy for a car traveling at 35 miles per hour.

6. The table below shows the average sale price, p , of a house in Norfolk, Virginia in thousands of dollars during certain years since 2005, t .

t	2005	2006	2007	2008	2009	2010
p	158	145	132	130	138	142

- a. Write an equation for the curve of best fit. b. Predict the value of the house in 2015.

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Bell: _____

Homework 11: Projectile Motion &
Quadratic Regression

**** This is a 2-page document! ****

1. A pumpkin is launched directly upwards at 72 feet per second from a platform 24 feet high. The height h of the pumpkin (in feet), t seconds after launch can be represented by the function $h(t) = -16t^2 + 72t + 24$. Find the maximum height of the pumpkin and the time it takes to reach this point.
2. A squirrel is 24 feet up in a tree and tosses a nut out of the tree with an initial velocity of 8 feet per second. The nut's height, h (in feet) at time t seconds can be represented by the function $h(t) = -16t^2 + 8t + 24$. If the squirrel climbs down the tree in 2 seconds, does it reach the ground before the nut?
3. A quarterback throws a football from a height of 6 feet with an initial velocity of 20 feet per second. The height, h , of the ball at time t seconds can be represented by the function $h(t) = -16t^2 + 20t + 6$. If the ball is caught at 2 feet, find the range of the function.
4. A penny is dropped from the top of the Statue of Liberty, which is 305 feet tall. The height of the penny, h , at time t seconds can be represented by the function $h(t) = -16t^2 + 305$. After 4 seconds, find the height of the ball and the time remaining until it reaches the ground.

5. Ben and Josh went to the roof of their 40-foot-tall high school to throw their math books off the edge. The initial velocity of Ben's book was 60 feet per second while the initial velocity of Josh's book was 48 feet per second. The function $h(t) = -16t^2 + vt + s$ can be used to represent the situation where h is the height of the textbook at time t seconds, v is the initial upward velocity, and s is the starting point. Whose textbook reaches the ground first and by how many seconds?

6. The table below gives the average cost (to the nearest hundred) of tuition and fees, y , at a four-year college in years since 1991, x .

x	1991	1994	1998	2000	2005	2008
y	5,500	6,200	7,000	7,200	\$8,800	\$9,600

- a. Write an equation for the curve of best fit. _____
- b. During what year will tuition and fees reach \$15,000? _____

7. The table below shows the hourly temperatures, for a city in Ohio on February 10, 2012, as predicted by the weatherman at the local news station at certain times since 10 a.m..

Time	10 a.m.	11 a.m.	12 p.m.	1 p.m.	2 p.m.	3 p.m.
Temp ($^{\circ}F$)	18	20	22	23	25	22

- a. Write an equation for the curve of best fit. _____
- b. Predict the temperature at 7:00 p.m. _____

Unit 4 Test Study Guide

(Solving Quadratics & Complex Numbers)

Name: _____

Date: _____ Per: _____

Topic 1: Complex Numbers

Simplify each expression below.

1. $\sqrt{-324}$	2. $2\sqrt{-147}$	3. $(-4+7i)+(-3-8i)$	4. $(21+9i)-(13-2i)$
5. $(-3i^7)^4 \cdot i^{16}$	6. $(1+5i)(4-2i)$	7. $(7-i)^2$	
8. $\frac{8}{-12i}$		9. $\frac{3-6i}{4-3i}$	

Simplify, then name all sets to which the value belongs.

10. i^{59}	11. $(9 + 5i)(9 - 5i)$
--------------	------------------------

For questions 11 and 12, use the complex numbers to write an example of each property.

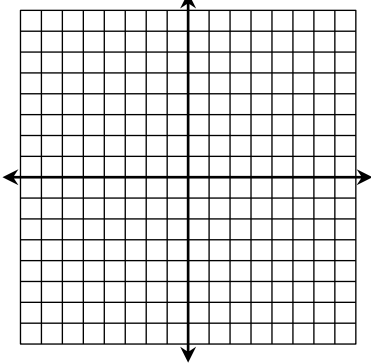
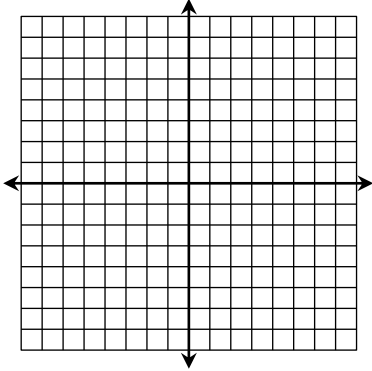
12. Associative Property of Multiplication	13. Distributive Property
14. Name the additive identity of $(-10 + 4i)$.	15. Name the multiplicative inverse of $7i$.

16. Name all sets that are closed under subtraction.

Topic 2: Standard, Vertex, and Factored Forms of Quadratic Functions**Complete the table below.**

Standard Form	Vertex Form (Identify Vertex & Axis of Symmetry)	Factored Form (Identify the Roots)
17. $f(x) = x^2 + 16x + 63$		
18.	$f(x) = (x + 2)^2 - 16$	

Topic 3: Solving Quadratics by Graphing

19. $x^2 + 10x + 24 = 0$ 	20. $-3x^2 - 6x = 0$ 
---	---

Topic 4: Solving Quadratics by Factoring

21. $x^2 - 11x + 18 = 0$	22. $2x^2 - 32x + 128 = 0$
--------------------------	----------------------------

23. $8x^2 + 10x = 0$

24. $7x^2 - 19x - 6 = 0$

Topic 5: Solving Quadratics by Square Roots

25. $16x^2 - 1 = 0$

26. $-3x^2 + 11 = 95$

27. $(x + 5)^2 = 4$

28. $(x - 8)^2 - 7 = 25$

Topic 6: Solving Quadratics by Completing the Square

29. $x^2 - 8x - 10 = 0$

30. $-4x^2 - 48x - 20 = 0$

Topic 7: Solving Quadratics by the Quadratic Formula

31. $-x^2 + 3x - 21 = 0$	32. $10x^2 + 8x - 1 = 0$
---------------------------------	---------------------------------

Topic 8: Discriminant/Choosing the Best Method

Find the discriminant of each equation and determine the number and type of roots. Then solve using the most appropriate method. Use each method once only.	
33. $-x^2 + 2x - 8 = 0$	Discriminant:
	#/Type of Roots:
	Method for Solving (Circle One) F / SR / CS / QF
	Solution(s):
34. $-2x^2 + 8 = x^2 - 28$	Discriminant:
	#/Type of Roots:
	Method for Solving (Circle One) F / SR / CS / QF
	Solution(s):

35. $-2x^2 - 5x - 4 = 0$	Discriminant:
	#/Type of Roots:
	Method for Solving (Circle One) F / SR / CS / QF
	Solution(s):
36. $4x^2 + 32x - 36 = 0$	Discriminant:
	#/Type of Roots:
	Method for Solving (Circle One) F / SR / CS / QF
	Solution(s):

Topic 9: Applications

- 37.** The chicken coup at the petting farm is 10 feet by 14 feet. The farm would like to double the current area by adding the same amount, x , to the length and width. What are the dimensions of the new enclosure? Round to the nearest hundredth of a foot.
- 38.** Find three positive consecutive odd integers such that the sum of the squares of the smallest integer and the largest integer is 129 more than the square of the median integer.

39. Natalie found a tennis ball outside a tennis court. She picked up the ball and threw it over the 12-foot fence into the court. The height (in feet) of the ball, h , at time t seconds is given by the function $h(t) = -16t^2 + 18t + 5$. Will she make it over the fence?

40. A punter kicked a football with an initial velocity of 84 feet per second. The height (in feet) of the ball, h , at time t seconds is given by the function $h(t) = -16t^2 + 84t + 2$. If the ball is caught by another player down the field at a height of 4 feet, how long was the ball was in the air?

41. Jack was so frustrated with his slow laptop that he threw it out of his second story window. The height, h , of the laptop at time t seconds can be given by the equation $h(t) = -16t^2 + 28t + 17$. Assuming the laptop hits the ground below, find the domain and range of the function.

42. The table below shows the altitude, in feet, of an airplane at certain times during its flight from Baltimore to Orlando since its departure at 11:00 a.m.. Use a quadratic model for the curve of best fit, then predict the altitude of the plane at 1:00 p.m.

Time	11:00 a.m.	11:30 a.m.	11:45 a.m.	12:00 pm.	12:30 p.m.	12:45 p.m.
Altitude	0	10,500	15,750	21,500	12,450	10,820

Name: _____

Date: _____ Per: _____

Unit 4 Test

Solving Quadratic Equations (& Complex Numbers)

1. Simplify the expression below.

$$\sqrt{-100} - 2\sqrt{-16}$$

- A. $-12i$
- B. $-2i$
- C. $12i$
- D. $2i$

2. Simplify the expression below.

$$-9 + \sqrt{-28}$$

- A. $-9 + 2i\sqrt{7}$
- B. $-9 - 2i\sqrt{7}$
- C. $i\sqrt{7}$
- D. $5i\sqrt{3}$

3. Simplify the expression below. Give your answer as a complex number in standard form.

$$(7 + i) - (6 - 4i)$$

4. Simplify the expression below. Give your answer as a complex number in standard form.

$$(4 - 5i)(-1 - 7i)$$

5. Simplify the expression below. Give your answer as a complex number in standard form.

$$(-7 + 3i)^2 + (-2 + 8i)$$

6. Simplify the expression below. Give your answer as a complex number in standard form.

$$\frac{-9 + 5i}{3 - i}$$

7. Check all values that are equivalent to $-i$.

☐ i^{14}	☐ i^{42}
☐ i^{19}	☐ i^{57}
☐ i^{25}	☐ i^{64}
☐ i^{37}	☐ i^{71}

8. Which sets do not contain the value $(2 + 6i)(2 - 6i) - 50$? Check all that apply.

☐ Real	☐ Whole
☐ Irrational	☐ Natural
☐ Rational	☐ Complex
☐ Integers	☐ Pure Imaginary

9. Which statement shows the associative property of multiplication?

- A. $(-4 + 17i) \cdot 1 = (-4 + 17i)$
 B. $3i + (-1 + 9i) = (-1 + 9i) + 3i$
 C. $(-5 + 3i) \cdot 6i = -5 \cdot 6i + 3i \cdot 6i$
 D. $((1 - i) \cdot 8i) \cdot -2i = (1 - i) \cdot (8i \cdot -2i)$

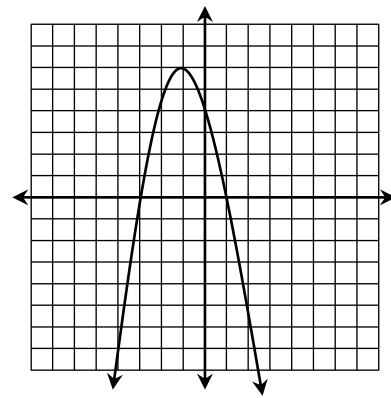
10. Which of the following is the additive identity of $5 + 2i$?

- A. 0
 B. 1
 C. $-(5 + 2i)$
 D. $5 - 2i$

11. Which of the following sets are closed under multiplication? Check all that apply.

☐	Real
☐	Irrational
☐	Rational
☐	Integers
☐	Whole
☐	Natural
☐	Complex
☐	Pure Imaginary

12. What is the apparent solution set for the equation associated with the following graph?



- A. $\{-3, 1\}$
 B. $\{-1, 3\}$
 C. $\{-3, 1, 4\}$
 D. $\{-1, 3, 4\}$

13. Which of the following gives the factored form and roots of the equation given below?

$$f(x) = 2(x + 4)^2 - 2$$

- A. $f(x) = 2(x + 3)(x + 5)$; $\{3, 5\}$
 B. $f(x) = 2(x + 3)(x + 5)$; $\{-3, -5\}$
 C. $f(x) = 2(x + 2)(x + 4)$; $\{2, 4\}$
 D. $f(x) = 2(x + 2)(x + 4)$; $\{-4, -2\}$

14. Solve by **factoring**:

$$x^2 + 9x - 83 = 4x + 1$$

15. Solve by **factoring**:

$$25x^2 + 1 = 10$$

16. Solve by **factoring**:

$$6x^2 = 3 - 7x$$

17. Solve by **square roots**:

$$2x^2 + 11 = 131$$

18. Solve by **square roots**:

$$\frac{1}{3}x^2 - 10 = -37$$

19. Solve by **square roots**:

$$(2x + 3)^2 = 36$$

20. Solve by **completing the square**:

$$x^2 + 10x + 7 = 0$$

21. Solve by **completing the square**:

$$-2x^2 + 6x + 7 = 2x - 3$$

22. Solve by **the quadratic formula**:

$$-x^2 + 16 = 6x + 1$$

23. Which functions have at least one real root? Check all that apply.

☐ $f(x) = -x^2 + 10x - 14$	☐ $f(x) = -x^2 + 4$
☐ $f(x) = 2x^2 - 12x + 18$	☐ $f(x) = -3x^2 + x - 2$
☐ $f(x) = x^2 - 3x + 25$	☐ $f(x) = 10x^2 - 4x$

For questions 24-27, find the discriminant of the equation and determine the number and type of roots, then solve the equation. Each method may only be used once!

24. $3x^2 + 15x + 19 = 0$

Discriminant:

/ Type of Roots:

Method (Circle one):

F SR CS QF

Solution(s):

25. $9x^2 + 24x = 0$	Discriminant:
	# / Type of Roots:
	Method (Circle one): ☐ F ☐ SR ☐ CS ☐ QF
	Solution(s):
26. $-x^2 - 4x + 16 = 0$	Discriminant:
	# / Type of Roots:
	Method (Circle one): ☐ F ☐ SR ☐ CS ☐ QF
	Solution(s):
27. $4x^2 + 28 = 0$	Discriminant:
	# / Type of Roots:
	Method (Circle one): ☐ F ☐ SR ☐ CS ☐ QF
	Solution(s):

28. The dimensions of a square are altered so that 8 inches is added to one side while 3 inches is subtracted from the other. The area of the resulting rectangle is 126 in^2 . What was the original side length of the square?

29. A juggler tosses a ball into the air. The ball's height, h , and time t seconds can be represented by the equation $h(t) = -16t^2 + 40t + 4$. Suppose the juggler missed and the ball hit the ground. Find the maximum height of the ball and the time it took to reach the ground. Round all answers to the nearest hundredth.

Maximum Height:

Time to reach ground:

30. The table below gives the average temperature of a lake in months since April in a certain year. Write an equation for the curve of best fit using a quadratic model, then predict the average temperature of the lake during December of the same year. Round all answers to the nearest hundredth.

Month	April	May	June	July	August
Temp ($^{\circ}\text{F}$)	59	68	71	78	74

Equation:

Temperature:

Name:

Date:

Topic:

Class:

Main Ideas/Questions

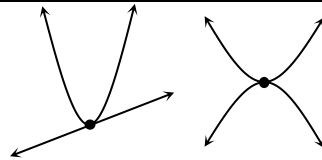
Notes/Examples

NONLINEAR SYSTEM

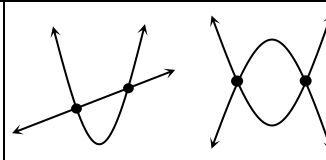
of Equations

- The **solution set** to a system of equations is the set of points in which the equations _____.
- There can be **one solution**, **multiple solutions**, or **no solution**. See the examples below.

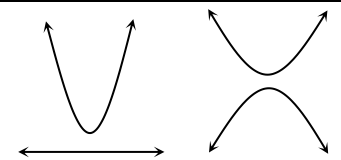
ONE SOLUTION



MULTIPLE SOLUTIONS



NO SOLUTION



SOLVE

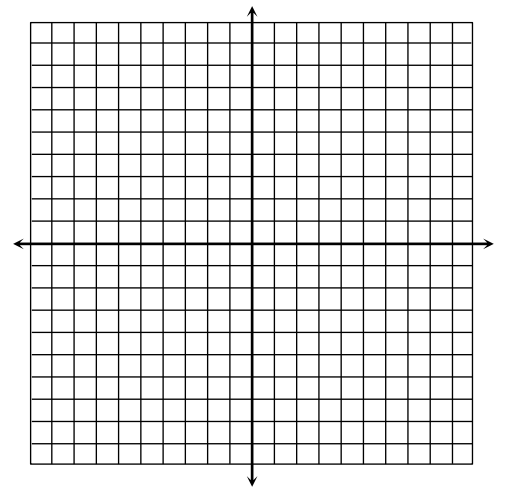
by Graphing

SET 1:

Linear/Quadratic

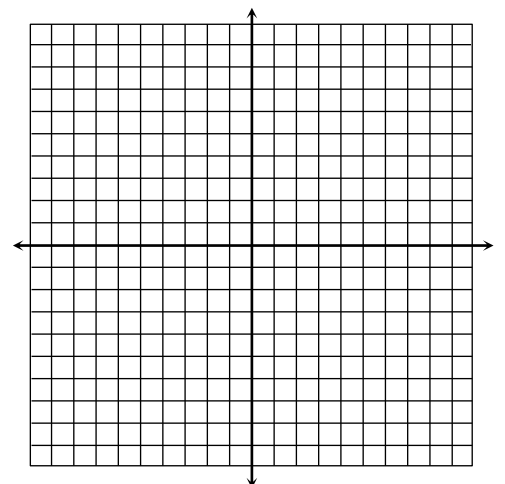
Solve the system by graphing each equation.

1.
$$\begin{cases} 4x - y = 1 \\ y = -2x^2 + 8x - 3 \end{cases}$$

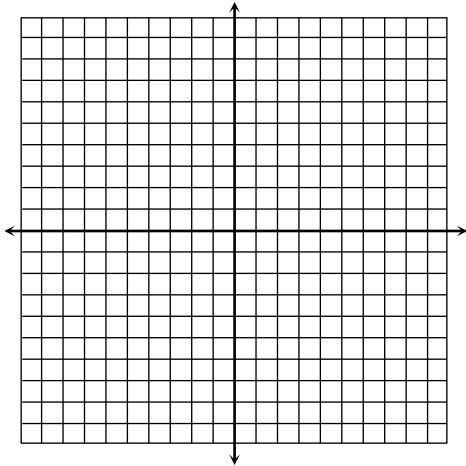
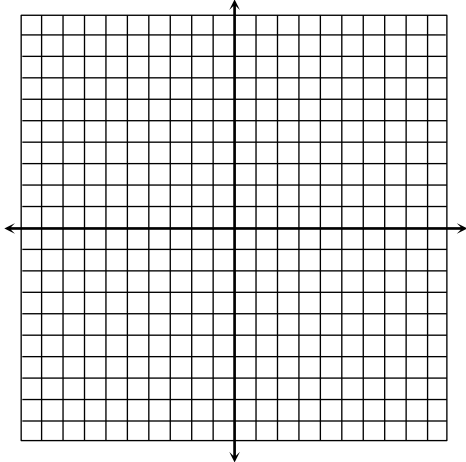
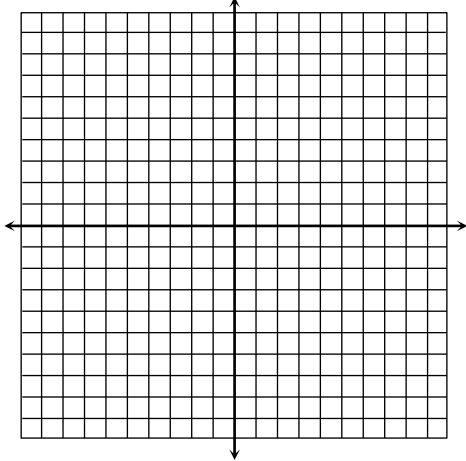
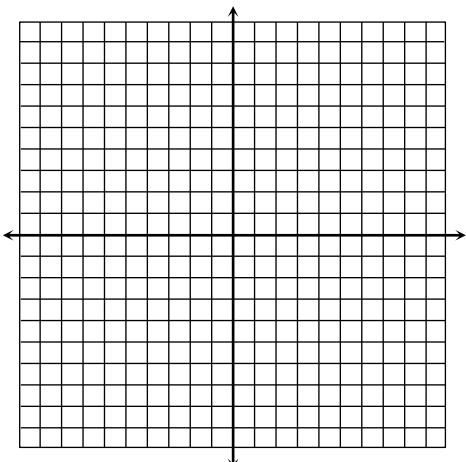


Solution(s): _____

2.
$$\begin{cases} y = x^2 - 4x - 5 \\ 2x + y = 3 \end{cases}$$



Solution(s): _____

	3. $\begin{cases} y = -x^2 - 12x - 32 \\ x - 3y = -21 \end{cases}$ Solution(s): _____	
SET 2: Quadratic/ Quadratic	4. $\begin{cases} y = 2x^2 + 8x - 1 \\ y = x^2 + 2x - 9 \end{cases}$ Solution(s): _____	
	5. $\begin{cases} y = -x^2 - 6x - 2 \\ y = x^2 - 4x + 8 \end{cases}$ Solution(s): _____	
	6. $\begin{cases} y = -3x^2 + 12x - 1 \\ y = x^2 + 4x + 3 \end{cases}$ Solution(s): _____	

Name: _____

Unit 4: Solving Quadratic Equations

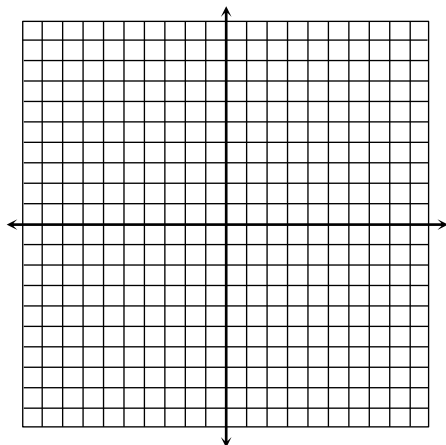


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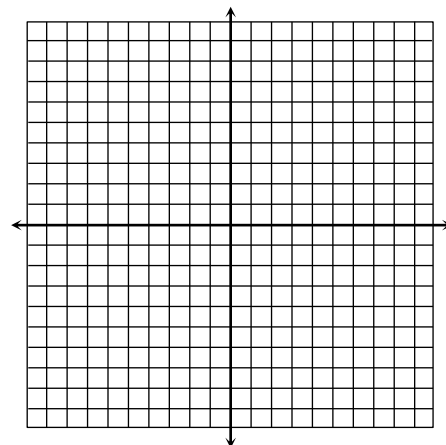
Homework 12: Solving Nonlinear Systems Graphically

**** This is a 2-page document! ******Solve each system by graphing. Be sure to identify the solution(s).**

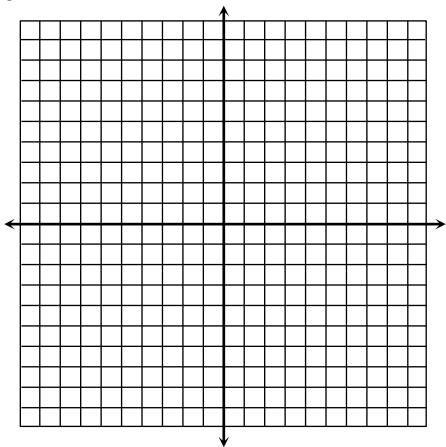
1.
$$\begin{cases} y = 4x + 5 \\ y = x^2 + 8x + 9 \end{cases}$$



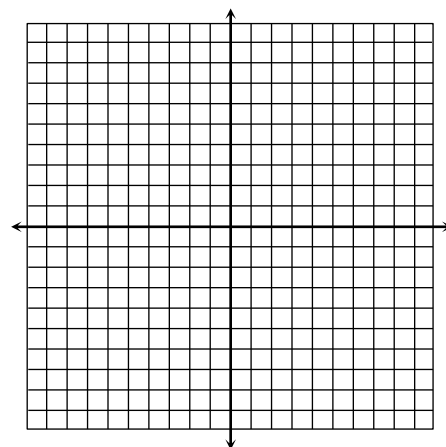
2.
$$\begin{cases} y = -x^2 - 4x + 2 \\ y = x^2 + 8x + 12 \end{cases}$$



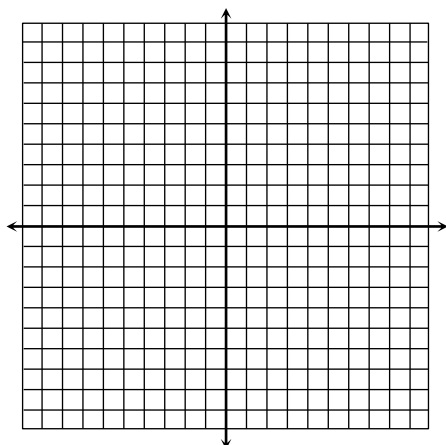
3.
$$\begin{cases} y = x^2 + 12x + 26 \\ y = -x^2 - 4x - 6 \end{cases}$$



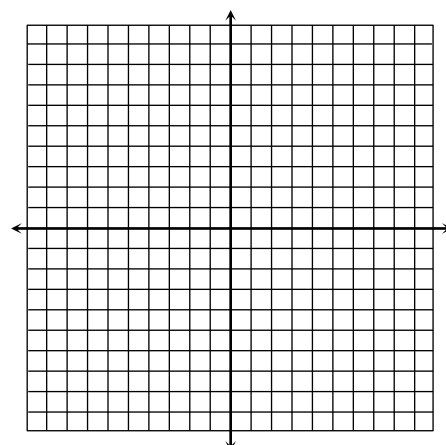
4.
$$\begin{cases} y = -x^2 + 10x - 28 \\ x - y = 7 \end{cases}$$



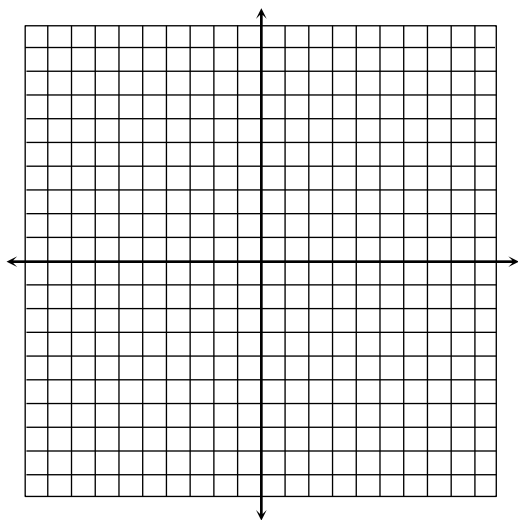
5.
$$\begin{cases} y = x^2 - 2x - 7 \\ y = -3x - 5 \end{cases}$$



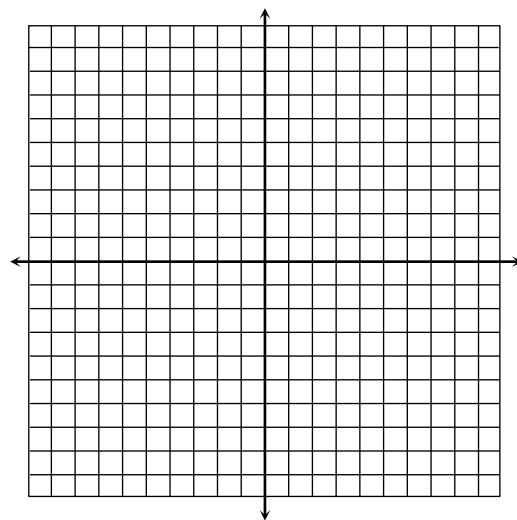
6.
$$\begin{cases} y = x^2 - 6x + 10 \\ y = -2x^2 + 4x \end{cases}$$



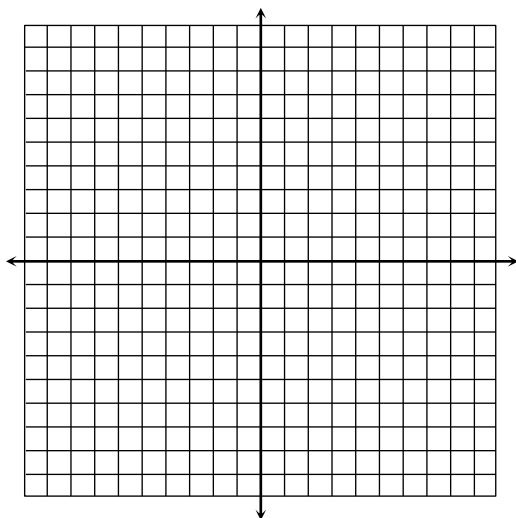
7.
$$\begin{cases} 2x + y = 5 \\ y = 2x^2 - 4x - 7 \end{cases}$$



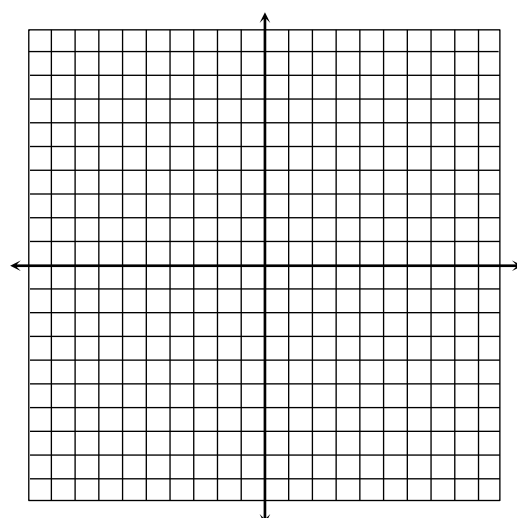
8.
$$\begin{cases} y = x^2 + 6 \\ y = -x^2 + 4x + 2 \end{cases}$$



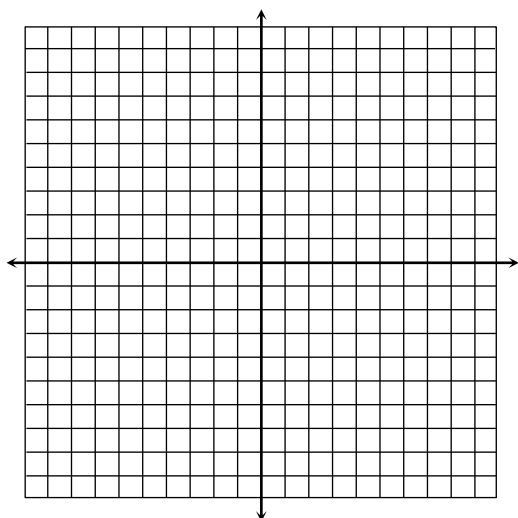
9.
$$\begin{cases} y = -x^2 + 6x - 6 \\ y = -3x^2 + 24x - 46 \end{cases}$$



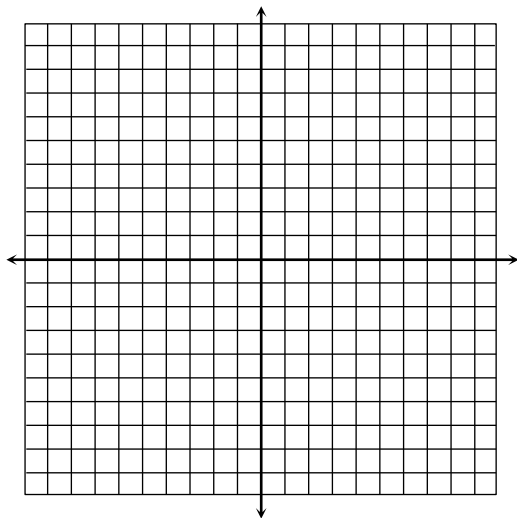
10.
$$\begin{cases} y = -x^2 + 2x + 1 \\ 4x - y = -2 \end{cases}$$



11.
$$\begin{cases} y = -x^2 - 12x - 31 \\ x + y = 9 \end{cases}$$



12.
$$\begin{cases} y = x^2 - 4x + 7 \\ y = -2x^2 + 20x - 41 \end{cases}$$



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
NONLINEAR SYSTEMS: <i>Solving Algebraically</i> *Remember: A nonlinear system can have one solution, multiple solutions, or no solution.	Follow the steps below to solve the system algebraically by substitution.	
	Steps	Example
	STEP 1: Isolate a variable in one of the equations.	$\begin{cases} 2x + y = 1 \\ y = x^2 - 8x + 6 \end{cases}$
	STEP 2: Substitute the expression equal to your isolated variable into the other equation and solve.	
	STEP 3: Substitute your solution(s) into either original equation to complete the ordered pairs.	
	STEP 4: Identify the solution(s).	
MORE EXAMPLES	Solve each system.	
	1. $\begin{cases} y = -2x^2 - 12x - 10 \\ 4x + y = -2 \end{cases}$	
	2. $\begin{cases} x - y = 8 \\ y = x^2 + 6x + 2 \end{cases}$	

$$3. \begin{cases} 2x - y = -4 \\ y = -x^2 + 4x + 7 \end{cases}$$

$$4. \begin{cases} y = x^2 + 12x + 24 \\ y = -x^2 - 2x + 4 \end{cases}$$

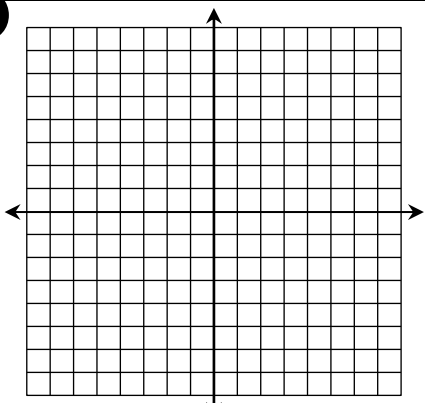
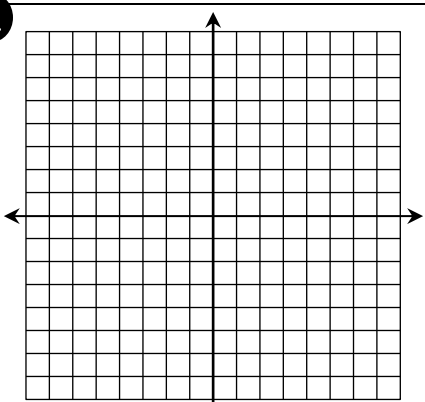
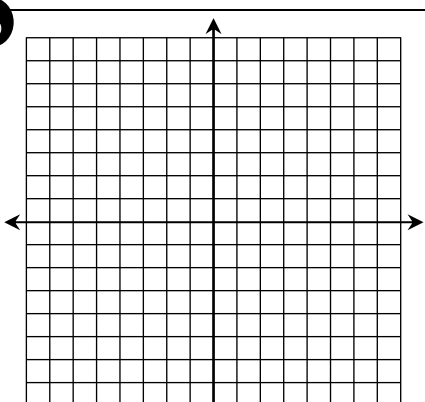
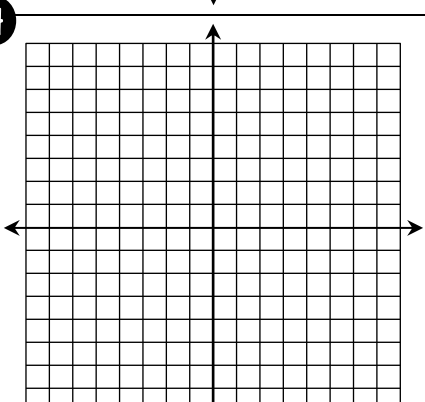
$$5. \begin{cases} y = 2x^2 - 8x - 1 \\ y = -2x^2 + 16x - 37 \end{cases}$$

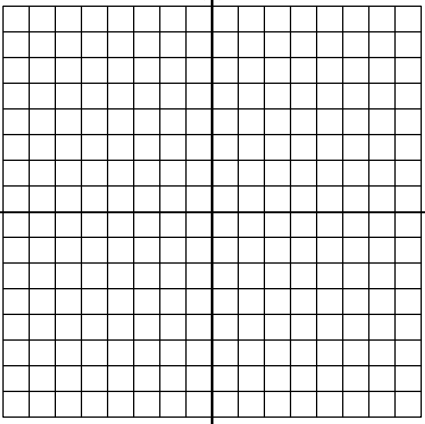
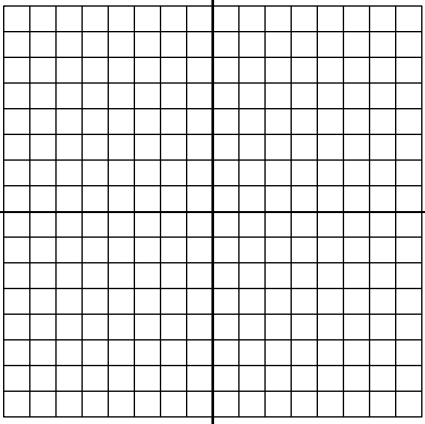
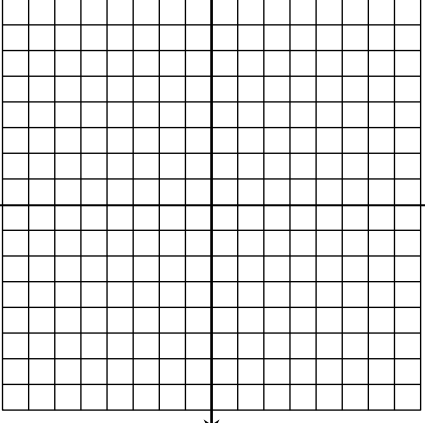
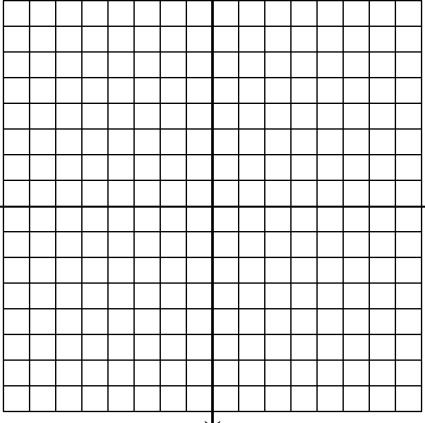
$$6. \begin{cases} y = x^2 - 4x - 11 \\ y = -x^2 - 12x - 30 \end{cases}$$

$$7. \begin{cases} y = 3x^2 - 12x + 1 \\ y = -x^2 + 8x - 15 \end{cases}$$

GRAPHING *vs.* SUBSTITUTION

Directions: Solve each given system of equations by graphing and substitution.
If done correctly, your answers should be the same!

	Graphing Method	Substitution Method
1		$\begin{cases} y = -x^2 - 2x + 5 \\ y = -4x + 6 \end{cases}$
2		$\begin{cases} y = x^2 + 8x + 13 \\ y = -x^2 - 2x + 5 \end{cases}$
3		$\begin{cases} y = x^2 - 4x \\ y = -2x^2 - 12x - 10 \end{cases}$
4		$\begin{cases} 2x - y = 5 \\ y = x^2 - 4x - 5 \end{cases}$

	Graphing Method	Substitution Method
5		$\begin{cases} y = 2x^2 - 12x + 11 \\ y = -x^2 + 6x - 4 \end{cases}$
6		$\begin{cases} x - y = 8 \\ y = x^2 + 2x - 6 \end{cases}$
7		$\begin{cases} y = 2x^2 + 10x + 15 \\ y = -x^2 - 2x + 3 \end{cases}$
8		$\begin{cases} y = x^2 - 8x + 9 \\ 3x + y = 5 \end{cases}$

Name: _____

Unit 4: Solving Quadratic Equations

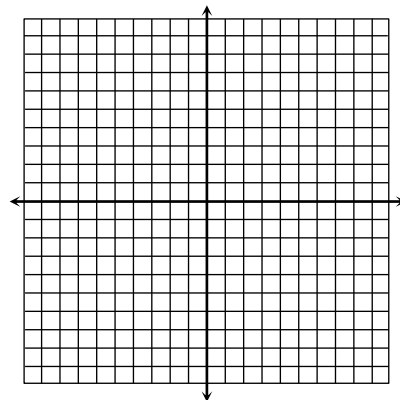


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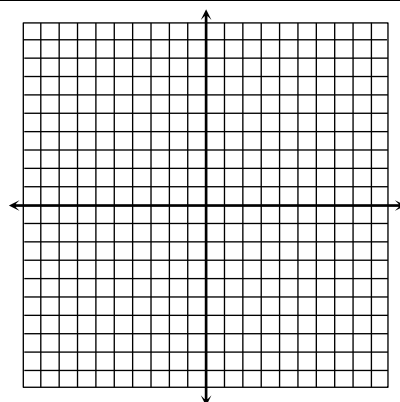
Homework 13: Solving Nonlinear Systems Algebraically

**** This is a 2-page document! ******Solve each system algebraically. Then verify your solution with a graph.**

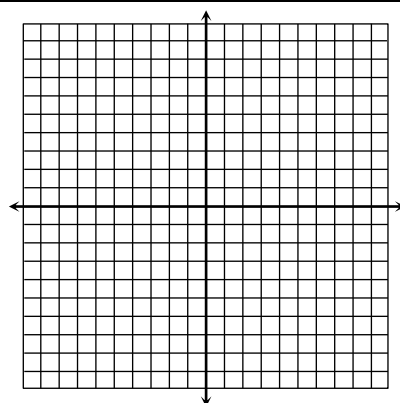
1.
$$\begin{cases} y = x^2 - 6x + 2 \\ 2x - y = 5 \end{cases}$$



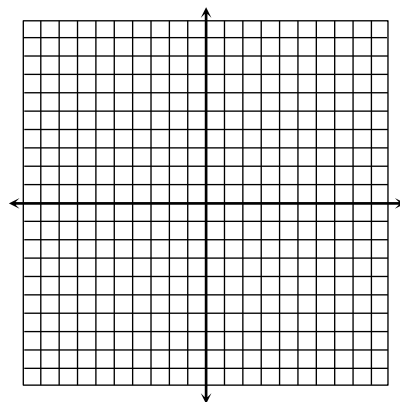
2.
$$\begin{cases} y = 3x^2 + 30x + 70 \\ y = -x^2 - 2x + 6 \end{cases}$$



3.
$$\begin{cases} x + y = -1 \\ y = x^2 - 10x + 21 \end{cases}$$



4.
$$\begin{cases} y = 2x^2 - 8x - 1 \\ y = -x^2 - 2x - 1 \end{cases}$$



Solve each system algebraically.

5.
$$\begin{cases} y = x^2 + 12x + 40 \\ y = -x^2 + 8x - 9 \end{cases}$$

6.
$$\begin{cases} y = -x^2 - 4x + 3 \\ 2x - y = -8 \end{cases}$$

7.
$$\begin{cases} y = 2x^2 - 4x + 2 \\ y = 8x - 16 \end{cases}$$

8.
$$\begin{cases} y = -x^2 - 10x - 23 \\ y = (x + 3)^2 \end{cases}$$

9.
$$\begin{cases} y = 3x^2 - 18x + 17 \\ y = -x^2 + 2x + 1 \end{cases}$$

10.
$$\begin{cases} y = x^2 + 14x + 35 \\ 3x - y = 1 \end{cases}$$

Name: _____

Algebra 2

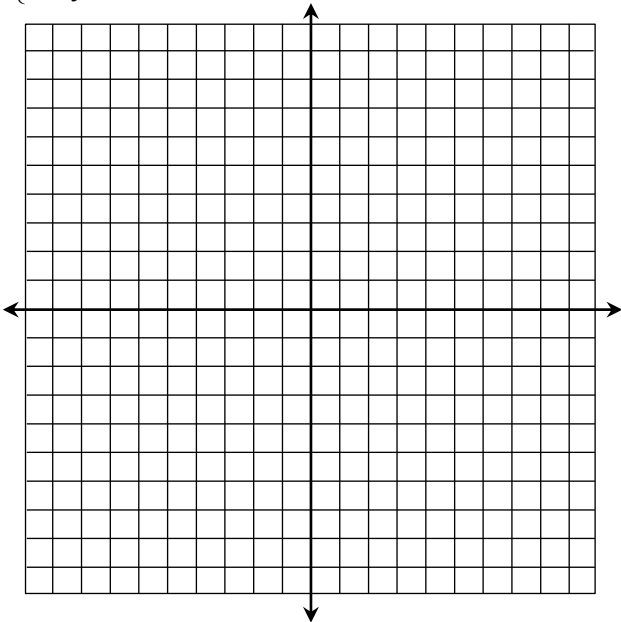
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Unit 4: Solving Quadratic Equations

Quiz 4-4: Linear-Quadratic and Quadratic-Quadratic-Systems

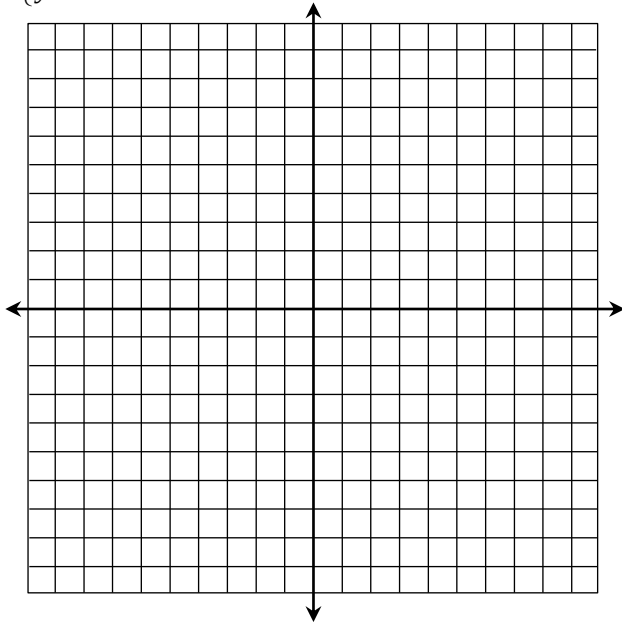
Solve by graphing:

1. $\begin{cases} y = x^2 - 10x + 17 \\ x + y = -1 \end{cases}$



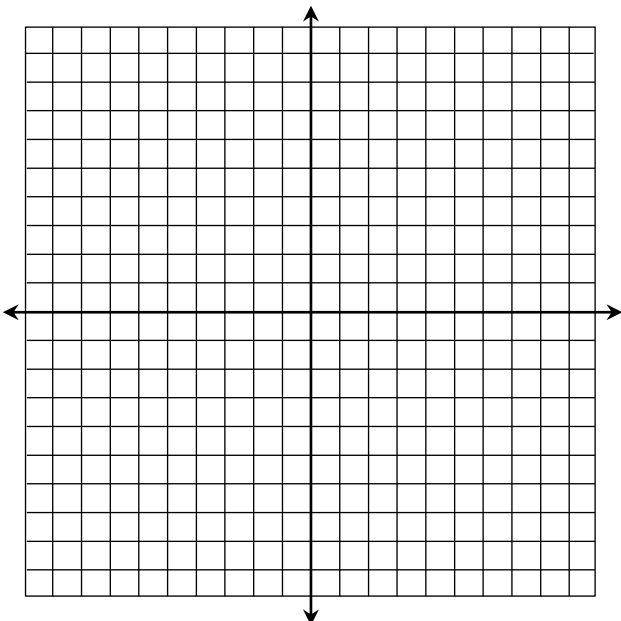
Solution(s):

2. $\begin{cases} y = x^2 + 4x - 3 \\ y = -2x^2 + 4x - 5 \end{cases}$



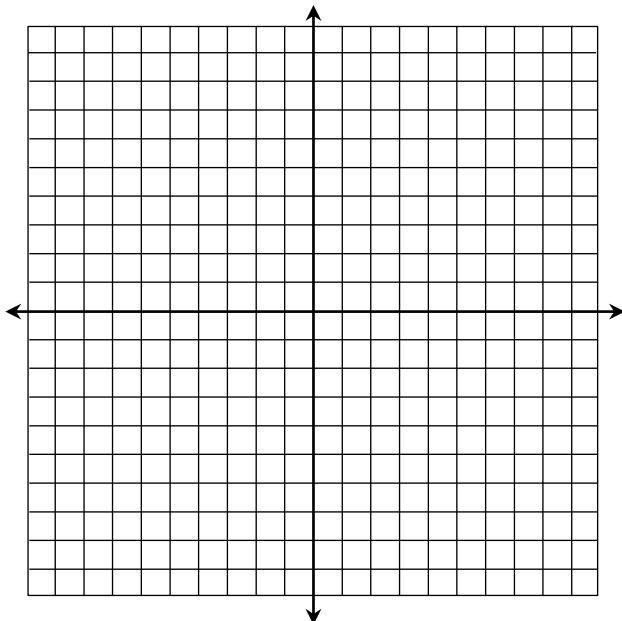
Solution(s):

3. $\begin{cases} y = -x^2 + 6 \\ 2x - y = -7 \end{cases}$



Solution(s):

4. $\begin{cases} y = 3x^2 + 12x + 7 \\ y = -x^2 - 8x - 17 \end{cases}$



Solution(s):

Solve algebraically:

5.
$$\begin{cases} y = x^2 + 4x + 2 \\ x - y = 2 \end{cases}$$

6.
$$\begin{cases} y = x^2 - 8x + 8 \\ y = -x^2 + 10x - 20 \end{cases}$$

Solution(s):

7.
$$\begin{cases} y = -3x^2 - 10x - 5 \\ y = x^2 + 6x + 11 \end{cases}$$

Solution(s):

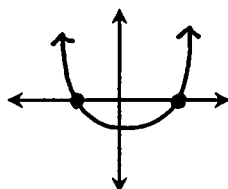
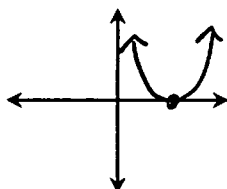
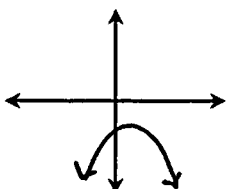
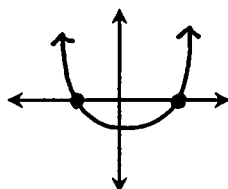
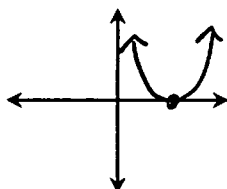
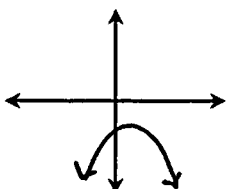
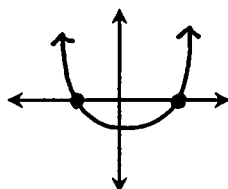
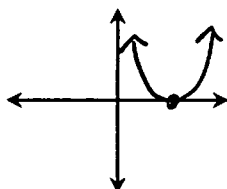
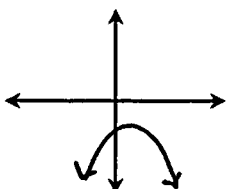
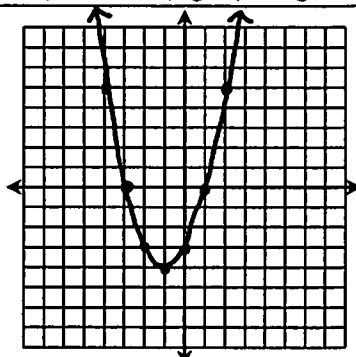
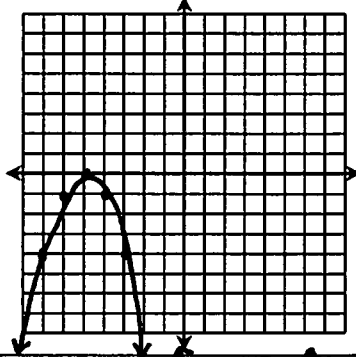
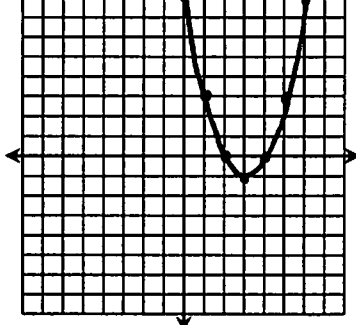
8.
$$\begin{cases} y = 2x^2 + 11x - 4 \\ 3x - y = -6 \end{cases}$$

Solution(s):

Solution(s):

Name:	Date:
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Topic:	Class:
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Main Ideas/Questions	Notes/Examples																																																
<i>Solutions to a</i> QUADRATIC EQUATION	- The solutions to a quadratic equation are the points at which the function intersects the <u>X - axis</u>. - The solutions are also referred to as <u>zeros</u>, <u>roots</u> or <u>X - intercepts</u>																																																
NUMBER OF SOLUTIONS	<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 33%; padding: 5px; text-align: center;"> 2 SOLUTIONS  </td><td style="width: 33%; padding: 5px; text-align: center;"> 1 SOLUTION  </td><td style="width: 33%; padding: 5px; text-align: center;"> NO REAL SOLUTIONS  </td></tr> </table>	2 SOLUTIONS 	1 SOLUTION 	NO REAL SOLUTIONS 																																													
2 SOLUTIONS 	1 SOLUTION 	NO REAL SOLUTIONS 																																															
<i>Find Solutions</i> BY GRAPHING SOLUTIONS: 1. <u>$x = \{-3, 1\}$</u> 2. <u>$x = \{-5\}$</u> 3. <u>$x = \{2, 4\}$</u>	Directions: Find the solutions to each quadratic equation by graphing. 1. $f(x) = x^2 + 2x - 3$ <table style="display: inline-table; border-collapse: collapse;"> <tr><th>X</th><th>y</th></tr> <tr><td>-4</td><td>5</td></tr> <tr><td>-3</td><td>0</td></tr> <tr><td>-2</td><td>-3</td></tr> <tr><td>-1</td><td>-4</td></tr> <tr><td>0</td><td>-3</td></tr> <tr><td>1</td><td>0</td></tr> <tr><td>2</td><td>5</td></tr> </table> (-3, 0) (1, 0)  2. $f(x) = -x^2 - 10x - 25$ <table style="display: inline-table; border-collapse: collapse;"> <tr><th>X</th><th>y</th></tr> <tr><td>-8</td><td>-9</td></tr> <tr><td>-7</td><td>-4</td></tr> <tr><td>-6</td><td>-1</td></tr> <tr><td>-5</td><td>0</td></tr> <tr><td>-4</td><td>-1</td></tr> <tr><td>-3</td><td>-4</td></tr> <tr><td>-2</td><td>-9</td></tr> </table> (-5, 0)  3. $f(x) = x^2 - 6x + 8$ <table style="display: inline-table; border-collapse: collapse;"> <tr><th>X</th><th>y</th></tr> <tr><td>0</td><td>8</td></tr> <tr><td>1</td><td>3</td></tr> <tr><td>2</td><td>0</td></tr> <tr><td>3</td><td>-1</td></tr> <tr><td>4</td><td>0</td></tr> <tr><td>5</td><td>3</td></tr> <tr><td>6</td><td>8</td></tr> </table> (2, 0) (4, 0) 	X	y	-4	5	-3	0	-2	-3	-1	-4	0	-3	1	0	2	5	X	y	-8	-9	-7	-4	-6	-1	-5	0	-4	-1	-3	-4	-2	-9	X	y	0	8	1	3	2	0	3	-1	4	0	5	3	6	8
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SOLUTIONS:

4. $X = \{-7, -4\}$

5. $X = \{-3, 3\}$

6. $X = \{-3, 0\}$

7. $X = \{\emptyset\}$

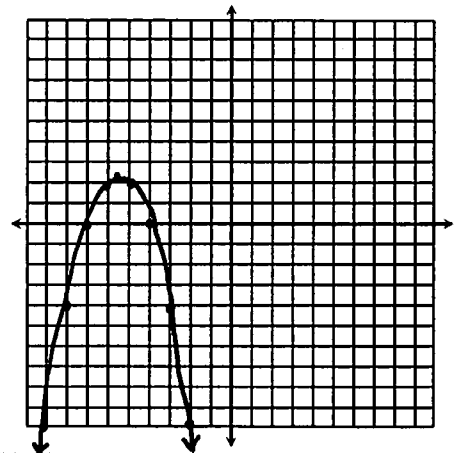
(no real solutions)

4. $f(x) = -x^2 - 11x - 28$

X	y
-8	-4
-7	0
-6	2
-5.5	2.25
-5	2
-4	0
-3	-4

$(-7, 0)$

$(-4, 0)$

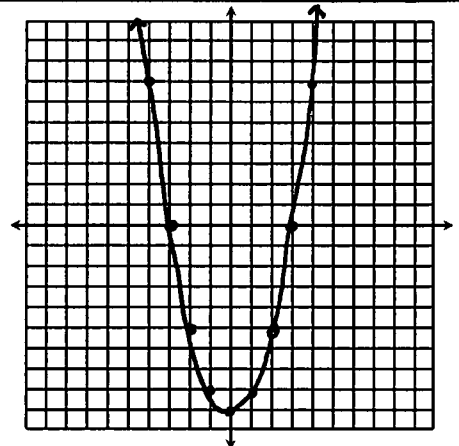


5. $f(x) = x^2 - 9$

X	y
-3	0
-2	-5
-1	-8
0	-9
1	-8
2	-5
3	0

$(-3, 0)$

$(3, 0)$

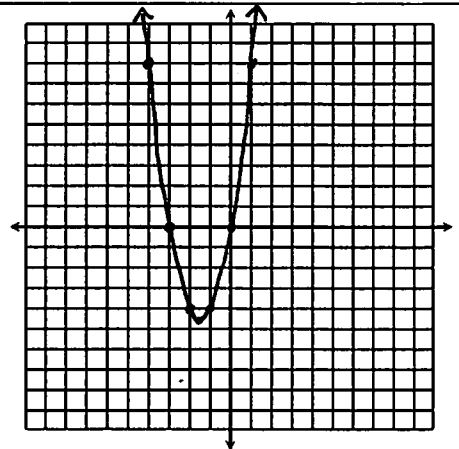


6. $f(x) = 2x^2 + 6x$

X	y
-4	8
-3	0
-2	-4
-1.5	-4.5
-1	-4
0	0
1	8

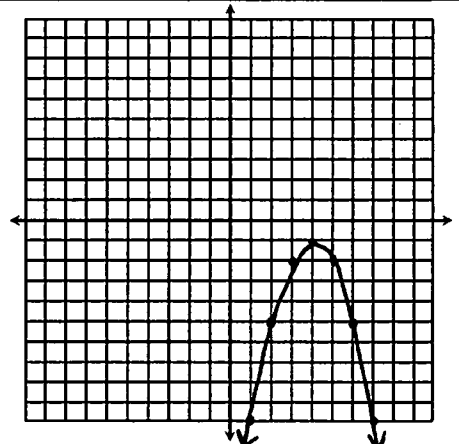
$(-3, 0)$

$(0, 0)$



7. $f(x) = -x^2 + 8x - 17$

X	y
1	-10
2	-5
3	-2
4	-1
5	-2
6	-5
7	-10



Factoring Refresher

Quadratic equations can be solved by factoring as well.
Use the problems below to refresh your factoring skills ☺

SET 1 Greatest Common Factor	
1. $12k - 18$ $6(2k - 3)$	2. $40x^8y + 64x^4y$ $8x^4y(5x^4 + 8)$
3. $14m^6 - 35m^3 - 7m^2$ $7m^2(2m^4 - 5m - 1)$	4. $3ab^2 - 5a^2b + 8ab$ $ab(3b - 5a + 8)$

**** When factoring, always look for a GCF first! ****

SET 2 Trinomials ($ax^2 + bx + c$, where $a = 1$)	
5. $x^2 + 14x + 45$ $(x + 9)(x + 5)$	6. $w^2 - 15w + 26$ $(w - 13)(w - 2)$
7. $c^2 + 2c - 48$ $(c + 8)(c - 6)$	8. $n^2 - n - 72$ $(n - 9)(n + 8)$
9. $a^2 + 12a + 36$ $(a + 6)(a + 6) = \boxed{(a + 6)^2}$	10. $n^2 - 2n + 1$ $(n - 1)(n - 1) = \boxed{(n - 1)^2}$
11. $2k^2 - 16k - 40$ $2(k^2 - 8k - 20)$ $\boxed{2(k - 10)(k + 2)}$	12. $5z^2 - 25z + 30$ $5(z^2 - 5z + 6)$ $\boxed{5(z - 2)(z - 3)}$

SET 3 Trinomials ($ax^2 + bx + c$, where $a \neq 1$)	
13. $2x^2 - 15x + 18$ $x^2 - 15x + 36$ $(x - \frac{12}{2})(x - \frac{3}{2})$ $\boxed{(x - 6)(2x - 3)}$	14. $5p^2 + 22p - 48$ $p^2 + 22p - 240$ $(p + \frac{30}{5})(p - \frac{8}{5})$ $\boxed{(p + 6)(5p - 8)}$

15. $3h^2 - 19h - 40$ $h^2 - 19h - 120$ $(h - \frac{24}{3})(h + \frac{5}{3})$ $(h - 8)(3h + 5)$	16. $12x^2 + 5x - 2$ $x^2 + 5x - 24$ $(x + \frac{8}{12})(x - \frac{3}{12})$ $(x + \frac{2}{3})(x - \frac{1}{4})$ $(3x + 2)(4x - 1)$
17. $6y^2 - 5y - 21$ $y^2 - 5y - 126$ $(y - \frac{14}{6})(y + \frac{9}{6})$ $(y - \frac{7}{3})(y + \frac{3}{2})$ $(3y - 7)(2y + 3)$	18. $16m^2 + 60m - 54$ $2(8m^2 + 30m - 27)$ $2(m^2 + 30m - 216)$ $2(m + \frac{36}{8})(m - \frac{6}{8})$ $2(m + \frac{9}{2})(m - \frac{3}{4})$ $2(2m + 9)(4m - 3)$
19. $9x^2 - 12x + 4$ $x^2 - 12x + 36$ $(x - \frac{6}{9})(x - \frac{6}{9})$ $(x - \frac{2}{3})(x - \frac{2}{3})$ $(3x - 2)(3x - 2) = (3x - 2)^2$	20. $8y^2 + 56y + 96$ $8(y^2 + 7y + 12)$ $8(y + 3)(y + 4)$

SET 4 Difference of Squares	
21. $x^2 - 4$ $(x + 2)(x - 2)$	22. $y^2 - 81$ $(y + 9)(y - 9)$
23. $9x^2 - 25y^4$ $(3x - 5y^2)(3x + 5y^2)$	24. $m^2n^2 - 49$ $(mn - 7)(mn + 7)$
25. $72r^6 - 2s^2$ $2(36r^6 - s^2)$ $2(6r^3 - s)(6r^3 + s)$	26. $2a^2b - 32a^4b$ $2a^2b(1 - 16a^2)$ $2a^2b(1 - 4a)(1 + 4a)$

Name: _____

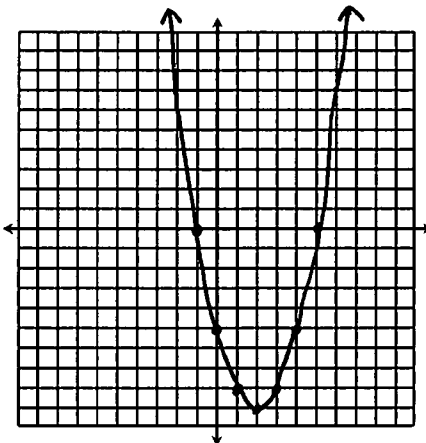
Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 1: Solving Quadratics by Graphing
& Factoring Review**** This is a 2-page document! ******Directions:** Find the solutions of each quadratic equation by graphing.

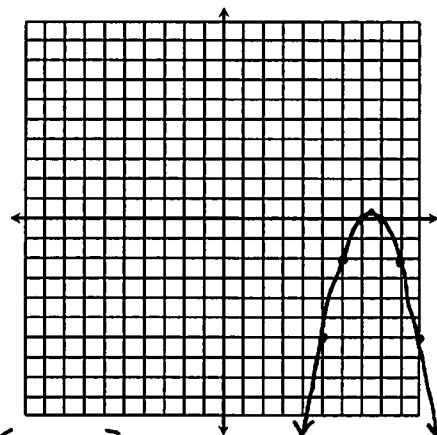
1. $f(x) = x^2 - 4x - 5$

X	Y
-1	0
0	-5
1	-8
2	-9
3	-8
4	-5
5	0

Solutions: $X = \{-1, 5\}$

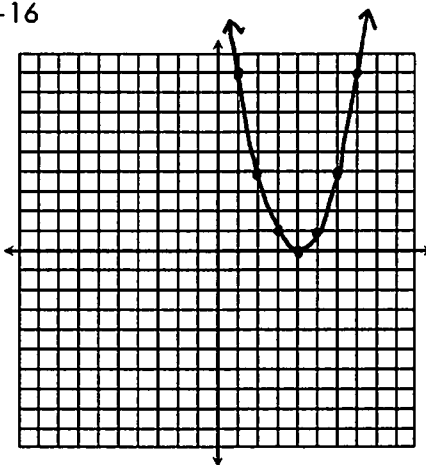
2. $f(x) = -x^2 + 15x - 56$

X	Y
5	-6
6	-2
7	0
7.5	0.25
8	0
9	-2
10	-6

Solutions: $X = \{7, 8\}$

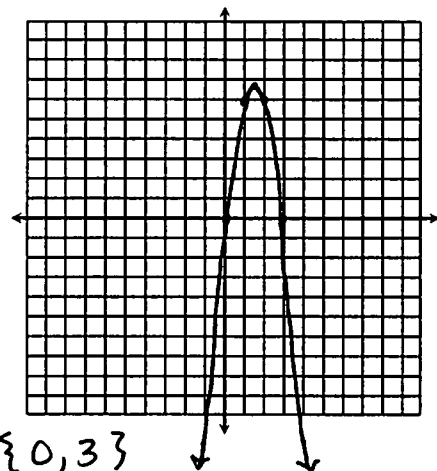
3. $f(x) = x^2 - 8x + 16$

X	Y
1	9
2	4
3	1
4	0
5	1
6	4
7	9

Solutions: $X = \{4\}$

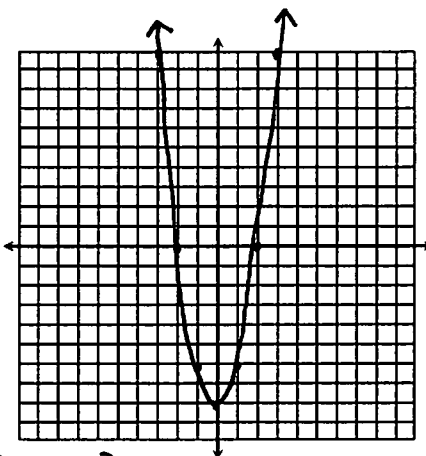
4. $f(x) = -3x^2 + 9x$

X	Y
-1	-12
0	0
1	6
1.5	6.75
2	6
3	0
4	-12

Solutions: $X = \{0, 3\}$

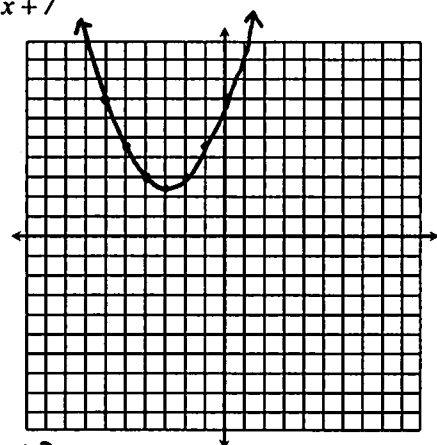
5. $f(x) = 2x^2 - 8$

X	Y
-3	10
-2	0
-1	-6
0	-8
1	-6
2	0
3	10

Solutions: $X = \{-2, 2\}$

6. $f(x) = \frac{1}{2}x^2 + 3x + 7$

X	Y
-6	7
-5	4.5
-4	3
-3	2.5
-2	3
-1	4.5
0	7

Solutions: $X = \{\emptyset\}$ No real solutions

Directions: Factor each expression. Be sure to check for a GCF first.

7. $x^2 + 10x + 24$

$$(x+4)(x+6)$$

8. $n^2 + 12n - 85$

$$(n+17)(n-5)$$

9. $6p^2 - 24p + 24$

$$6(p^2 - 4p + 4)$$

$$6(p-2)(p-2)$$

$$6(p-2)^2$$

10. $3a^2 - 3a - 36$

$$3(a^2 - a - 12)$$

$$3(a-4)(a+3)$$

11. $2x^2 - 15x - 8$

$$x^2 - 15x - 16$$

$$(x - \frac{16}{2})(x + \frac{1}{2})$$

$$(x-8)(2x+1)$$

12. $9m^2 + 24m + 7$

$$m^2 + 24m + 63$$

$$(m + \frac{21}{9})(m + \frac{3}{9})$$

$$(m + \frac{7}{3})(m + \frac{1}{3})$$

$$(3m+7)(3m+1)$$

13. $16w^2 + 8w + 1$

$$w^2 + 8w + 16$$

$$(w + \frac{4}{16})(w + \frac{4}{16})$$

$$(w + \frac{1}{4})(w + \frac{1}{4})$$

$$(4w+1)(4w+1) = (4w+1)^2$$

14. $20y^2 - 34y + 6$

$$2(10y^2 - 17y + 3)$$

$$2(y^2 - 17y + 30)$$

$$2(y - \frac{15}{10})(y - \frac{2}{10})$$

$$2(y - \frac{3}{2})(y - \frac{1}{5}) = 2(2y-3)(5y-1)$$

15. $k^2 - 25$

$$(k+5)(k-5)$$

16. $9m^2 - 49$

$$(3m+7)(3m-7)$$

17. $64x^2y - y$

$$y(64x^2 - 1)$$

$$y(8x+1)(8x-1)$$

18. $20c^2 - 45d^2$

$$5(4c^2 - 9d^2)$$

$$5(2c+3d)(2c-3d)$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
SOLVING QUADRATICS <i>By Factoring</i>	① MOVE ALL TERMS to one side and set the EQUAL TO 0 .
	② FACTOR!
	③ SET EACH FACTOR EQUAL TO 0 and SOLVE for each x-value.
	④ Write your answer as a SOLUTION SET . $x = \{_, _\}$

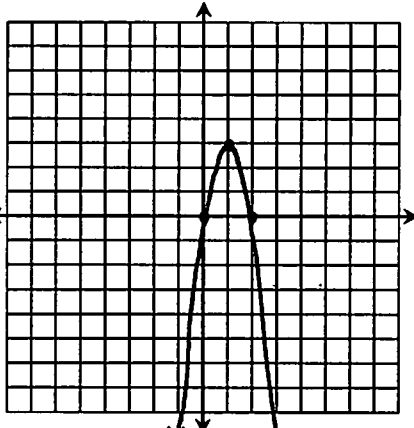
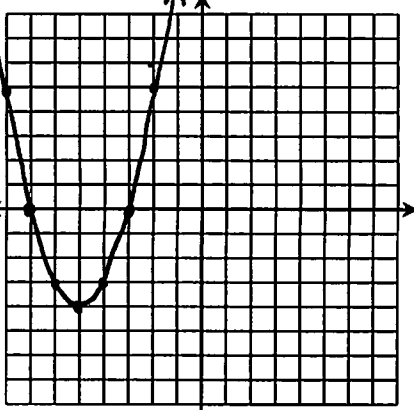
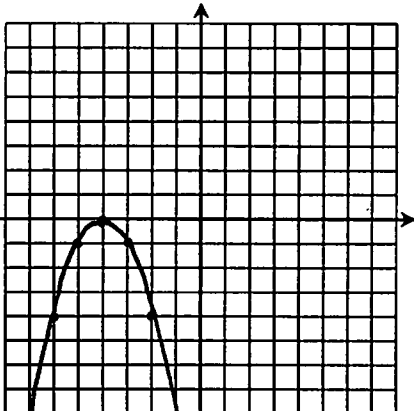
Directions: Solve each equation by factoring.

1. $x^2 + 9x = 0$ $\begin{array}{l l} x & (x+9) = 0 \\ \hline x=0 & x+9=0 \\ & x=-9 \end{array}$ $x = \{-9, 0\}$	2. $4x^2 - 8x = 0$ $\begin{array}{l l} 4x & (x-2) = 0 \\ \hline 4x=0 & x-2=0 \\ x=0 & x=2 \end{array}$ $x = \{0, 2\}$
3. $x^2 - 4 = 0$ $\begin{array}{l l} (x+2) & (x-2) = 0 \\ \hline x+2=0 & x-2=0 \\ x=-2 & x=2 \end{array}$ $x = \{-2, 2\}$	4. $9x^2 - 25 = 0$ $\begin{array}{l l} (3x+5) & (3x-5) = 0 \\ \hline 3x+5=0 & 3x-5=0 \\ 3x=-5 & 3x=5 \\ x=-\frac{5}{3} & x=\frac{5}{3} \end{array}$ $x = \{-\frac{5}{3}, \frac{5}{3}\}$
5. $x^2 - 10x + 21 = 0$ $\begin{array}{l l} (x-7) & (x-3) = 0 \\ \hline x-7=0 & x-3=0 \\ x=7 & x=3 \end{array}$ $x = \{3, 7\}$	6. $2x^2 - 2x - 24 = 0$ $2(x^2 - x - 12) = 0$ $\begin{array}{l l l} 2 & (x-4) & (x+3) = 0 \\ \hline 2 \neq 0 & x-4=0 & x+3=0 \\ & x=4 & x=-3 \end{array}$ $x = \{-3, 4\}$
7. $3x^2 - 13x - 30 = 0$ $x^2 - 13x - 90 = 0$ $(x - \frac{18}{3})(x + \frac{5}{3}) = 0$ $\begin{array}{l l} (x-6) & (3x+5) = 0 \\ \hline x-6=0 & 3x+5=0 \\ x=6 & x=-\frac{5}{3} \end{array}$ $x = \{-\frac{5}{3}, 6\}$	8. $6x^2 + 5x - 4 = 0$ $x^2 + 5x - 24 = 0$ $(x + \frac{8}{6})(x - \frac{3}{6}) = 0$ $(x + \frac{4}{3})(x - \frac{1}{2}) = 0$ $\begin{array}{l l} (3x+4) & (2x-1) = 0 \\ \hline 3x+4=0 & 2x-1=0 \\ x=-\frac{4}{3} & x=\frac{1}{2} \end{array}$ $x = \{-\frac{4}{3}, \frac{1}{2}\}$

9. $2x^2 = 5x$ $2x^2 - 5x = 0$ $x(2x - 5) = 0$ <table><tr><td>$x=0$</td><td>$2x-5=0$ $x=\frac{5}{2}$</td></tr></table> $x = \{0, \frac{5}{2}\}$	$x=0$	$2x-5=0$ $x=\frac{5}{2}$	10. $7x^2 + 8x = x^2$ $6x^2 + 8x = 0$ $2x(3x + 4) = 0$ <table><tr><td>$2x=0$ $x=0$</td><td>$3x+4=0$ $x=-\frac{4}{3}$</td></tr></table> $x = \{-\frac{4}{3}, 0\}$	$2x=0$ $x=0$	$3x+4=0$ $x=-\frac{4}{3}$	
$x=0$	$2x-5=0$ $x=\frac{5}{2}$					
$2x=0$ $x=0$	$3x+4=0$ $x=-\frac{4}{3}$					
11. $2x^2 - 50 = x^2 - 1$ $x^2 - 49 = 0$ $(x+7)(x-7) = 0$ <table><tr><td>$x+7=0$ $x=-7$</td><td>$x-7=0$ $x=7$</td></tr></table> $x = \{-7, 7\}$	$x+7=0$ $x=-7$	$x-7=0$ $x=7$	12. $3x^2 = 81 - x^2$ $4x^2 - 81 = 0$ $(2x+9)(2x-9) = 0$ <table><tr><td>$2x+9=0$ $x=-\frac{9}{2}$</td><td>$2x-9=0$ $x=\frac{9}{2}$</td></tr></table> $x = \{-\frac{9}{2}, \frac{9}{2}\}$	$2x+9=0$ $x=-\frac{9}{2}$	$2x-9=0$ $x=\frac{9}{2}$	
$x+7=0$ $x=-7$	$x-7=0$ $x=7$					
$2x+9=0$ $x=-\frac{9}{2}$	$2x-9=0$ $x=\frac{9}{2}$					
13. $x^2 + 2x = 80$ $x^2 + 2x - 80 = 0$ $(x+10)(x-8) = 0$ <table><tr><td>$x+10=0$ $x=-10$</td><td>$x-8=0$ $x=8$</td></tr></table> $x = \{-10, 8\}$	$x+10=0$ $x=-10$	$x-8=0$ $x=8$	14. $x^2 + 11x + 29 = x + 4$ $x^2 + 10x + 25 = 0$ $(x+5)(x+5) = 0$ <table><tr><td>$x+5=0$ $x=-5$</td><td>$x+5=0$ $x=-5$</td></tr></table> $x = \{-5\}$	$x+5=0$ $x=-5$	$x+5=0$ $x=-5$	
$x+10=0$ $x=-10$	$x-8=0$ $x=8$					
$x+5=0$ $x=-5$	$x+5=0$ $x=-5$					
15. $2x^2 - 9x - 65 = x^2 - x$ $x^2 - 8x - 65 = 0$ $(x-13)(x+5) = 0$ <table><tr><td>$x-13=0$ $x=13$</td><td>$x+5=0$ $x=-5$</td></tr></table> $x = \{-5, 13\}$	$x-13=0$ $x=13$	$x+5=0$ $x=-5$	16. $4x^2 + 41x + 98 = 2 - 3x$ $4x^2 + 44x + 96 = 0$ $4(x^2 + 11x + 24) = 0$ $4(x+8)(x+3) = 0$ <table><tr><td>$4 \neq 0$</td><td>$x+8=0$ $x=-8$</td><td>$x+3=0$ $x=-3$</td></tr></table> $x = \{-8, -3\}$	$4 \neq 0$	$x+8=0$ $x=-8$	$x+3=0$ $x=-3$
$x-13=0$ $x=13$	$x+5=0$ $x=-5$					
$4 \neq 0$	$x+8=0$ $x=-8$	$x+3=0$ $x=-3$				
17. $2x^2 = x + 21$ $2x^2 - x - 21 = 0$ $x^2 - x - 42 = 0$ $(x-\frac{7}{2})(x+\frac{6}{2}) = 0$ <table><tr><td>$(2x-7)(x+3) = 0$ $2x-7=0$ $x=\frac{7}{2}$</td><td>$x+3=0$ $x=-3$</td></tr></table> $x = \{-3, \frac{7}{2}\}$	$(2x-7)(x+3) = 0$ $2x-7=0$ $x=\frac{7}{2}$	$x+3=0$ $x=-3$	18. $4x^2 - 11x = x - 9$ $4x^2 - 12x + 9 = 0$ $x^2 - 3x + \frac{9}{4} = 0$ $(x-\frac{3}{2})(x-\frac{3}{2}) = 0$ <table><tr><td>$(2x-3)(2x-3) = 0$ $2x-3=0$ $x=\frac{3}{2}$</td><td>$2x-3=0$ $x=\frac{3}{2}$</td></tr></table> $x = \{\frac{3}{2}\}$	$(2x-3)(2x-3) = 0$ $2x-3=0$ $x=\frac{3}{2}$	$2x-3=0$ $x=\frac{3}{2}$	
$(2x-7)(x+3) = 0$ $2x-7=0$ $x=\frac{7}{2}$	$x+3=0$ $x=-3$					
$(2x-3)(2x-3) = 0$ $2x-3=0$ $x=\frac{3}{2}$	$2x-3=0$ $x=\frac{3}{2}$					

REPRESENTING QUADRATIC FUNCTIONS

STANDARD FORM	VERTEX FORM	FACTORED FORM	GRAPH
$f(x) = x^2 - 8x + 15$	$f(x) = (x^2 - 8x) + 15$ $\downarrow$ $-\frac{8}{2} = -4 ; (-4)^2 = 16$ $f(x) = (x^2 - 8x + 16) + 15 - 16$ $f(x) = (x - 4)^2 - 1$ Vertex: $(4, -1)$ Axis: $x = 4$	$f(x) = (x - 3)(x - 5)$ Roots: $x = \{3, 5\}$	
$f(x) = -2x^2 - 4x + 6$	$f(x) = -2(x^2 + 2x) + 6$ $\downarrow$ $\frac{2}{2} = 1 ; 1^2 = 1$ $f(x) = -2(x^2 + 2x + 1) + 6 + 2$ $f(x) = -2(x + 1)^2 + 8$ Vertex: $(-1, 8)$ Axis: $x = -1$	$f(x) = -2(x^2 + 2x - 3)$ $f(x) = -2(x + 3)(x - 1)$ Roots: $x = \{-3, 1\}$	
$f(x) = -(x - 7)^2 + 1$ $f(x) = -(x - 7)(x - 7) + 1$ $f(x) = -(x^2 - 14x + 49) + 1$ $f(x) = -x^2 + 14x - 48$	$f(x) = -(x - 7)^2 + 1$ Vertex: $(7, 1)$ Axis: $x = 7$	$f(x) = -(x^2 - 14x + 48)$ $f(x) = -(x - 8)(x - 6)$ Roots: $x = \{6, 8\}$	

STANDARD FORM	VERTEX FORM	FACTORED FORM	GRAPH
$f(x) = -3(x-1)^2 + 3$ $f(x) = -3(x-1)(x-1) + 3$ $f(x) = -3(x^2 - 2x + 1) + 3$ $f(x) = -3x^2 + 6x$	$f(x) = -3(x-1)^2 + 3$ Vertex: $(1, 3)$ Axis: $x = 1$	$f(x) = -3x(x-2)$ Roots: $x = \{0, 2\}$	
$f(x) = x^2 + 10x + 21$	$f(x) = (x^2 + 10x) + 21$ $\downarrow$ $\frac{10}{2} = 5; 5^2 = 25$ $f(x) = (x^2 + 10x + 25) + 21 - 25$ $f(x) = (x+5)^2 - 4$ Vertex: $(-5, -4)$ Axis: $x = -5$	$f(x) = (x+3)(x+7)$ Roots: $x = \{-7, -3\}$	
$f(x) = -(x+4)^2$ $f(x) = -(x+4)(x+4)$ $f(x) = -(x^2 + 8x + 16)$ $f(x) = -x^2 - 8x - 16$	$f(x) = -(x^2 + 8x) - 16$ $\downarrow$ $\frac{8}{2} = 4; 4^2 = 16$ $f(x) = -(x^2 + 8x + 16) - 16 + 16$ $f(x) = -(x+4)^2 + 0$ $f(x) = -(x+4)^2$ Vertex: $(-4, 0)$ Axis: $x = -4$	$f(x) = -(x+4)^2$	

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 2: Solving Quadratics by Factoring

**** This is a 2-page document! ******Directions:** Find the solutions of each quadratic equation factoring.

1. $x^2 - 4x = 0$

$$\begin{array}{l|l} x(x-4) = 0 & \\ \hline x=0 & x-4=0 \\ & x=4 \end{array}$$

$x = \{0, 4\}$

2. $10x^2 = 6x$

$$\begin{array}{l|l} 10x^2 - 6x = 0 & \\ \hline 2x(5x-3) = 0 & \\ \hline 2x=0 & 5x-3=0 \\ x=0 & x=\frac{3}{5} \end{array}$$

$x = \{0, \frac{3}{5}\}$

3. $x^2 - 36 = 0$

$$\begin{array}{l|l} (x+6)(x-6) = 0 & \\ \hline x+6=0 & x-6=0 \\ x=-6 & x=6 \end{array}$$

$x = \{-6, 6\}$

4. $16x^2 = 81$

$$\begin{array}{l|l} 16x^2 - 81 = 0 & \\ \hline (4x+9)(4x-9) = 0 & \\ \hline 4x+9=0 & 4x-9=0 \\ x=-\frac{9}{4} & x=\frac{9}{4} \end{array}$$

$x = \{-\frac{9}{4}, \frac{9}{4}\}$

5. $x^2 + 12x + 27 = 0$

$$\begin{array}{l|l} (x+3)(x+9) = 0 & \\ \hline x+3=0 & x+9=0 \\ x=-3 & x=-9 \end{array}$$

$x = \{-9, -3\}$

6. $x^2 - 4x = 77$

$$\begin{array}{l|l} x^2 - 4x - 77 = 0 & \\ \hline (x-11)(x+7) = 0 & \\ \hline x-11=0 & x+7=0 \\ x=11 & x=-7 \end{array}$$

$x = \{-7, 11\}$

7. $x^2 = x + 30$

$$\begin{array}{l|l} x^2 - x - 30 = 0 & \\ \hline (x-6)(x+5) = 0 & \\ \hline x-6=0 & x+5=0 \\ x=6 & x=-5 \end{array}$$

$x = \{-5, 6\}$

8. $x^2 + 8x - 65 = x - 5$

$$\begin{array}{l|l} x^2 + 7x - 60 = 0 & \\ \hline (x+12)(x-5) = 0 & \\ \hline x+12=0 & x-5=0 \\ x=-12 & x=5 \end{array}$$

$x = \{-12, 5\}$

9. $2x^2 + 6x - 80 = 0$

$$\begin{array}{l|l} 2(x^2 + 3x - 40) = 0 & \\ \hline 2(x+8)(x-5) = 0 & \\ \hline 2 \neq 0 & \begin{array}{l|l} x+8=0 & x-5=0 \\ \hline x=-8 & x=5 \end{array} \end{array}$$

$x = \{-8, 5\}$

10. $5x^2 + 11x + 2 = 0$

$$\begin{array}{l|l} x^2 + 11x + 10 = 0 & \\ \hline (x+1)(x+10) = 0 & \\ \hline \frac{5}{5} & \frac{5}{5} \\ \hline (5x+1)(x+2) = 0 & \\ \hline 5x+1=0 & x+2=0 \\ x=-\frac{1}{5} & x=-2 \end{array}$$

$x = \{-2, -\frac{1}{5}\}$

11. $8x^2 - 2x = 15$ $8x^2 - 2x - 15 = 0$ $x^2 - 2x - 120 = 0$ $(x - 12)(x + 10) = 0$ $(x - \frac{3}{2})(x + \frac{5}{4}) = 0$ $(2x - 3)(4x + 5) = 0$ $(2x-3)(4x+5)=0$ $2x-3=0 \quad 4x+5=0$ $x=\frac{3}{2} \quad x=-\frac{5}{4}$ $x = \{-\frac{5}{4}, \frac{3}{2}\}$	12. $12x^2 - 26x - 7 = 3x + 1$ $12x^2 - 29x - 8 = 0$ $x^2 - 29x - 96 = 0$ $(x - 32)(x + 3) = 0$ $(x - \frac{8}{3})(x + \frac{1}{4}) = 0$ $(3x - 8)(4x + 1) = 0$ $3x - 8 = 0 \quad 4x + 1 = 0$ $x = \frac{8}{3} \quad x = -\frac{1}{4}$ $x = \{\frac{8}{3}, -\frac{1}{4}\}$
Directions: Identify the roots of each quadratic function below.	
13. $f(x) = (x+5)(x+6)$ $x+5=0 \quad x+6=0$ $x=-5 \quad x=-6$ $x = \{-6, -5\}$	14. $f(x) = -2(x+9)(x-1)$ $-2 \neq 0 \quad x+9=0 \quad x-1=0$ $x=-9 \quad x=1$ $x = \{-9, 1\}$
15. $f(x) = x^2 - 5x + 6$ $f(x) = (x-2)(x-3)$ $x-2=0 \quad x-3=0$ $x=2 \quad x=3$ $x = \{2, 3\}$	16. $f(x) = x^2 - 16x + 64$ $f(x) = (x-8)(x-8)$ $x-8=0 \quad x-8=0$ $x=8 \quad x=8$ $x = \{8\}$
17. $f(x) = (x-1)^2 - 25$ $f(x) = (x-1)(x-1) - 25$ $f(x) = x^2 - 2x + 1 - 25$ $f(x) = x^2 - 2x - 24$ $f(x) = (x-6)(x+4)$ $x-6=0 \quad x+4=0$ $x=6 \quad x=-4$ $x = \{-4, 6\}$	18. $f(x) = -(x-2)^2 + 4$ $f(x) = -(x-2)(x-2) + 4$ $f(x) = -(x^2 - 4x + 4) + 4$ $f(x) = -x^2 + 4x$ $f(x) = -x(x-4)$ $-x=0 \quad x-4=0$ $x=0 \quad x=4$ $x = \{0, 4\}$
19. Give the vertex form and factored form of $f(x) = x^2 - 12x + 35$. What specific information does each form provide? <u>Vertex form:</u> $f(x) = (x^2 - 12x) + 35$ $\downarrow$ $-\frac{12}{2} = -6; (-6)^2 = 36$ $f(x) = (x^2 - 12x + 36) + 35 - 36$ $f(x) = (x-6)^2 - 1$ Vertex: $(6, -1)$ Axis: $x=6$ <u>Factored form:</u> $f(x) = (x-7)(x-5)$ Roots: $x = \{5, 7\}$	

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
RADICALS <i>Review</i>	- A radical is any expression with a square root, cube root, etc. - Radical Symbol: $\sqrt{\quad}$ - List the first 20 perfect squares: <u>1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400</u>
SIMPLIFYING <i>Radicals</i>	Case 1: Perfect Squares – Take the square root of the number. Case 2: Non-Perfect Squares – Break the radical down using the largest perfect square. Take the square root of the perfect square, and leave the "leftover" under the radical symbol. For both cases: If there is a coefficient outside the radical, multiply this with any number you take out.

Directions: Write each expression below in simplest radical form.

1. $\sqrt{49}$ 7	2. $\sqrt{225}$ 15	3. $2\sqrt{144}$ $2 \cdot 12 = $ 24
4. $\sqrt{\frac{4}{9}}$ $\frac{2}{3}$	5. $\sqrt{\frac{1}{81}}$ $\frac{1}{9}$	6. $\sqrt{\frac{36}{121}}$ $\frac{6}{11}$
7. $\sqrt{112}$ $\sqrt{16} \cdot \sqrt{7}$ $= 4\sqrt{7}$	8. $\sqrt{245}$ $\sqrt{49} \cdot \sqrt{5}$ $= 7\sqrt{5}$	9. $5\sqrt{72}$ $5 \cdot \sqrt{36} \cdot \sqrt{2}$ $5 \cdot 6\sqrt{2} = $ $30\sqrt{2}$
10. $9\sqrt{45}$ $9 \cdot \sqrt{9} \cdot \sqrt{5}$ $9 \cdot 3\sqrt{5} = $ $27\sqrt{5}$	11. $\sqrt{250}$ $\sqrt{25} \cdot \sqrt{10}$ $= 5\sqrt{10}$	12. $3\sqrt{578}$ $3 \cdot \sqrt{289} \cdot \sqrt{2}$ $3 \cdot 17 \cdot \sqrt{2} = $ $51\sqrt{2}$

SOLVING QUADRATICS

by Square Roots

If a quadratic equation can be written in the form $x^2 = n$, then the quadratic can be solved using the **Square Root Property**:

If $x^2 = n$, then $x = \pm\sqrt{n}$

- ① **REWRITE** the equation as $x^2 = n$
- ② Take the **SQUARE ROOT** of both sides.
- ③ **SIMPLIFY** the radical. Place " $\pm$ " to indicate both answers.

Directions: Solve each equation using the square roots method.

13. $x^2 - 64 = 0$
 $x^2 = 64$
 $x = \pm\sqrt{64}$
 $x = \{\pm 8\}$

14. $7x^2 + 8 = 15$
 $7x^2 = 7$
 $x^2 = 1$
 $x = \pm\sqrt{1}$
 $x = \{\pm 1\}$

15. $81x^2 + 5 = 21$
 $81x^2 = 16$
 $x^2 = \frac{16}{81}$
 $x = \pm\sqrt{\frac{16}{81}}$
 $x = \{\pm \frac{4}{9}\}$

16. $8x^2 + 1 = 17$
 $8x^2 = 16$
 $x^2 = 2$
 $x = \{\pm\sqrt{2}\}$

17. $2x^2 - 9 = 55$
 $2x^2 = 64$
 $x^2 = 32$
 $x = \pm\sqrt{32}$
 $x = \pm\sqrt{16 \cdot 2}$
 $x = \{\pm 4\sqrt{2}\}$

18. $9x^2 + 3 = 111$
 $9x^2 = 108$
 $x^2 = 12$
 $x = \pm\sqrt{12}$
 $x = \pm\sqrt{4 \cdot 3}$
 $x = \{\pm 2\sqrt{3}\}$

19. $4 - 3x^2 = -77$
 $-3x^2 = -81$
 $x^2 = 27$
 $x = \pm\sqrt{27}$
 $x = \pm\sqrt{9 \cdot 3}$
 $x = \{\pm 3\sqrt{3}\}$

20. $5x^2 + 10 = 310$
 $5x^2 = 300$
 $x^2 = 60$
 $x = \pm\sqrt{60}$
 $x = \pm\sqrt{4 \cdot 15}$
 $x = \{\pm 2\sqrt{15}\}$

21. $-\frac{1}{2}x^2 + 1 = -39$
 $-\frac{1}{2}x^2 = -40$
 $x^2 = 80$
 $x = \pm\sqrt{80}$
 $x = \pm\sqrt{16 \cdot 5}$
 $x = \{\pm 4\sqrt{5}\}$

22. $(x+2)^2 = 49$
 $x+2 = \pm\sqrt{49}$
 $x+2 = \pm 7$
 $x = -2 \pm 7$
 $x = \{-9, 5\}$

23. $3(x-1)^2 = 18$
 $(x-1)^2 = 6$
 $x-1 = \pm\sqrt{6}$
 $x = \{1 \pm \sqrt{6}\}$

24. $-2(x+7)^2 = -40$
 $(x+7)^2 = 20$
 $x+7 = \pm\sqrt{20}$
 $x+7 = \pm\sqrt{4 \cdot 5}$
 $x+7 = \pm 2\sqrt{5}$
 $x = \{-7 \pm 2\sqrt{5}\}$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 3: Solving Quadratics by
Square Roots**** This is a 2-page document! ******Directions:** Solve each equation by square roots. Give all solutions in simplest radical form.

1. $x^2 - 100 = 0$

$x^2 = 100$

$x = \pm \sqrt{100}$

$x = \{\pm 10\}$

2. $25x^2 = 4$

$x^2 = \frac{4}{25}$

$x = \pm \sqrt{\frac{4}{25}}$

$x = \{\pm \frac{2}{5}\}$

3. $5x^2 - 11 = 234$

$5x^2 = 245$

$x^2 = 49$

$x = \pm \sqrt{49}$

$x = \{\pm 7\}$

4. $\frac{1}{3}x^2 + 14 = 26$

$\frac{1}{3}x^2 = 12$

$x^2 = 36$

$x = \pm \sqrt{36}$

$x = \{\pm 6\}$

5. $2x^2 - 3 = -3$

$2x^2 = 0$

$x^2 = 0$

$x = \pm \sqrt{0}$

$x = \{0\}$

6. $9x^2 - 1 = 63$

$9x^2 = 64$

$x^2 = \frac{64}{9}$

$x = \pm \sqrt{\frac{64}{9}}$

$x = \{\pm \frac{8}{3}\}$

7. $x^2 - 1 = 39$

$x^2 = 40$

$x = \pm \sqrt{40}$

$x = \pm \sqrt{4} \sqrt{10}$

$x = \{\pm 2\sqrt{10}\}$

8. $5x^2 = 90$

$x^2 = 18$

$x = \pm \sqrt{18}$

$x = \pm \sqrt{9} \sqrt{2}$

$x = \{\pm 3\sqrt{2}\}$

9. $4x^2 - 30 = 22$

$4x^2 = 52$

$x^2 = 13$

$x = \{\pm \sqrt{13}\}$

10. $3x^2 + 4 = 166$

$3x^2 = 162$

$x^2 = 54$

$x = \pm \sqrt{54}$

$x = \pm \sqrt{9} \sqrt{6}$

$x = \{\pm 3\sqrt{6}\}$

11. $10 - 2x^2 = -198$

$$\frac{-2x^2}{-2} = \frac{-208}{-2}$$

$$x^2 = 104$$

$$x = \pm \sqrt{104}$$

$$x = \pm \sqrt{4} \sqrt{26}$$

$$x = \{\pm 2\sqrt{26}\}$$

12. $\frac{1}{4}x^2 - 7 = 25$

$$\frac{1}{4}x^2 = 32$$

$$x^2 = 128$$

$$x = \pm \sqrt{64} \sqrt{2}$$

$$x = \{\pm 8\sqrt{2}\}$$

13. $(x-4)^2 = 9$

$$x-4 = \pm \sqrt{9}$$

$$x-4 = \pm 3$$

$$x = 4 \pm 3$$

$$x = \{1, 7\}$$

14. $(x+1)^2 = 75$

$$x+1 = \pm \sqrt{75}$$

$$x+1 = \pm \sqrt{25} \sqrt{3}$$

$$x+1 = \pm 5\sqrt{3}$$

$$x = \{-1 \pm 5\sqrt{3}\}$$

15. $2(x+3)^2 = 56$

$$(x+3)^2 = 28$$

$$x+3 = \pm \sqrt{28}$$

$$x+3 = \pm \sqrt{4} \sqrt{7}$$

$$x+3 = \pm 2\sqrt{7}$$

$$x = \{3 \pm 2\sqrt{7}\}$$

16. $\frac{3}{2}(x-2)^2 = 72$

$$(x-2)^2 = 48$$

$$x-2 = \pm \sqrt{48}$$

$$x-2 = \pm \sqrt{16} \sqrt{3}$$

$$x-2 = \pm 4\sqrt{3}$$

$$x = \{2 \pm 4\sqrt{3}\}$$

17. Which of the equations below can be solved by square roots? How would you solve the other equation? Explain your reasoning. Give the solutions to both equations.

Equation A: $x^2 - 9x = 0$

$$x(x-9) = 0$$

$$x=0 ; x=9$$

$$x = \{0, 9\}$$

Equation B: $x^2 - 9 = 0$

$$x^2 = 9$$

$$x = \pm \sqrt{9}$$

$$x = \{\pm 3\}$$

This must be factored
since there is an x -term.

This can be solved
with square roots
since there is no
 x -term.

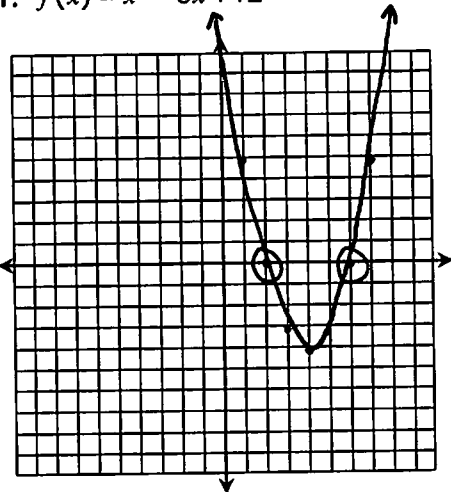
Name: _____ Algebra 2

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Quiz 4-1: Graphing, Factoring, & Square Roots Methods

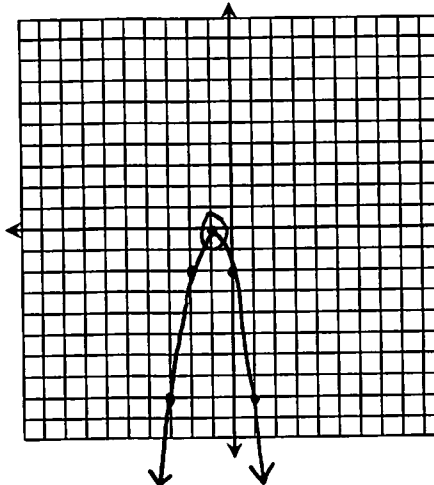
For questions 1-2, find the roots of the function by graphing.

1. $f(x) = x^2 - 8x + 12$



X	Y
1	5
2	0
3	-3
4	-4
5	-3
6	0
7	5

2. $f(x) = -2x^2 - 4x - 2$



1. $X = \{2, 6\}$

2. $X = \{-1\}$

X	Y
-4	-18
-3	-8
-2	-2
-1	0
0	-2
1	-8
2	-18

For questions 3-12, solve by factoring.

3. $x^2 + 5x - 36 = 0$

$$\begin{array}{l} (x+9)(x-4) = 0 \\ \hline x+9=0 \quad x-4=0 \\ x=-9 \quad x=4 \end{array}$$

4. $x^2 + 61 = 1 - 17x$

$$\begin{array}{l} x^2 + 17x + 60 = 0 \\ (x+12)(x+5) = 0 \\ \hline x+12=0 \quad x+5=0 \\ x=-12 \quad x=-5 \end{array}$$

3. $X = \{-9, 4\}$

4. $X = \{-12, -5\}$

5. $X = \{8, 14\}$

6. $X = \{-7, 3\}$

7. $X = \{-\frac{5}{3}, \frac{1}{4}\}$

8. $X = \{\frac{2}{5}\}$

5. $3x^2 + 112 = 2x^2 + 22x$

$x^2 - 22x + 112 = 0$

$$\begin{array}{l} (x-14)(x-8) = 0 \\ \hline x-14=0 \quad x-8=0 \\ x=14 \quad x=8 \end{array}$$

6. $2x^2 + 7x = 42 - x$

$2x^2 + 8x - 42 = 0$

$2(x^2 + 4x - 21) = 0$

$$\begin{array}{l} 2(x+7)(x-3) = 0 \\ \hline 2 \neq 0 \quad x+7=0 \quad x-3=0 \\ x=-7 \quad x=3 \end{array}$$

7. $12x^2 + 17x = 5$

$12x^2 + 17x - 5 = 0$

$x^2 + 17x - 60 = 0$

$$\begin{array}{l} (x+\frac{20}{12})(x-\frac{3}{12}) = 0 \end{array}$$

$$\begin{array}{l} (3x+5)(4x-1) = 0 \\ \hline 3x+5=0 \quad 4x-1=0 \\ x=-\frac{5}{3} \quad x=\frac{1}{4} \end{array}$$

8. $25x^2 = 20x - 4$

$25x^2 - 20x + 4 = 0$

$x^2 - 20x + 100 = 0$

$$\begin{array}{l} (x-\frac{10}{25})(x-\frac{10}{25}) = 0 \end{array}$$

$$\begin{array}{l} (5x-2)(5x-2) = 0 \\ \hline 5x-2=0 \quad 5x-2=0 \\ x=\frac{2}{5} \quad x=\frac{2}{5} \end{array}$$

$$9. \quad x^2 + 5x = 0$$

$$\begin{array}{c|c} x & (x+5) = 0 \\ \hline x=0 & x+5=0 \\ & x=-5 \end{array}$$

$$11. \quad x^2 = 100$$

$$x^2 - 100 = 0$$

$$\begin{array}{c|c} (x-10) & (x+10) = 0 \\ \hline x-10=0 & x+10=0 \\ x=10 & x=-10 \end{array}$$

$$10. \quad 8x^2 + 30x = 2x$$

$$8x^2 + 28x = 0$$

$$\begin{array}{c|c} 4x & (2x+7) = 0 \\ \hline 4x=0 & 2x+7=0 \\ x=0 & x=-\frac{7}{2} \end{array}$$

$$12. \quad 9x^2 - 6 = 10$$

$$9x^2 - 16 = 0$$

$$\begin{array}{c|c} (3x+4) & (3x-4) = 0 \\ \hline 3x+4=0 & 3x-4=0 \\ x=-\frac{4}{3} & x=\frac{4}{3} \end{array}$$

$$9. \quad x = \{-5, 0\}$$

$$10. \quad x = \{-\frac{7}{2}, 0\}$$

$$11. \quad x = \{-10, 10\}$$

$$12. \quad x = \{-\frac{4}{3}, \frac{4}{3}\}$$

For questions 13-16, solve using the square roots method. Write answers in simplest radical form.

$$13. \quad x^2 - 121 = 0$$

$$x^2 = 121$$

$$x = \pm \sqrt{121}$$

$$x = \pm 11$$

$$14. \quad 4x^2 - 64 = 17$$

$$4x^2 = 81$$

$$x^2 = \frac{81}{4}$$

$$x = \pm \sqrt{\frac{81}{4}} ; x = \pm \frac{9}{2}$$

$$13. \quad x = \{\pm 11\}$$

$$14. \quad x = \{\pm \frac{9}{2}\}$$

$$15. \quad x = \{7 \pm 2\sqrt{15}\}$$

$$16. \quad x = \{-2 \pm 6\sqrt{3}\}$$

$$15. \quad (x-7)^2 = 60$$

$$x-7 = \pm \sqrt{60}$$

$$x-7 = \pm \sqrt{4 \cdot 15}$$

$$x-7 = \pm 2\sqrt{15}$$

$$x = 7 \pm 2\sqrt{15}$$

$$16. \quad \frac{1}{4}(x+2)^2 = 27$$

$$(x+2)^2 = 108$$

$$x+2 = \pm \sqrt{108}$$

$$x+2 = \pm \sqrt{36 \cdot 3}$$

$$x+2 = \pm 6\sqrt{3}$$

$$x = -2 \pm 6\sqrt{3}$$

17. Write the function below in factored form and identify the axis of symmetry, vertex, and roots.

$$f(x) = -(x+3)^2 + 16$$

$$= -(x+3)(x+3) + 16$$

$$= -(x^2 + 6x + 9) + 16$$

$$= -x^2 - 6x - 9 + 16$$

$$= -x^2 - 6x + 7$$

$$= -1(x^2 + 6x - 7)$$

$$= -1(x+7)(x-1)$$

Factored Form: $f(x) = -1(x+7)(x-1)$	
Axis of Symmetry: $x = -3$	Vertex: $(-3, 16)$
Roots: $x = \{-7, 1\}$	

18. Write the function below in vertex form and identify the axis of symmetry, vertex, and roots.

$$f(x) = 2(x+9)(x+5)$$

$$= 2(x^2 + 14x + 45)$$

$$= 2x^2 + 28x + 90$$

$$= 2(x^2 + 14x) + 90$$

$$\quad \quad \quad \downarrow$$

$$\quad \quad \quad 7^2 = 49$$

$$= 2(x^2 + 14x + 49) + 90 - 98$$

$$= 2(x+7)^2 - 8$$

Vertex Form: $f(x) = 2(x+7)^2 - 8$	
Axis of Symmetry: $x = -7$	Vertex: $(-7, -8)$
Roots: $x = \{-9, -5\}$	

Name:	Date:
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Main Ideas/Questions	Notes/Examples						
IMAGINARY NUMBERS	- Equations such as $x^2 + 1 = 0$ have no real solution, so mathematicians defined the imaginary numbers to represent their solutions. - The Imaginary unit, i, is defined as $i = \sqrt{-1}$. This is useful when working with square roots of negative numbers. - A pure imaginary number is written in the form bi, where b is the real number and i is the imaginary part.						
<i>Simplifying</i> NEGATIVE SQUARE ROOTS	Step 1: Rewrite $\sqrt{-a}$ as $\sqrt{-1 \cdot a}$ Step 2: Break a down if it is not a perfect square. Step 3: Simplify the radical, recalling that $\sqrt{-1} = i$. <table><tr><td>1. $\sqrt{-9}$ $\sqrt{-1 \cdot 9}$ $= \boxed{3i}$</td><td>2. $\sqrt{-196}$ $\sqrt{-1 \cdot 196}$ $= \boxed{14i}$</td><td>3. $\sqrt{-5}$ $\sqrt{-1 \cdot 5}$ $= \boxed{i\sqrt{5}}$</td></tr><tr><td>4. $\sqrt{-80}$ $\sqrt{-1 \cdot 80}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16 \cdot 5}$ $= \boxed{4i\sqrt{5}}$</td><td>5. $\sqrt{-32}$ $\sqrt{-1 \cdot 32}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16} \sqrt{2}$ $= \boxed{4i\sqrt{2}}$</td><td>6. $\sqrt{-192}$ $\sqrt{-1 \cdot 192}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{64} \sqrt{3}$ $= \boxed{8i\sqrt{3}}$</td></tr></table>	1. $\sqrt{-9}$ $\sqrt{-1 \cdot 9}$ $= \boxed{3i}$	2. $\sqrt{-196}$ $\sqrt{-1 \cdot 196}$ $= \boxed{14i}$	3. $\sqrt{-5}$ $\sqrt{-1 \cdot 5}$ $= \boxed{i\sqrt{5}}$	4. $\sqrt{-80}$ $\sqrt{-1 \cdot 80}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16 \cdot 5}$ $= \boxed{4i\sqrt{5}}$	5. $\sqrt{-32}$ $\sqrt{-1 \cdot 32}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16} \sqrt{2}$ $= \boxed{4i\sqrt{2}}$	6. $\sqrt{-192}$ $\sqrt{-1 \cdot 192}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{64} \sqrt{3}$ $= \boxed{8i\sqrt{3}}$
1. $\sqrt{-9}$ $\sqrt{-1 \cdot 9}$ $= \boxed{3i}$	2. $\sqrt{-196}$ $\sqrt{-1 \cdot 196}$ $= \boxed{14i}$	3. $\sqrt{-5}$ $\sqrt{-1 \cdot 5}$ $= \boxed{i\sqrt{5}}$					
4. $\sqrt{-80}$ $\sqrt{-1 \cdot 80}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16 \cdot 5}$ $= \boxed{4i\sqrt{5}}$	5. $\sqrt{-32}$ $\sqrt{-1 \cdot 32}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{16} \sqrt{2}$ $= \boxed{4i\sqrt{2}}$	6. $\sqrt{-192}$ $\sqrt{-1 \cdot 192}$ $\downarrow \quad \downarrow$ $i \quad \sqrt{64} \sqrt{3}$ $= \boxed{8i\sqrt{3}}$					
QUADRATIC EQUATIONS <i>with Imaginary Solutions</i>	Solve using the square roots method. Write all answers in simplest form. <table><tr><td>7. $x^2 + 81 = 0$ $x^2 = -81$ $x = \pm \sqrt{-81}$ $x = \pm \sqrt{-1 \cdot 81}$ $\boxed{x = \pm 9i}$</td><td>8. $2x^2 + 9 = 1$ $2x^2 = -8$ $x^2 = -4$ $x = \pm \sqrt{-4}$ $x = \pm \sqrt{-1 \cdot 4}$ $\boxed{x = \pm 2i}$</td></tr><tr><td>9. $4x^2 + 15 = -9$ $4x^2 = -24$ $x^2 = -6$ $x = \pm \sqrt{-6}$ $x = \pm \sqrt{-1 \cdot 6}$ $\boxed{x = \pm i\sqrt{6}}$</td><td>10. $x^2 + 13 = 1$ $x^2 = -12$ $x = \pm \sqrt{-12}$ $x = \pm \sqrt{-1 \cdot 12}$ $x = \pm i \sqrt{4 \cdot 3}$ $\boxed{x = \pm 2i\sqrt{3}}$</td></tr></table>	7. $x^2 + 81 = 0$ $x^2 = -81$ $x = \pm \sqrt{-81}$ $x = \pm \sqrt{-1 \cdot 81}$ $\boxed{x = \pm 9i}$	8. $2x^2 + 9 = 1$ $2x^2 = -8$ $x^2 = -4$ $x = \pm \sqrt{-4}$ $x = \pm \sqrt{-1 \cdot 4}$ $\boxed{x = \pm 2i}$	9. $4x^2 + 15 = -9$ $4x^2 = -24$ $x^2 = -6$ $x = \pm \sqrt{-6}$ $x = \pm \sqrt{-1 \cdot 6}$ $\boxed{x = \pm i\sqrt{6}}$	10. $x^2 + 13 = 1$ $x^2 = -12$ $x = \pm \sqrt{-12}$ $x = \pm \sqrt{-1 \cdot 12}$ $x = \pm i \sqrt{4 \cdot 3}$ $\boxed{x = \pm 2i\sqrt{3}}$		
7. $x^2 + 81 = 0$ $x^2 = -81$ $x = \pm \sqrt{-81}$ $x = \pm \sqrt{-1 \cdot 81}$ $\boxed{x = \pm 9i}$	8. $2x^2 + 9 = 1$ $2x^2 = -8$ $x^2 = -4$ $x = \pm \sqrt{-4}$ $x = \pm \sqrt{-1 \cdot 4}$ $\boxed{x = \pm 2i}$						
9. $4x^2 + 15 = -9$ $4x^2 = -24$ $x^2 = -6$ $x = \pm \sqrt{-6}$ $x = \pm \sqrt{-1 \cdot 6}$ $\boxed{x = \pm i\sqrt{6}}$	10. $x^2 + 13 = 1$ $x^2 = -12$ $x = \pm \sqrt{-12}$ $x = \pm \sqrt{-1 \cdot 12}$ $x = \pm i \sqrt{4 \cdot 3}$ $\boxed{x = \pm 2i\sqrt{3}}$						

	11. $3x^2 - 5 = -446$ $3x^2 = -441$ $x^2 = \sqrt{-147}$ $x = \pm \sqrt{-1 \cdot 147}$ $x = \pm i \sqrt{49 \cdot 3}$ $x = \pm 7i\sqrt{3}$	12. $-\frac{2}{3}x^2 - 1 = 17$ $-\frac{2}{3}x^2 = 18$ $x^2 = -27$ $x = \pm \sqrt{-27}$ $x = \pm \sqrt{-1 \cdot 27}$ $x = \pm i \sqrt{9 \cdot 3}$ $x = \pm 3i\sqrt{3}$
POWERS OF i	$i^1 = \frac{\sqrt{-1}}{} = i$ $i^2 = \frac{(\sqrt{-1})^2}{} = -1$ $i^3 = \frac{i^2 \cdot i}{} = -1 \cdot i = -i$ $i^4 = \frac{(i^2)^2}{} = (-1)^2 = 1$	$i^5 = \frac{(i^2)^2 \cdot i}{} = 1 \cdot i = i$ $i^6 = \frac{(i^2)^3}{} = (-1)^3 = -1$ $i^7 = \frac{(i^2)^3 \cdot i}{} = -1 \cdot i = -i$ $i^8 = \frac{(i^2)^4}{} = (-1)^4 = 1$
TIP: Break apart into a power of i^2 to simplify!	13. i^{15} $(i^2)^7 \cdot i$ $= (-1)^7 \cdot i = -1 \cdot i = \boxed{-i}$	14. i^{62} $(i^2)^{31} = (-1)^{31}$ $= \boxed{-1}$
	15. i^{76} $(i^2)^{38} = (-1)^{38}$ $= \boxed{1}$	16. i^{43} $(i^2)^{21} \cdot i$ $= (-1)^{21} \cdot i = -1 \cdot i = \boxed{-i}$
FINDING PRODUCTS	Simplify each expression. 17. $4i \cdot 7i$ $28i^2$ $= 28(-1) = \boxed{-28}$	18. $(-4i)(2i)(-9i)$ $72i^3$ $= 72(-i)$ $= -72i$
	19. $(2i)^3 \cdot (5i)$ $8i^3 \cdot 5i = 40i^4$ $= 40(1)$ $= 40$	20. $(i\sqrt{3})^2 \cdot (-8i)^2$ $3i^2 \cdot 64i^2$ $= 192i^4$ $= 192(1)$ $= 192$
WATCH OUT: Rewrite negative square roots as imaginary numbers BEFORE performing operations!	21. $\sqrt{-18} \cdot \sqrt{-10}$ $i\sqrt{18} \cdot i\sqrt{10}$ $= i^2 \sqrt{180}$ $= -1 \sqrt{36 \cdot 5}$ $= -6\sqrt{5}$	22. $\sqrt{-24} \cdot \sqrt{-3} \cdot \sqrt{-2}$ $i\sqrt{24} \cdot i\sqrt{3} \cdot i\sqrt{2}$ $= i^3 \sqrt{144}$ $= -i(12)$ $= -12i$

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 4: Pure Imaginary Numbers

**** This is a 2-page document! ******Directions:** Simplify the expressions below.

1. $\sqrt{-25}$

$$\sqrt{-1 \cdot 25}$$

$$= \boxed{5i}$$

2. $\sqrt{-324}$

$$\sqrt{-1 \cdot 324}$$

$$= \boxed{18i}$$

3. $\sqrt{-15}$

$$\sqrt{-1 \cdot 15}$$

$$= \boxed{i\sqrt{15}}$$

4. $\sqrt{-44}$

$$\sqrt{-1 \cdot 44}$$

$$= i\sqrt{4} \sqrt{11}$$

$$= \boxed{2i\sqrt{11}}$$

5. $\sqrt{-252}$

$$\sqrt{-1 \cdot 252}$$

$$= i\sqrt{36} \sqrt{7}$$

$$= \boxed{6i\sqrt{7}}$$

6. $\sqrt{-288}$

$$\sqrt{-1 \cdot 288}$$

$$= i\sqrt{144} \sqrt{2}$$

$$= \boxed{12i\sqrt{2}}$$

7. $\sqrt{-176}$

$$\sqrt{-1 \cdot 176}$$

$$= i\sqrt{16} \sqrt{11}$$

$$= \boxed{4i\sqrt{11}}$$

8. $\sqrt{-8} \cdot \sqrt{24}$

$$i\sqrt{8} \cdot \sqrt{24}$$

$$= i\sqrt{192}$$

$$= i\sqrt{64} \sqrt{3}$$

$$= \boxed{8i\sqrt{3}}$$

9. $\sqrt{-6} \cdot \sqrt{-12} \cdot \sqrt{-5}$

$$i\sqrt{6} \cdot i\sqrt{12} \cdot i\sqrt{5}$$

$$= i^3 \sqrt{360}$$

$$= -i\sqrt{36} \cdot \sqrt{10}$$

$$= \boxed{-6i\sqrt{10}}$$

10. i^{28}

$$(i^2)^{14} = (-1)^{14}$$

$$= \boxed{1}$$

11. i^{49}

$$(i^2)^{24} \cdot i$$

$$= (-1)^{24} \cdot i$$

$$= 1 \cdot i = \boxed{i}$$

12. i^{86}

$$(i^2)^{43} = (-1)^{43}$$

$$= \boxed{-1}$$

13. $8i \cdot -9i \cdot 6i$

$$-432 i^3$$

$$= -432 \cdot i^2 \cdot i$$

$$= -432 \cdot (-1) \cdot i$$

$$= \boxed{432i}$$

14. $(-4i)^3 \cdot 2i$

$$-64i^3 \cdot 2i$$

$$= -128 i^4$$

$$= -128(1)$$

$$= \boxed{-128}$$

15. $(2i)^5 \cdot (i\sqrt{6})^2$

$$32i^5 \cdot 6i^2$$

$$= 192 i^7$$

$$= 192(i^2)^3 \cdot i$$

$$= 192(-1) \cdot i$$

$$= \boxed{-192i}$$

Directions: Solve using the square roots method. Write all answers in simplest form.

16. $x^2 + 4 = 0$

$$x^2 = -4$$

$$x = \pm \sqrt{-4}$$

$$x = \pm \sqrt{-1 \cdot 4}$$

$$x = \pm 2i$$

17. $-3x^2 = 150$

$$x^2 = -50$$

$$x = \pm \sqrt{-50}$$

$$x = \pm \sqrt{-1 \cdot 50}$$

$$x = \pm i \sqrt{25 \cdot 2}$$

$$x = \pm 5i\sqrt{2}$$

18. $2x^2 + 71 = -163$

$$2x^2 = -234$$

$$x^2 = -117$$

$$x = \pm \sqrt{-117}$$

$$x = \pm \sqrt{-1 \cdot 117}$$

$$x = \pm i \sqrt{9 \cdot 13}$$

$$x = \pm 3i\sqrt{13}$$

19. $18 - x^2 = 39$

$$-x^2 = 21$$

$$x^2 = -21$$

$$x = \pm \sqrt{-21}$$

$$x = \pm \sqrt{-1 \cdot 21}$$

$$x = \pm i\sqrt{21}$$

20. $-4x^2 - 3 = 481$

$$-4x^2 = 484$$

$$x^2 = -121$$

$$x = \pm \sqrt{-121}$$

$$x = \pm \sqrt{-1 \cdot 121}$$

$$x = \pm 11i$$

21. $-\frac{4}{3}x^2 - 15 = 201$

$$-\frac{4}{3}x^2 = 216$$

$$x^2 = -162$$

$$x = \pm \sqrt{-162}$$

$$x = \pm \sqrt{-1 \cdot 162}$$

$$x = \pm i \sqrt{81 \cdot 2}$$

$$x = \pm 9i\sqrt{2}$$

Directions: The solutions of a quadratic equation are given below. Give an example of an equation with the given solutions.

22. $\pm 4i$

$$x = \pm 4i$$

$$x = \pm \sqrt{-16}$$

$$x^2 = -16$$

$$x^2 + 16 = 0$$

23. $\pm i\sqrt{7}$

$$x = \pm i\sqrt{7}$$

$$x = \pm \sqrt{-7}$$

$$x^2 = -7$$

$$x^2 + 7 = 0$$

24. $\pm 3i\sqrt{2}$

$$x = \pm 3i\sqrt{2}$$

$$x = \pm \sqrt{-18}$$

$$x^2 = -18$$

$$x^2 + 18 = 0$$

25. $\pm 8i\sqrt{3}$

$$x = \pm 8i\sqrt{3}$$

$$x = \pm \sqrt{-192}$$

$$x^2 = -192$$

$$x^2 + 192 = 0$$

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Topic:	Class:

Main Ideas/Questions	Notes/Examples	
COMPLEX NUMBERS $\mathbb{C}$	- A complex number is a number written in the form <u>$a+bi$</u> where <u>a</u> is the real part and <u>bi</u> is the imaginary part. - Examples: <u>$6-3i$, $-4-7i$, $8+2i$, $-5+i$</u>	
ADDING & SUBTRACTING <i>Complex Numbers</i>	Directions: Simplify the expressions below. Final answers must be in $a+bi$ form.	
	1. $(-11+3i) + (9+2i)$ $-2+5i$	2. $(4+i) + (7-5i)$ $11-4i$
MULTIPLYING <i>Complex Numbers</i> **Recall** $i^2 = -1$	3. $(7-2i) - (2+6i)$ $5-8i$	4. $6i - (14-i) + (5-3i)$ $-9+4i$
	Directions: Simplify the expressions below. Final answers must be in $a+bi$ form.	
	5. $2i(8-3i)$ $16i-6i^2$ $16i-6(-1)$ $6+16i$	6. $-i(-2+10i)$ $2i-10i^2$ $2i-10(-1)$ $10+2i$
	7. $(7+i)(4-i)$ $28-7i+4i-i^2$ $28-3i-(-1)$ $29-3i$	8. $(2-4i)(-5-3i)$ $-10-6i+20i+12i^2$ $-10+14i+12(-1)$ $-10+14i-12$ $-22+14i$
	9. $(6-2i)^2$ $(6-2i)(6-2i)$ $36-12i-12i+4i^2$ $36-24i+4(-1)$ $36-24i-4$ $32-24i$	10. $(1+7i)(9+3i) - (4+2i)$ $9+3i+63i+21i^2 - 4-2i$ $5+64i+21(-1)$ $5+64i-21$ $-16+64i$

COMPLEX CONJUGATES

Two complex numbers in the form $a + bi$ and $a - bi$ are called complex conjugates. The product of two conjugates is always a real number.

11. $(8+i)(8-i)$

$$64 - 8i + 8i - i^2$$

$$64 - (-1)$$

$$64 + 1 = \boxed{65}$$

12. $(3-i)(3+i)$

$$9 + 3i - 3i - i^2$$

$$9 - (-1)$$

$$9 + 1 = \boxed{10}$$

13. $(5-4i)(5+4i)$

$$25 + 20i - 20i - 16i^2$$

$$25 - 16(-1)$$

$$25 + 16 = \boxed{41}$$

14. $(-7+2i)(-7-2i)$

$$49 + 14i - 14i - 4i^2$$

$$49 - 4(-1)$$

$$49 + 4 = \boxed{53}$$

QUADRATIC EQUATIONS

with Complex
Solutions

Solve using the square roots method. Write all answers in simplest form.

15. $(x+2)^2 = -81$

$$x+2 = \pm \sqrt{-81}$$

$$x+2 = \pm \sqrt{-1 \cdot 81}$$

$$x+2 = \pm 9i$$

$$\boxed{x = -2 \pm 9i}$$

16. $(x-6)^2 = -32$

$$x-6 = \pm \sqrt{-32}$$

$$x-6 = \pm \sqrt{-1 \cdot 32}$$

$$x-6 = \pm i \sqrt{16} \sqrt{2}$$

$$x-6 = \pm 4i \sqrt{2}$$

$$\boxed{x = 6 \pm 4i\sqrt{2}}$$

17. $-(x+4)^2 = 75$

$$(x+4)^2 = -75$$

$$x+4 = \pm \sqrt{-75}$$

$$x+4 = \pm \sqrt{-1 \cdot 75}$$

$$x+4 = \pm i \sqrt{25} \sqrt{3}$$

$$x+4 = \pm 5i \sqrt{3}$$

$$\boxed{x = -4 \pm 5i\sqrt{3}}$$

18. $2(x-1)^2 = -56$

$$(x-1)^2 = -28$$

$$x-1 = \pm \sqrt{-28}$$

$$x-1 = \pm \sqrt{-1 \cdot 28}$$

$$x-1 = \pm i \sqrt{4} \sqrt{7}$$

$$x-1 = \pm 2i \sqrt{7}$$

$$\boxed{x = 1 \pm 2i\sqrt{7}}$$

19. $-\frac{1}{2}(x-7)^2 = 27$

$$(x-7)^2 = -54$$

$$x-7 = \pm \sqrt{-54}$$

$$x-7 = \pm \sqrt{-1 \cdot 54}$$

$$x-7 = \pm i \sqrt{9} \sqrt{6}$$

$$x-7 = \pm 3i \sqrt{6}$$

$$\boxed{x = 7 \pm 3i\sqrt{6}}$$

20. $\frac{3}{4}(x+3)^2 = -96$

$$(x+3)^2 = -128$$

$$x+3 = \pm \sqrt{-128}$$

$$x+3 = \pm \sqrt{-1 \cdot 128}$$

$$x+3 = \pm i \sqrt{64} \sqrt{2}$$

$$x+3 = \pm 8i \sqrt{2}$$

$$\boxed{x = -3 \pm 8i\sqrt{2}}$$

Name: _____

Unit 4: Solving Quadratic Equations

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Homework 5: Complex Numbers

**** This is a 2-page document! ****

Simplify each expression below.

1. $(11-7i) + (2-5i)$

$13 - 12i$

2. $(-6+10i) - (1-2i)$

$-7 + 12i$

3. $(7+3i) + (-1-14i)$

$6 - 11i$

4. $(-3+8i) - (1+8i)$

-4

5. $-5i(-2-i)$

$10i + 5i^2$

$10i + 5(-1)$

$-5 + 10i$

6. $8i(3-6i)$

$24i - 48i^2$

$24i - 48(-1)$

$48 + 24i$

7. $(7+i)(4+3i)$

$28 + 21i + 4i + 3i^2$

$28 + 25i + 3(-1)$

$28 + 25i - 3$

$25 + 25i$

8. $(-3-8i)(1-2i)$

$-3 + 6i - 8i + 16i^2$

$-3 - 2i + 16(-1)$

$-3 - 2i - 16$

$-19 - 2i$

9. $(-7+i)^2$

$(-7+i)(-7+i)$

$49 - 7i - 7i + i^2$

$49 - 14i + (-1)$

$48 - 14i$

10. $(5-2i)(2+7i) - 2i(3i)^2$

$10 + 35i - 4i - 14i^2 - 2i(9i^2)$

$10 + 31i - 14(-1) - 2i(-9)$

$10 + 31i + 14 + 18i$

$24 + 49i$

11. $(3+5i)(3-5i)$

$9 - 15i + 15i - 25i^2$

$9 - 25(-1)$

$9 + 25$

$= 34$

12. $(-2-9i)(-2+9i)$

$4 - 18i + 18i - 81i^2$

$4 - 81(-1)$

$4 + 81 = 85$

Solve using the square roots method. Write all answers in simplest form.

13. $(x-1)^2 = -16$

$$x-1 = \pm \sqrt{-16}$$

$$x-1 = \pm \sqrt{-1 \cdot 16}$$

$$x-1 = \pm 4i$$

$$\boxed{x = 1 \pm 4i}$$

14. $(x+4)^2 = -90$

$$x+4 = \pm \sqrt{-90}$$

$$x+4 = \pm \sqrt{-1 \cdot 90}$$

$$x+4 = \pm i \sqrt{9} \sqrt{10}$$

$$x+4 = \pm 3i \sqrt{10}$$

$$\boxed{x = -4 \pm 3i\sqrt{10}}$$

15. $(x+9)^2 = -24$

$$x+9 = \pm \sqrt{-24}$$

$$x+9 = \pm \sqrt{-1 \cdot 24}$$

$$x+9 = \pm i \sqrt{4} \sqrt{6}$$

$$x+9 = \pm 2i \sqrt{6}$$

$$\boxed{x = -9 \pm 2i\sqrt{6}}$$

16. $-(x-5)^2 = 108$

$$(x-5)^2 = -108$$

$$x-5 = \pm \sqrt{-108}$$

$$x-5 = \pm \sqrt{-1 \cdot 108}$$

$$x-5 = \pm i \sqrt{36} \sqrt{3}$$

$$x-5 = \pm 6i \sqrt{3}$$

$$\boxed{x = 5 \pm 6i\sqrt{3}}$$

17. $-3(x+2)^2 = 24$

$$(x+2)^2 = -8$$

$$x+2 = \pm \sqrt{-8}$$

$$x+2 = \pm \sqrt{-1 \cdot 8}$$

$$x+2 = \pm i \sqrt{4} \sqrt{2}$$

$$x+2 = \pm 2i \sqrt{2}$$

$$\boxed{x = -2 \pm 2i\sqrt{2}}$$

18. $5(x-3)^2 = -225$

$$(x-3)^2 = -45$$

$$x-3 = \pm \sqrt{-45}$$

$$x-3 = \pm \sqrt{-1 \cdot 45}$$

$$x-3 = \pm i \sqrt{9} \sqrt{5}$$

$$x-3 = \pm 3i \sqrt{5}$$

$$\boxed{x = 3 \pm 3i\sqrt{5}}$$

19. $\frac{1}{3}(x-7)^2 = -28$

$$(x-7)^2 = -84$$

$$x-7 = \pm \sqrt{-84}$$

$$x-7 = \pm \sqrt{-1 \cdot 84}$$

$$x-7 = \pm i \sqrt{4} \sqrt{21}$$

$$x-7 = \pm 2i \sqrt{21}$$

$$\boxed{x = 7 \pm 2i\sqrt{21}}$$

20. $-\frac{5}{2}(x+1)^2 = 30$

$$(x+1)^2 = -12$$

$$x+1 = \pm \sqrt{-12}$$

$$x+1 = \pm \sqrt{-1 \cdot 12}$$

$$x+1 = \pm i \sqrt{4} \sqrt{3}$$

$$x+1 = \pm 2i \sqrt{3}$$

$$\boxed{x = -1 \pm 2i\sqrt{3}}$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions	Notes/Examples	
DIVIDING <i>Complex Numbers</i> WATCH OUT! <i>"i" can not be in the denominator of a complex number.</i>	CASE 1	THE DENOMINATOR IS A MONOMIAL: Multiply both the numerator and denominator by "i"
	1. $\frac{10 \cdot i}{2i \cdot i} = \frac{10i}{2i^2}$ $= \frac{10i}{-2}$ $= \boxed{-5i}$	2. $\frac{4 \cdot i}{-9i \cdot i} = \frac{4i}{-9i^2}$ $= \boxed{\frac{4i}{9}}$
	3. $\frac{-7 \cdot i}{5i \cdot i} = \frac{-7i}{5i^2}$ $= \frac{-7i}{-5}$ $= \boxed{\frac{7i}{5}}$	4. $\frac{(28-8i) \cdot i}{4i \cdot i} = \frac{28i-8i^2}{4i^2}$ $= \frac{28i+8}{-4}$ $= \boxed{-2-7i}$
	5. $\frac{(1+4i) \cdot i}{2i \cdot i} = \frac{i+4i^2}{2i^2}$ $= \frac{i-4}{-2}$ $= \boxed{\frac{4-i}{2}}$	6. $\frac{(-8-2i) \cdot i}{6i \cdot i} = \frac{-8i-2i^2}{6i^2}$ $= \frac{-8i+2}{-6}$ $= \boxed{\frac{-1+4i}{3}}$
BINOMIAL <i>Denominators</i>	CASE 2	THE DENOMINATOR IS A BINOMIAL: Multiply both the numerator and denominator by the conjugate of the denominator.
	7. $\frac{(-2)(5+i)}{(5-i)(5+i)}$ $= \frac{-10-2i}{25+5i-5i-i^2}$ $= \frac{-10-2i}{26}$ $= \boxed{\frac{-5-i}{13}}$	8. $\frac{(5i)(6-2i)}{(6+2i)(6-2i)}$ $= \frac{30i-10i^2}{36-12i+12i-4i^2}$ $= \frac{10+30i}{40}$ $= \boxed{\frac{1+3i}{4}}$

$$\begin{aligned}
 9. \frac{3i}{(4-3i)(4+3i)} \\
 &= \frac{12i + 9i^2}{16 + 12i - 12i - 9i^2} \\
 &= \frac{-9 + 12i}{25}
 \end{aligned}$$

$$\begin{aligned}
 10. \frac{2}{(-3-5i)(-3+5i)} \\
 &= \frac{-6 + 10i}{9 - 15i + 15i - 25i^2} \\
 &= \frac{-6 + 10i}{34} \\
 &= \frac{-3 + 5i}{17}
 \end{aligned}$$

$$\begin{aligned}
 11. \frac{7i}{(-4+8i)(-4-8i)} \\
 &= \frac{-28i - 56i^2}{16 + 32i - 32i - 64i^2} \\
 &= \frac{56 - 28i}{80} \\
 &= \frac{14 - 7i}{20}
 \end{aligned}$$

$$\begin{aligned}
 12. \frac{(7+3i)(2-i)}{(2+i)(2-i)} \\
 &= \frac{14 - 7i + 6i - 3i^2}{4 - 2i + 2i - i^2} \\
 &= \frac{14 - i + 3}{4 + 1} \\
 &= \frac{17 - i}{5}
 \end{aligned}$$

$$\begin{aligned}
 13. \frac{(1+8i)(2+4i)}{(2-4i)(2+4i)} \\
 &= \frac{2 + 4i + 16i + 32i^2}{4 + 8i - 8i - 16i^2} \\
 &= \frac{2 + 20i - 32}{4 + 16} \\
 &= \frac{-30 + 20i}{20} = \frac{-3 + 2i}{2}
 \end{aligned}$$

$$\begin{aligned}
 14. \frac{(-3+7i)(-9-i)}{(-9+i)(-9-i)} \\
 &= \frac{27 + 3i - 63i - 7i^2}{81 + 9i - 9i - i^2} \\
 &= \frac{27 - 60i + 7}{81 + 1} \\
 &= \frac{34 - 60i}{82} \\
 &= \frac{17 - 30i}{41}
 \end{aligned}$$

$$\begin{aligned}
 15. \frac{(9+6i)(5-5i)}{(5+5i)(5-5i)} \\
 &= \frac{45 - 45i + 30i - 30i^2}{25 - 25i + 25i - 25i^2} \\
 &= \frac{45 - 15i + 30}{25 + 25} \\
 &= \frac{75 - 15i}{50} \\
 &= \frac{15 - 3i}{10}
 \end{aligned}$$

$$\begin{aligned}
 16. \frac{(-8-4i)(-4+3i)}{(-4-3i)(-4+3i)} \\
 &= \frac{32 - 24i + 16i - 12i^2}{16 - 12i + 12i - 9i^2} \\
 &= \frac{32 - 8i + 12}{16 + 9} \\
 &= \frac{44 - 8i}{25}
 \end{aligned}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples										
ORGANIZING <i>Number Sets</i>	Pure imaginary numbers and real numbers make up the complex numbers. The complex numbers and its subsets are organized as: Complex Numbers (C) Pure Imaginary (i) Real Numbers (R) Irrational (I) Rational (Q) Integers (Z) Whole (W) Natural (N)										
CLASSIFYING <i>Numbers</i>	Directions: Simplify, then name all sets to which each value belongs. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $\sqrt{-64} = \boxed{8i}$ $\boxed{i, C}$ </td><td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $10i \cdot -2i$ $-20i^2 = \boxed{20}$ $\boxed{N, W, Z, Q, R, C}$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 3. $\sqrt{\frac{1}{9}} = \boxed{\frac{1}{3}}$ $\boxed{Q, R, C}$ </td><td style="padding: 5px; vertical-align: top;"> 4. $(15 - 8i) - (9 - 3i)$ $\boxed{6 - 5i}$ $\boxed{C}$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 5. $(4 - i)(9 + 5i)$ $36 + 20i - 9i - 5i^2$ $= \boxed{41 + 11i}$ $\boxed{C}$ </td><td style="padding: 5px; vertical-align: top;"> 6. i^{41} $(i^2)^{20} \cdot i = (-1)^{20} \cdot i = \boxed{i}$ $\boxed{i, C}$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 7. $(-2i)^3 \cdot (i\sqrt{7})^2$ $-8i^3 \cdot 7i^2 = -56i^5$ $= -56(i^2)^2 \cdot i$ $= \boxed{-56i}$ $\boxed{i, C}$ </td><td style="padding: 5px; vertical-align: top;"> 8. $(3 - 5i)(3 + 5i)$ $9 + 15i - 15i - 25i^2$ $= 9 + 25 = \boxed{34}$ $\boxed{N, W, Z, Q, R, C}$ </td></tr> <tr> <td style="padding: 5px; vertical-align: top;"> 9. $i^{16} (i^2)^8 = (-1)^8 = \boxed{1}$ $\boxed{N, W, Z, Q, R, C}$ </td><td style="padding: 5px; vertical-align: top;"> 10. $\sqrt{150}$ $\sqrt{25} \cdot \sqrt{6}$ $= \boxed{5\sqrt{6}}$ $\boxed{I, R, C}$ </td></tr> </table>	1. $\sqrt{-64} = \boxed{8i}$ $\boxed{i, C}$	2. $10i \cdot -2i$ $-20i^2 = \boxed{20}$ $\boxed{N, W, Z, Q, R, C}$	3. $\sqrt{\frac{1}{9}} = \boxed{\frac{1}{3}}$ $\boxed{Q, R, C}$	4. $(15 - 8i) - (9 - 3i)$ $\boxed{6 - 5i}$ $\boxed{C}$	5. $(4 - i)(9 + 5i)$ $36 + 20i - 9i - 5i^2$ $= \boxed{41 + 11i}$ $\boxed{C}$	6. i^{41} $(i^2)^{20} \cdot i = (-1)^{20} \cdot i = \boxed{i}$ $\boxed{i, C}$	7. $(-2i)^3 \cdot (i\sqrt{7})^2$ $-8i^3 \cdot 7i^2 = -56i^5$ $= -56(i^2)^2 \cdot i$ $= \boxed{-56i}$ $\boxed{i, C}$	8. $(3 - 5i)(3 + 5i)$ $9 + 15i - 15i - 25i^2$ $= 9 + 25 = \boxed{34}$ $\boxed{N, W, Z, Q, R, C}$	9. $i^{16} (i^2)^8 = (-1)^8 = \boxed{1}$ $\boxed{N, W, Z, Q, R, C}$	10. $\sqrt{150}$ $\sqrt{25} \cdot \sqrt{6}$ $= \boxed{5\sqrt{6}}$ $\boxed{I, R, C}$
1. $\sqrt{-64} = \boxed{8i}$ $\boxed{i, C}$	2. $10i \cdot -2i$ $-20i^2 = \boxed{20}$ $\boxed{N, W, Z, Q, R, C}$										
3. $\sqrt{\frac{1}{9}} = \boxed{\frac{1}{3}}$ $\boxed{Q, R, C}$	4. $(15 - 8i) - (9 - 3i)$ $\boxed{6 - 5i}$ $\boxed{C}$										
5. $(4 - i)(9 + 5i)$ $36 + 20i - 9i - 5i^2$ $= \boxed{41 + 11i}$ $\boxed{C}$	6. i^{41} $(i^2)^{20} \cdot i = (-1)^{20} \cdot i = \boxed{i}$ $\boxed{i, C}$										
7. $(-2i)^3 \cdot (i\sqrt{7})^2$ $-8i^3 \cdot 7i^2 = -56i^5$ $= -56(i^2)^2 \cdot i$ $= \boxed{-56i}$ $\boxed{i, C}$	8. $(3 - 5i)(3 + 5i)$ $9 + 15i - 15i - 25i^2$ $= 9 + 25 = \boxed{34}$ $\boxed{N, W, Z, Q, R, C}$										
9. $i^{16} (i^2)^8 = (-1)^8 = \boxed{1}$ $\boxed{N, W, Z, Q, R, C}$	10. $\sqrt{150}$ $\sqrt{25} \cdot \sqrt{6}$ $= \boxed{5\sqrt{6}}$ $\boxed{I, R, C}$										

PROPERTIES OF COMPLEX NUMBERS

The properties of real numbers (Commutative, Associative, Identity, Inverse, and Distributive) can also be applied to complex numbers.

Directions: For questions 11 – 15, write an example of the property using the complex numbers.

- | | |
|---|---|
| 11. Commutative Property of Addition: | <u>$3+4i = 4i+3$</u> |
| 12. Associative Property of Multiplication: | <u>$2 \cdot [3 \cdot (4+5i)] = (2 \cdot 3) \cdot (4+5i)$</u> |
| 13. Distributive Property: | <u>$5(6-7i) = 30-35i$</u> |
| 14. Identity Property of Multiplication: | <u>$7i \cdot 1 = 7i$</u> |
| 15. Inverse Property of Addition: | <u>$(3-8i) + (-3+8i) = 0$</u> |

Directions: For questions 16 – 21, name the property that justifies the statement.

- | | |
|--|--|
| 16. $2i(7-5i) = 2i \cdot 7 - 2i \cdot 5i$
<div style="text-align: center; margin-top: 10px;">Distributive Property</div> | 17. $\frac{1}{3i} \cdot 3i = 1$
<div style="text-align: center; margin-top: 10px;">Inverse Property of Multiplication</div> |
| 18. $(-5+4i)+9i = -5+(4i+9i)$
<div style="text-align: center; margin-top: 10px;">Associative Property of Addition</div> | 19. $(-12+i)+0 = (-12+i)$
<div style="text-align: center; margin-top: 10px;">Identity Property of Addition</div> |
| 20. $(-6i \cdot 2i) \cdot 8i = 8i \cdot (-6i \cdot 2i)$
<div style="text-align: center; margin-top: 10px;">Commutative Property of Multiplication</div> | 21. $(7-2i) + -(7-2i) = 0$
<div style="text-align: center; margin-top: 10px;">Inverse Property of Addition</div> |

22. Name the additive and multiplicative identity of $6i$.

Add. identity : 0
Mult. identity : 1

23. Name the additive and multiplicative inverse of $(-8-3i)$.

Add. Inverse: $(8+3i)$ Mult. Inverse: $\frac{1}{(-8-3i)}$

Directions: For questions 24 – 28 answer true or false. If false, provide a counterexample.

- | | |
|---|--|
| 24. The set of complex numbers is closed under addition. | <u>True</u> |
| 25. The set of complex numbers is closed under division. | <u>True</u> |
| 26. All complex numbers are pure imaginary numbers. | <u>False; $7+2i$</u> |
| 27. A number cannot be both irrational and complex. | <u>False; $\sqrt{-47}+3 = 3+i\sqrt{47}$</u> |
| 28. The product of a real number and a complex number is a real number. | <u>False; $2(3+2i) = 6+4i$</u> |

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 6: Dividing Complex Numbers;
Classifying & Properties**** This is a 2-page document! ****

Simplify each expression below.

$$1. \frac{6 \cdot i}{i \cdot i} = \frac{6i}{i^2} = \frac{6i}{-1} = \boxed{-6i}$$

$$2. \frac{-4 \cdot i}{7i \cdot i} = \frac{-4i}{7i^2} = \frac{-4i}{-7} = \boxed{\frac{4i}{7}}$$

$$3. \frac{18 \cdot i}{9i \cdot i} = \frac{18i}{9i^2} = \frac{18i}{-9} = \boxed{-2i}$$

$$4. \frac{(8+4i) \cdot i}{2i \cdot i} = \frac{8i+4i^2}{2i^2} = \frac{-4+8i}{-2} = \boxed{2-4i}$$

$$5. \frac{(-35-15i) \cdot i}{5i \cdot i} = \frac{-35i-15i^2}{5i^2} = \frac{15-35i}{-5} = \boxed{-3+7i}$$

$$6. \frac{(2+i) \cdot i}{3i \cdot i} = \frac{2i+i^2}{3i^2} = \frac{-1+2i}{-3} = \boxed{\frac{1-2i}{3}}$$

$$7. \frac{6i(1+i)}{(1-i)(1+i)} = \frac{6i+6i^2}{1+i-i-i^2} = \frac{-6+6i}{2} = \boxed{-3+3i}$$

$$8. \frac{10(6-2i)}{(6+2i)(6-2i)} = \frac{60-20i}{36-12i+12i-4i^2} = \frac{60-20i}{40} = \boxed{\frac{3-i}{2}}$$

$$9. \frac{2i(-9-7i)}{(-9+7i)(-9-7i)} = \frac{-18i-14i^2}{81+63i-63i-49i^2} = \frac{14-18i}{130} = \boxed{\frac{7-9i}{65}}$$

$$10. \frac{(4+10i)(3-i)}{(3+i)(3-i)} = \frac{12-4i+30i-10i^2}{9-3i+3i-i^2} = \frac{22+26i}{10} = \boxed{\frac{11+13i}{5}}$$

11. $\frac{(4-i)(4+5i)}{(4-5i)(4+5i)} = \frac{16+20i-4i-5i^2}{16+20i-20i-25i^2}$ $= \frac{21+16i}{41}$	12. $\frac{(5-8i)(-1+4i)}{(-1-4i)(-1+4i)} = \frac{-5+20i+8i-32i^2}{1-4i+4i-16i^2}$ $= \frac{27+28i}{17}$
--	--

Simplify, then name all sets to which each value belongs.

13. i^{67}
 $(i^2)^{33} \cdot i = (-1)^{33} \cdot i$
 $= -i ; i, C$

14. $(-8i^9)^2$
 $64i^{18} = 64(i^2)^9$
 $= 64(-1)^9$
 $= -64 ; Z, Q, R, C$

15. $(11-7i)-(-4-2i)$
 $15-5i ; C$

16. $\sqrt{-63}$
 $\sqrt{-1 \cdot 63} = i\sqrt{9} \sqrt{7}$
 $= 3i\sqrt{7} ; i, C$

17. $(-9+4i)(-9-4i)$
 $= 81+36i-36i-16i^2$
 $= 81+16 = 97 ; N, W, Z, Q, R, C$

18. $\sqrt{\frac{49}{25}} = \frac{7}{5} ; Q, R, C$

For questions 19-20, write an example of each property using the complex numbers.

19. Commutative Property of Multiplication
 $3 \cdot (4+2i) = (4+2i) \cdot 3$

20. Distributive Property
 $7(5+9i) = 35+63i$

21. Name the additive inverse of $5+i$.
 $(-5-i)$

22. Name the multiplicative identity of $-12i$.
 $\frac{1}{-12i}$

For questions 23-24, answer true/false. If false, give a counterexample.

23. The set of complex numbers is closed under subtraction.
True

24. The set of irrational numbers is closed under multiplication.
False ; $\sqrt{6} \cdot \sqrt{6} = \sqrt{36} = 6$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 4: Solving Quadratic Equations

Quiz 4-2: Complex Numbers, Equations with Complex Solutions

Simplify each expression.

$$1. \sqrt{-64}$$

$$\sqrt{-1 \cdot 64}$$

$$8i$$

$$2. 2\sqrt{-108}$$

$$2\sqrt{-1 \cdot 108}$$

$$2i\sqrt{36} \sqrt{3}$$

$$2i \cdot 6 \sqrt{3} = 12i\sqrt{3}$$

$$3. (-5i)(-3i)$$

$$15i^2$$

$$15(-1) = -15$$

$$4. (i\sqrt{7})^2 \cdot 2i$$

$$7i^2 \cdot 2i$$

$$14i^3 = 14(-i)$$

$$5. (8+3i)+(-6-12i)$$

$$6. (5-9i)-(1+4i)$$

$$7. (9+5i)(4-2i)$$

$$36 - 18i + 20i - 10i^2$$

$$36 + 2i + 10$$

$$46 + 2i$$

$$8. (-1-3i)^2$$

$$(-1-3i)(-1-3i)$$

$$1 + 3i + 3i + 9i^2$$

$$1 + 6i - 9 = -8 + 6i$$

$$9. \frac{(20+8i) \cdot i}{4i \cdot i}$$

$$\frac{20i + 8i^2}{4i^2}$$

$$\frac{-8 + 20i}{-4} = 2 - 5i$$

$$10. \frac{(-4-4i)(10+2i)}{(10-2i)(10+2i)}$$

$$\frac{-40 - 8i - 40i - 8i^2}{100 + 20i - 20i - 4i^2} = \frac{-40 - 48i + 8}{104} = \frac{-32 - 48i}{104}$$

$$= \frac{-4 - 6i}{13}$$

Simplify, then name all sets to which each value belongs.

$$11. i^{49}$$

$$(i^2)^{24} \cdot i$$

$$(-1)^{24} \cdot i$$

$$= i$$

$$12. (6+i)(6-i)$$

$$36 - 6i + 6i - i^2$$

$$36 + 1$$

$$37$$

$$1. \underline{8i}$$

$$2. \underline{12i\sqrt{3}}$$

$$3. \underline{-15}$$

$$4. \underline{-14i}$$

$$5. \underline{2 - 9i}$$

$$6. \underline{4 - 13i}$$

$$7. \underline{46 + 2i}$$

$$8. \underline{-8 + 6i}$$

$$9. \underline{2 - 5i}$$

$$10. \underline{\frac{-4 - 6i}{13}}$$

$$11. \underline{i}$$

$$\text{Sets: } \underline{i, C}$$

$$12. \underline{37}$$

$$\text{Sets: } \underline{N, W, Z, Q, R, C}$$

Solve using the square root property. Give all solutions in simplest form.

13. $x^2 + 100 = 0$

$$x^2 = -100$$

$$x = \pm \sqrt{-100}$$

$$x = \pm \sqrt{-1 \cdot 100}$$

$$x = \pm 10i$$

14. $2x^2 + 27 = -13$

$$2x^2 = -40$$

$$x^2 = -20$$

$$x = \pm \sqrt{-20}$$

$$x = \pm \sqrt{-1 \cdot 20}$$

$$x = \pm i\sqrt{4} \sqrt{5}$$

$$x = \pm 2i\sqrt{5}$$

15. $11 - 4x^2 = 187$

$$-4x^2 = 176$$

$$x^2 = -44$$

$$x = \pm \sqrt{-44}$$

$$x = \pm \sqrt{-1 \cdot 44}$$

$$x = \pm i\sqrt{4} \sqrt{11}$$

$$x = \pm 2i\sqrt{11}$$

16. $\frac{4}{3}x^2 - 6 = -70$

$$\frac{4}{3}x^2 = -64$$

$$x^2 = -48$$

$$x = \pm \sqrt{-48}$$

$$x = \pm \sqrt{-1 \cdot 48}$$

$$x = \pm i\sqrt{16} \sqrt{3}$$

$$x = \pm 4i\sqrt{3}$$

17. $(x-5)^2 = -16$

$$x-5 = \pm \sqrt{-16}$$

$$x-5 = \pm \sqrt{-1 \cdot 16}$$

$$x-5 = \pm 4i$$

$$x = 5 \pm 4i$$

18. $-(x+2)^2 = 54$

$$(x+2)^2 = -54$$

$$x+2 = \pm \sqrt{-54}$$

$$x+2 = \pm \sqrt{-1 \cdot 54}$$

$$x+2 = \pm i\sqrt{9} \sqrt{6}$$

$$x+2 = \pm 3i\sqrt{6}$$

$$x = -2 \pm 3i\sqrt{6}$$

19. $3(x+1)^2 = -240$

$$(x+1)^2 = -80$$

$$x+1 = \pm \sqrt{-80}$$

$$x+1 = \pm \sqrt{-1 \cdot 80}$$

$$x+1 = \pm i\sqrt{16} \sqrt{5}$$

$$x+1 = \pm 4i\sqrt{5}$$

$$x = -1 \pm 4i\sqrt{5}$$

20. $-\frac{1}{2}(x-7)^2 = 96$

$$(x-7)^2 = -192$$

$$x-7 = \pm \sqrt{-192}$$

$$x-7 = \pm \sqrt{-1 \cdot 192}$$

$$x-7 = \pm i\sqrt{64} \sqrt{3}$$

$$x-7 = \pm 8i\sqrt{3}$$

$$x = 7 \pm 8i\sqrt{3}$$

13. $x = \{\pm 10i\}$

14. $x = \{\pm 2i\sqrt{5}\}$

15. $x = \{\pm 2i\sqrt{11}\}$

16. $x = \{\pm 4i\sqrt{3}\}$

17. $x = \{5 \pm 4i\}$

18. $x = \{-2 \pm 3i\sqrt{6}\}$

19. $x = \{-1 \pm 4i\sqrt{5}\}$

20. $x = \{7 \pm 8i\sqrt{3}\}$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																					
REVIEW: Perfect Square Trinomials	Factor the following trinomials: $x^2 + 4x + 4$ $(x+2)(x+2) = (x+2)^2$ $x^2 - 14x + 49$ $(x-7)(x-7) = (x-7)^2$ $x^2 - 20x + 100$ $(x-10)(x-10) = (x-10)^2$ $x^2 + 16x + 64$ $(x+8)(x+8) = (x+8)^2$ These are called perfect square trinomials. If you have a perfect square trinomial, you can solve the quadratic equation by square roots! Solve the equations below by factoring and using the square root property. 1. $x^2 + 14x + 49 = 4$ $(x+7)(x+7) = 4$ $(x+7)^2 = 4$ $x+7 = \pm \sqrt{4}$ $x = -7 \pm 2$ $x = -7+2$ $x = -5$ $x = -7-2$ $x = -9$ $x = \{-9, -5\}$ 2. $x^2 - 6x + 9 = 20$ $(x-3)(x-3) = 20$ $(x-3)^2 = 20$ $x-3 = \pm \sqrt{20}$ $x-3 = \pm \sqrt{4} \sqrt{5}$ $x-3 = \pm 2\sqrt{5}$ $x = \{3 \pm 2\sqrt{5}\}$																					
COMPLETING THE SQUARE Method	We can take any quadratic equation, create a perfect square trinomial, and solve it in a similar way. This method is called completing the square. <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th style="width: 10%;"></th> <th style="width: 45%; text-align: center;">Steps</th> <th style="width: 45%; text-align: center;">Example</th> </tr> </thead> <tbody> <tr> <td style="text-align: center; font-weight: bold;">1</td> <td>Rewrite as $ax^2 + bx = c$.</td> <td style="text-align: center;"> $x^2 + 8x - 34 = 0$ $x^2 + 8x = 34$ $\downarrow$ </td> </tr> <tr> <td style="text-align: center; font-weight: bold;">2</td> <td>Divide both sides by "a" to create $x^2 + bx = c$.</td> <td style="text-align: center;"> $\frac{8}{2} = 4$; $4^2 = 16$ </td> </tr> <tr> <td style="text-align: center; font-weight: bold;">3</td> <td>Take half of b, square it, then add this to both sides of the equation.</td> <td style="text-align: center;"> $x^2 + 8x + 16 = 34 + 16$ $(x+4)^2 = 50$ </td> </tr> <tr> <td style="text-align: center; font-weight: bold;">4</td> <td>Factor the perfect square trinomial.</td> <td style="text-align: center;"> $x+4 = \pm \sqrt{50}$ </td> </tr> <tr> <td style="text-align: center; font-weight: bold;">5</td> <td>Take the square root of both sides and solve for both cases.</td> <td style="text-align: center;"> $x+4 = \pm \sqrt{25} \sqrt{2}$ </td> </tr> <tr> <td style="text-align: center; font-weight: bold;">6</td> <td>Simplify all irrational or complex solutions.</td> <td style="text-align: center;"> $x = \{-4 \pm 5\sqrt{2}\}$ </td> </tr> </tbody> </table>		Steps	Example	1	Rewrite as $ax^2 + bx = c$.	$x^2 + 8x - 34 = 0$ $x^2 + 8x = 34$ $\downarrow$	2	Divide both sides by "a" to create $x^2 + bx = c$.	$\frac{8}{2} = 4$; $4^2 = 16$	3	Take half of b, square it, then add this to both sides of the equation.	$x^2 + 8x + 16 = 34 + 16$ $(x+4)^2 = 50$	4	Factor the perfect square trinomial.	$x+4 = \pm \sqrt{50}$	5	Take the square root of both sides and solve for both cases.	$x+4 = \pm \sqrt{25} \sqrt{2}$	6	Simplify all irrational or complex solutions.	$x = \{-4 \pm 5\sqrt{2}\}$
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6	Simplify all irrational or complex solutions.	$x = \{-4 \pm 5\sqrt{2}\}$																				

RATIONAL

Solutions

Solve by completing the square. Give all solutions in simplest form.

3. $x^2 - 18x + 56 = 0$

$$x^2 - 18x = -56$$

$$\downarrow$$

$$-\frac{18}{2} = -9 ; (-9)^2 = 81$$

$$x^2 - 18x + 81 = -56 + 81$$

$$(x-9)^2 = 25$$

$$x-9 = \pm \sqrt{25}$$

$$x-9 = \pm 5$$

$$x = 9 \pm 5$$

$$x = \{4, 14\}$$

4. $\frac{2x^2}{2} - \frac{16x}{2} = \frac{-30}{2}$

$$x^2 - 8x = -15$$

$$\downarrow$$

$$-\frac{8}{2} = -4 ; (-4)^2 = 16$$

$$x^2 - 8x + 16 = -15 + 16$$

$$(x-4)^2 = 1$$

$$x-4 = \pm \sqrt{1}$$

$$x-4 = \pm 1$$

$$x = 4 \pm 1$$

$$x = \{3, 5\}$$

5. $\frac{4x^2}{4} - \frac{8x}{4} = \frac{-3}{4}$

$$x^2 - 2x = -\frac{3}{4}$$

$$\downarrow$$

$$-\frac{2}{2} = -1 ; (-1)^2 = 1$$

$$x^2 - 2x + 1 = -\frac{3}{4} + 1$$

$$(x-1)^2 = \frac{1}{4}$$

$$x-1 = \pm \sqrt{\frac{1}{4}}$$

$$x-1 = \pm \frac{1}{2}$$

$$x = 1 \pm \frac{1}{2}$$

$$x = \{\frac{1}{2}, \frac{3}{2}\}$$

6. $3x^2 + 10x + 8 = 0$

$$\frac{3x^2}{3} + \frac{10x}{3} = \frac{-8}{3}$$

$$x^2 + \frac{10}{3}x = \frac{-8}{3}$$

$$\downarrow$$

$$\frac{10}{3} \cdot \frac{1}{2} = \frac{10}{6} = \frac{5}{3} ; (\frac{5}{3})^2 = \frac{25}{9}$$

$$x^2 + \frac{10}{3}x + \frac{25}{9} = \frac{-8}{3} + \frac{25}{9}$$

$$(x + \frac{5}{3})^2 = \frac{1}{9}$$

$$x + \frac{5}{3} = \pm \sqrt{\frac{1}{9}}$$

$$x = -\frac{5}{3} \pm \frac{1}{3}$$

$$x = -\frac{4}{3}, -\frac{6}{3}$$

$$x = \{-2, \frac{4}{3}\}$$

IRRATIONAL

Solutions

7. $x^2 + 16x - 21 = -5$

$$x^2 + 16x = 16$$

$$\downarrow$$

$$\frac{16}{2} = 8 ; 8^2 = 64$$

$$x^2 + 16x + 64 = 16 + 64$$

$$(x+8)^2 = 80$$

$$x+8 = \pm \sqrt{80}$$

$$x+8 = \pm \sqrt{16} \sqrt{5}$$

$$x+8 = \pm 4\sqrt{5}$$

$$x = \{-8 \pm 4\sqrt{5}\}$$

8. $\frac{3x^2}{3} - \frac{30x}{3} = \frac{69}{3}$

$$x^2 - 10x = 23$$

$$\downarrow$$

$$-\frac{10}{2} = -5 ; (-5)^2 = 25$$

$$x^2 - 10x + 25 = 23 + 25$$

$$(x-5)^2 = 48$$

$$x-5 = \pm \sqrt{48}$$

$$x-5 = \pm \sqrt{16} \sqrt{3}$$

$$x-5 = \pm 4\sqrt{3}$$

$$x = \{5 \pm 4\sqrt{3}\}$$

COMPLEX
Solutions

$$\begin{aligned}
 9. \quad x^2 - 6x - 31 &= 0 \\
 x^2 - 6x &= 31 \\
 \downarrow \\
 -\frac{6}{2} &= -3 ; (-3)^2 = 9 \\
 x^2 - 6x + 9 &= 31 + 9 \\
 (x-3)^2 &= 40 \\
 x-3 &= \pm \sqrt{40} \\
 x-3 &= \pm \sqrt{4} \sqrt{10} \\
 x-3 &= \pm 2\sqrt{10} \\
 \boxed{x = \{3 \pm 2\sqrt{10}\}}
 \end{aligned}$$

$$\begin{aligned}
 10. \quad 2x^2 + 4x - 57 &= 87 \\
 \frac{2x^2 + 4x}{2} &= \frac{144}{2} \\
 x^2 + 2x &= 72 \\
 \downarrow \\
 \frac{2}{2} &= 1 ; 1^2 = 1 \\
 x^2 + 2x + 1 &= 72 + 1 \\
 (x+1)^2 &= 73 \\
 x+1 &= \pm \sqrt{73} \\
 \boxed{x = \{-1 \pm \sqrt{73}\}}
 \end{aligned}$$

$$\begin{aligned}
 11. \quad x^2 + 12x + 43 &= 0 \\
 x^2 + 12x &= -43 \\
 \downarrow \\
 \frac{12}{2} &= 6 ; 6^2 = 36 \\
 x^2 + 12x + 36 &= -43 + 36 \\
 (x+6)^2 &= -7 \\
 x+6 &= \pm \sqrt{-7} \\
 x+6 &= \pm i\sqrt{7} \\
 \boxed{x = \{-6 \pm i\sqrt{7}\}}
 \end{aligned}$$

$$\begin{aligned}
 12. \quad x^2 - 18x + 97 &= 0 \\
 x^2 - 18x &= -97 \\
 \downarrow \\
 -\frac{18}{2} &= -9 ; (-9)^2 = 81 \\
 x^2 - 18x + 81 &= -97 + 81 \\
 (x-9)^2 &= -16 \\
 x-9 &= \pm \sqrt{-16} \\
 x-9 &= \pm i\sqrt{16} \\
 x-9 &= \pm 4i \\
 \boxed{x = \{9 \pm 4i\}}
 \end{aligned}$$

$$\begin{aligned}
 13. \quad 4x^2 + 76 &= 16x \\
 \frac{4x^2 - 16x}{4} &= \frac{-76}{4} \\
 x^2 - 4x &= -19 \\
 \downarrow \\
 -\frac{4}{2} &= -2 ; (-2)^2 = 4 \\
 x^2 - 4x + 4 &= -19 + 4 \\
 (x-2)^2 &= -15 \\
 x-2 &= \pm \sqrt{-15} \\
 x-2 &= \pm i\sqrt{15} \\
 \boxed{x = \{2 \pm i\sqrt{15}\}}
 \end{aligned}$$

$$\begin{aligned}
 14. \quad 3x^2 - 6x + 80 &= -4 \\
 \frac{3x^2 - 6x}{3} &= \frac{-84}{3} \\
 x^2 - 2x &= -28 \\
 \downarrow \\
 -\frac{2}{2} &= -1 ; (-1)^2 = 1 \\
 x^2 - 2x + 1 &= -28 + 1 \\
 (x-1)^2 &= -27 \\
 x-1 &= \pm \sqrt{-27} \\
 x-1 &= \pm i\sqrt{9} \sqrt{3} \\
 x-1 &= \pm 3i\sqrt{3} \\
 \boxed{x = \{1 \pm 3i\sqrt{3}\}}
 \end{aligned}$$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 7: Solving Quadratics by
Completing the Square**** This is a 2-page document! ******Directions:** Solve each equation by completing the square. Give all solutions in simplest form.

1. $x^2 + 4x + 22 = 67$

$$x^2 + 4x = 45$$

$$\downarrow$$

 $2^2 = 4$

$$x^2 + 4x + 4 = 45 + 4$$

$$(x+2)^2 = 49$$

$$x+2 = \pm\sqrt{49}$$

$$x+2 = \pm 7$$

$$x = -2 \pm 7$$

$$x = \{-9, 5\}$$

2. $x^2 = 16x - 63$

$$x^2 - 16x = -63$$

$$\downarrow$$

 $(-8)^2 = 64$

$$x^2 - 16x + 64 = -63 + 64$$

$$(x-8)^2 = 1$$

$$x-8 = \pm\sqrt{1}$$

$$x-8 = \pm 1$$

$$x = 8 \pm 1$$

$$x = \{7, 9\}$$

3. $2x^2 - x - 3 = 0$

$$x^2 - \frac{1}{2}x = \frac{3}{2}$$

$$\downarrow$$

 $(-\frac{1}{4})^2 = \frac{1}{16}$

$$x^2 - \frac{1}{2}x + \frac{1}{16} = \frac{3}{2} + \frac{1}{16}$$

$$(x - \frac{1}{4})^2 = \frac{25}{16}$$

$$x - \frac{1}{4} = \pm\sqrt{\frac{25}{16}}$$

$$x - \frac{1}{4} = \pm\frac{5}{4}$$

$$x = \frac{1}{4} \pm \frac{5}{4}$$

$$x = \{-1, \frac{3}{2}\}$$

4. $4x^2 + 24x = -35$

$$x^2 + 6x = -\frac{35}{4}$$

$$\downarrow$$

 $3^2 = 9$

$$x^2 + 6x + 9 = -\frac{35}{4} + 9$$

$$(x+3)^2 = \frac{1}{4}$$

$$x+3 = \pm\sqrt{\frac{1}{4}}$$

$$x+3 = \pm\frac{1}{2}$$

$$x = -3 \pm \frac{1}{2}$$

$$x = \{-\frac{7}{2}, -\frac{5}{2}\}$$

5. $x^2 - 12x - 44 = 0$

$$x^2 - 12x = 44$$

$$\downarrow$$

 $(-6)^2 = 36$

$$x^2 - 12x + 36 = 44 + 36$$

$$(x-6)^2 = 80$$

$$x-6 = \pm\sqrt{80}$$

$$x-6 = \pm\sqrt{16}\sqrt{5}$$

$$x = \{6 \pm 4\sqrt{5}\}$$

6. $2x^2 - 16x + 8 = 0$

$$x^2 - 8x = -4$$

$$\downarrow$$

 $(-4)^2 = 16$

$$x^2 - 8x + 16 = -4 + 16$$

$$(x-4)^2 = 12$$

$$x-4 = \pm\sqrt{12}$$

$$x-4 = \pm\sqrt{4}\sqrt{3}$$

$$x = \{4 \pm 2\sqrt{3}\}$$

7. $x^2 + 10x + 1 = 24$

$$x^2 + 10x = 23$$

$$\downarrow$$

$$5^2 = 25$$

$$x^2 + 10x + 25 = 23 + 25$$

$$(x+5)^2 = 48$$

$$x+5 = \pm\sqrt{48}$$

$$x+5 = \pm\sqrt{16}\sqrt{3}$$

$$x+5 = \pm 4\sqrt{3}$$

$$x = \{-5 \pm 4\sqrt{3}\}$$

8. $3x^2 = 3 - 42x$

$$3x^2 + 42x = 3$$

$$x^2 + 14x = 1$$

$$\downarrow$$

$$7^2 = 49$$

$$x^2 + 14x + 49 = 1 + 49$$

$$(x+7)^2 = 50$$

$$x+7 = \pm\sqrt{50}$$

$$x+7 = \pm\sqrt{25}\sqrt{2}$$

$$x+7 = \pm 5\sqrt{2}$$

$$x = \{-7 \pm 5\sqrt{2}\}$$

9. $x^2 + 2x + 37 = 0$

$$x^2 + 2x = -37$$

$$\downarrow$$

$$1^2 = 1$$

$$x^2 + 2x + 1 = -37 + 1$$

$$(x+1)^2 = -36$$

$$x+1 = \pm\sqrt{36}$$

$$x+1 = \pm\sqrt{-1 \cdot 36}$$

$$x+1 = \pm 6i$$

$$x = \{-1 \pm 6i\}$$

10. $5x^2 - 30x + 85 = 0$

$$x^2 - 6x = -17$$

$$\downarrow$$

$$(-3)^2 = 9$$

$$x^2 - 6x + 9 = -17 + 9$$

$$(x-3)^2 = -8$$

$$x-3 = \pm\sqrt{-8}$$

$$x-3 = \pm\sqrt{-1 \cdot 8}$$

$$x-3 = \pm i\sqrt{4}\sqrt{2}$$

$$x-3 = \pm 2i\sqrt{2}$$

$$x = \{3 \pm 2i\sqrt{2}\}$$

11. $x^2 - 14x + 62 = -77$

$$x^2 - 14x = -139$$

$$\downarrow$$

$$(-7)^2 = 49$$

$$x^2 - 14x + 49 = -139 + 49$$

$$(x-7)^2 = -90$$

$$x-7 = \pm\sqrt{-90}$$

$$x-7 = \pm\sqrt{-1 \cdot 90}$$

$$x-7 = \pm i\sqrt{9}\sqrt{10}$$

$$x-7 = \pm 3i\sqrt{10}$$

$$x = \{7 \pm 3i\sqrt{10}\}$$

12. $2x^2 - 20x + 89 = -87$

$$2x^2 - 20x = -176$$

$$x^2 - 10x = -88$$

$$\downarrow$$

$$(-5)^2 = 25$$

$$x^2 - 10x + 25 = -88 + 25$$

$$(x-5)^2 = -63$$

$$x-5 = \pm\sqrt{-63}$$

$$x-5 = \pm\sqrt{-1 \cdot 63}$$

$$x-5 = \pm i\sqrt{9}\sqrt{7}$$

$$x = \{5 \pm 3i\sqrt{7}\}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
THE QUADRATIC FORMULA	① Write the equation in STANDARD FORM, $ax^2 + bx + c = 0$ ② Identify a, b, and c. SUBSTITUTE them into the formula. ③ SIMPLIFY! $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Directions: Solve each quadratic equation below using the quadratic formula.	
1. $x^2 + 2x - 63 = 0$ $a=1$ $b=2$ $c=-63$ $x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-63)}}{2(1)}$ $x = \frac{-2 \pm \sqrt{4+252}}{2}$ $x = \frac{-2 \pm \sqrt{256}}{2}$ $x = \frac{-2 \pm 16}{2}$ $x = \frac{14}{2}, \frac{-18}{2}$ $x = \{-9, 7\}$	2. $6x^2 - 11x - 10 = 0$ $a=6$ $b=-11$ $c=-10$ $x = \frac{11 \pm \sqrt{(-11)^2 - 4(6)(-10)}}{2(6)}$ $x = \frac{11 \pm \sqrt{121+240}}{12}$ $x = \frac{11 \pm \sqrt{361}}{12}$ $x = \frac{11 \pm 19}{12}$ $x = \frac{30}{12}, \frac{-8}{12}$ $x = \{-\frac{2}{3}, \frac{5}{2}\}$
3. $-x^2 + 2x = 1$ $-x^2 + 2x - 1 = 0$ $a=-1$ $b=2$ $c=-1$ $x = \frac{-2 \pm \sqrt{2^2 - 4(-1)(-1)}}{2(-1)}$ $x = \frac{-2 \pm \sqrt{4-4}}{-2}$ $x = \frac{-2 \pm \sqrt{0}}{-2}$ $x = \frac{-2}{-2}$ $x = \{1\}$	4. $-2x^2 + 3x = 4x - 15$ $-2x^2 - x + 15 = 0$ $a=-2$ $b=-1$ $c=15$ $x = \frac{1 \pm \sqrt{(-1)^2 - 4(-2)(15)}}{2(-2)}$ $x = \frac{1 \pm \sqrt{1+120}}{-4}$ $x = \frac{1 \pm \sqrt{121}}{-4}$ $x = \frac{1 \pm 11}{-4}$ $x = \{-3, \frac{5}{2}\}$
5. $9x^2 - 1 = 24$ $9x^2 - 25 = 0$ $a=9$ $b=0$ $c=-25$ $x = \frac{0 \pm \sqrt{0^2 - 4(9)(-25)}}{2(9)}$ $x = \frac{0 \pm \sqrt{900}}{18}$ $x = \pm \frac{30}{18}$ $x = \{\pm \frac{5}{3}\}$	6. $8x^2 = 10x$ $8x^2 - 10x = 0$ $a=8$ $b=-10$ $c=0$ $x = \frac{10 \pm \sqrt{(-10)^2 - 4(8)(0)}}{2(8)}$ $x = \frac{10 \pm \sqrt{100-0}}{16}$ $x = \frac{10 \pm 10}{16}$ $x = \frac{20}{16}, \frac{0}{16}$ $x = \{\frac{5}{4}, 0\}$

7. $x^2 - 9 = 4x$

$$x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-9)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{16 + 36}}{2}$$

$$x = \frac{4 \pm \sqrt{52}}{2} \leftarrow \sqrt{4} \cdot \sqrt{13}$$

$$x = \frac{4 \pm 2\sqrt{13}}{2}$$

$$x = \{2 \pm \sqrt{13}\}$$

8. $-2x^2 + 17 = 8x - 1$

$$-2x^2 - 8x + 18 = 0$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - 4(-2)(18)}}{2(-2)}$$

$$x = \frac{8 \pm \sqrt{64 + 144}}{-4}$$

$$x = \frac{8 \pm \sqrt{208}}{-4} \leftarrow \sqrt{16} \cdot \sqrt{13}$$

$$x = \frac{8 \pm 4\sqrt{13}}{-4}$$

$$x = \{-2 \pm \sqrt{13}\}$$

9. $-8x^2 + 7 = 4$

$$-8x^2 + 3 = 0$$

$$x = \frac{0 \pm \sqrt{0^2 - 4(-8)(3)}}{2(-8)}$$

$$x = \frac{0 \pm \sqrt{0 + 96}}{-16}$$

$$x = \frac{0 \pm \sqrt{96}}{-16} \leftarrow \sqrt{16} \cdot \sqrt{6}$$

$$x = \frac{0 \pm 4\sqrt{6}}{-16}$$

$$x = \left\{ \pm \frac{\sqrt{6}}{4} \right\}$$

10. $-x^2 + 10x = 8$

$$-x^2 + 10x - 8 = 0$$

$$x = \frac{-10 \pm \sqrt{10^2 - 4(-1)(-8)}}{2(-1)}$$

$$x = \frac{-10 \pm \sqrt{100 - 32}}{-2}$$

$$x = \frac{-10 \pm \sqrt{68}}{-2} \leftarrow \sqrt{4} \cdot \sqrt{17}$$

$$x = \frac{-10 \pm 2\sqrt{17}}{-2}$$

$$x = \{5 \pm \sqrt{17}\}$$

11. $-3x^2 + 8x - 7 = 0$

$$x = \frac{-8 \pm \sqrt{8^2 - 4(-3)(-7)}}{2(-3)}$$

$$x = \frac{-8 \pm \sqrt{64 - 84}}{-6}$$

$$x = \frac{-8 \pm \sqrt{-20}}{-6} \leftarrow i\sqrt{4} \sqrt{5}$$

$$x = \frac{-8 \pm 2i\sqrt{5}}{-6}$$

$$x = \left\{ \frac{4 \pm i\sqrt{5}}{3} \right\}$$

12. $5x^2 + 9 = 4x + 5$

$$5x^2 - 4x + 4 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(5)(4)}}{2(5)}$$

$$x = \frac{4 \pm \sqrt{16 - 80}}{10}$$

$$x = \frac{4 \pm \sqrt{-64}}{10}$$

$$x = \frac{4 \pm 8i}{10}$$

$$x = \left\{ \frac{2 \pm 4i}{5} \right\}$$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 8: The Quadratic Formula

**** This is a 2-page document! ******Directions:** Solve each equation by the quadratic formula.

1. $x^2 + 12x - 8 = 0$

$$X = \frac{-12 \pm \sqrt{12^2 - 4(1)(-8)}}{2(1)}$$

$$X = \frac{-12 \pm \sqrt{144 + 32}}{2}$$

$$X = \frac{-12 \pm \sqrt{176}}{2} \quad \leftarrow \sqrt{16} \sqrt{11}$$

$$X = \frac{-12 \pm 4\sqrt{11}}{2}$$

$$X = \{-6 \pm 2\sqrt{11}\}$$

2. $-2x^2 + 7x = 3$

$$-2x^2 + 7x - 3 = 0$$

$$X = \frac{-7 \pm \sqrt{(-7)^2 - 4(-2)(-3)}}{2(-2)}$$

$$X = \frac{-7 \pm \sqrt{49 - 24}}{-4}$$

$$X = \frac{-7 \pm \sqrt{25}}{-4}$$

$$X = \frac{-7 \pm 5}{-4}$$

$$X = \left\{\frac{1}{2}, 3\right\}$$

3. $-x^2 + 5x - 6 = 2$

$$-x^2 + 5x - 8 = 0$$

$$X = \frac{-5 \pm \sqrt{5^2 - 4(-1)(-8)}}{2(-1)}$$

$$X = \frac{-5 \pm \sqrt{25 - 32}}{-2}$$

$$X = \frac{-5 \pm \sqrt{-7}}{-2}$$

$$X = \frac{-5 \pm i\sqrt{7}}{-2}$$

$$X = \left\{\frac{5 \pm i\sqrt{7}}{2}\right\}$$

4. $13x^2 - 16x = x^2 - x$

$$12x^2 - 15x = 0$$

$$X = \frac{15 \pm \sqrt{(-15)^2 - 4(12)(0)}}{2(12)}$$

$$X = \frac{15 \pm \sqrt{225 - 0}}{24}$$

$$X = \frac{15 \pm \sqrt{225}}{24}$$

$$X = \frac{15 \pm 15}{24}$$

$$X = \left\{0, \frac{5}{4}\right\}$$

5. $8x^2 + 15x = 7x - 4$

$$8x^2 + 8x + 4 = 0$$

$$X = \frac{-8 \pm \sqrt{8^2 - 4(8)(4)}}{2(8)}$$

$$X = \frac{-8 \pm \sqrt{64 - 128}}{16}$$

$$X = \frac{-8 \pm \sqrt{-64}}{16}$$

$$X = \frac{-8 \pm 8i}{16}$$

$$X = \left\{-\frac{1 \pm i}{2}\right\}$$

6. $-x^2 + 18 = 93$

$$-x^2 - 75 = 0$$

$$X = \frac{0 \pm \sqrt{0^2 - 4(-1)(-75)}}{2(-1)}$$

$$X = \frac{0 \pm \sqrt{0 - 300}}{-2}$$

$$X = \frac{0 \pm \sqrt{-300}}{-2} \quad \leftarrow i\sqrt{100} \sqrt{3}$$

$$X = \frac{\pm 10i\sqrt{3}}{-2}$$

$$X = \{\pm 5i\sqrt{3}\}$$

7. $3x^2 + 5x = 8$

$$3x^2 + 5x - 8 = 0$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-8)}}{2(3)}$$

$$x = \frac{-5 \pm \sqrt{25 + 96}}{6}$$

$$x = \frac{-5 \pm \sqrt{121}}{6}$$

$$x = \frac{-5 \pm 11}{6}$$

$$x = \left\{ -\frac{8}{3}, 1 \right\}$$

8. $5x^2 + 1 = 4x - 7$

$$5x^2 - 4x + 8 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(5)(8)}}{2(5)}$$

$$x = \frac{4 \pm \sqrt{16 - 160}}{10}$$

$$x = \frac{4 \pm \sqrt{-144}}{10}$$

$$x = \frac{4 \pm 12i}{10}$$

$$x = \left\{ \frac{2 \pm 6i}{5} \right\}$$

9. $7x^2 - 10x + 9 = 5 - 2x$

$$7x^2 - 8x + 4 = 0$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - 4(7)(4)}}{2(7)}$$

$$x = \frac{8 \pm \sqrt{64 - 112}}{14}$$

$$x = \frac{8 \pm \sqrt{-48}}{14} \leftarrow i\sqrt{16}\sqrt{3}$$

$$x = \frac{8 \pm 4i\sqrt{3}}{14}$$

$$x = \left\{ \frac{4 \pm 2i\sqrt{3}}{7} \right\}$$

10. $-12x^2 + 21 = 0$

$$x = \frac{0 \pm \sqrt{0^2 - 4(-12)(21)}}{2(-12)}$$

$$x = \frac{0 \pm \sqrt{0 + 1008}}{-24}$$

$$x = \frac{0 \pm \sqrt{1008}}{-24} \leftarrow \sqrt{144}\sqrt{7}$$

$$x = \frac{\pm 12\sqrt{7}}{-24}$$

$$x = \left\{ \pm \frac{\sqrt{7}}{2} \right\}$$

11. $-2x^2 + 24 = 8x$

$$-2x^2 - 8x + 24 = 0$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - 4(-2)(24)}}{2(-2)}$$

$$x = \frac{8 \pm \sqrt{64 + 192}}{-4}$$

$$x = \frac{8 \pm \sqrt{256}}{-4}$$

$$x = \frac{8 \pm 16}{-4}$$

$$x = \{-6, 2\}$$

12. $3x^2 + x = -2x - 7$

$$3x^2 + 3x + 7 = 0$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(3)(7)}}{2(3)}$$

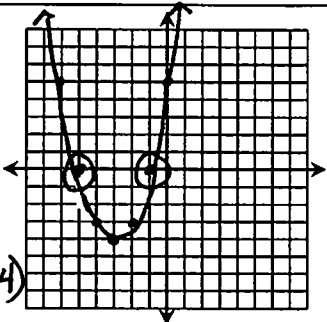
$$x = \frac{-3 \pm \sqrt{9 - 84}}{6}$$

$$x = \frac{-3 \pm \sqrt{-75}}{6} \leftarrow i\sqrt{25}\sqrt{3}$$

$$x = \left\{ \frac{-3 \pm 5i\sqrt{3}}{6} \right\}$$

METHODS FOR SOLVING QUADRATIC EQUATIONS

Directions: Solve the equation below using each method.
If a method can't be used, explain why.

$x^2 + 6x + 5 = 0$					
GRAPHING $x = \{-5, -1\}$	axis: $x = \frac{-b}{2(1)} ; x = -3$ Vertex: $(-3)^2 + 6(-3) + 5$ $9 - 18 + 5$ $-9 + 5 = -4 \rightarrow (-3, -4)$ 				
FACTORING $x = \{-5, -1\}$	$x^2 + 6x + 5 = 0$ $(x+5)(x+1) = 0$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>$x+5=0$</td><td>$x+1=0$</td></tr> <tr> <td>$x=-5$</td><td>$x=-1$</td></tr> </table>	$x+5=0$	$x+1=0$	$x=-5$	$x=-1$
$x+5=0$	$x+1=0$				
$x=-5$	$x=-1$				
SQUARE ROOTS	Cannot be solved using square roots since there is an x-term.				
COMPLETING THE SQUARE	$x^2 + 6x = -5$ $\downarrow$ $3^2 = 9$ $x^2 + 6x + 9 = -5 + 9$ $(x+3)^2 = 4$ $x+3 = \pm\sqrt{4}$ $x = -3 \pm 2$ $x = \{-5, -1\}$				
QUADRATIC FORMULA	$x = \frac{-b \pm \sqrt{b^2 - 4(1)(5)}}{2(1)}$ $x = \frac{-6 \pm \sqrt{36 - 20}}{2}$ $x = \frac{-6 \pm \sqrt{16}}{2}$ $x = \frac{-6 \pm 4}{2}$ $x = \frac{-2}{2}, \frac{-10}{2}$ $x = \{-5, -1\}$				

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
THE DISCRIMINANT	The discriminant of a quadratic equation is a numerical value that reveals both the number and type of roots a quadratic equation has.		Formula: $b^2 - 4ac$
	Value of d	# of Roots	Type of Roots
	$d > 0$ (a perfect square)	2	rational
	$d > 0$ (NOT a perfect square)	2	irrational
	$d = 0$	1	rational
	$d < 0$	2	imaginary
EXAMPLES	Find the value of the discriminant, then determine the number and type of roots.		
	1. $2x^2 + 12x - 13 = 0$ $a=2$ $b=12$ $c=-13$ $12^2 - 4(2)(-13)$ $144 + 104$ $= 248$ 2, irrational	2. $-x^2 + 16x - 64 = 0$ $a=-1$ $b=16$ $c=-64$ $16^2 - 4(-1)(-64)$ $256 - 256$ $= 0$ 1, rational	
	3. $-4x^2 + 84 = 10x$ $-4x^2 - 10x + 84 = 0$ $a=-4$ $b=-10$ $c=84$ $(-10)^2 - 4(-4)(84)$ $100 + 1344$ $= 1444$ 2, rational	4. $3x^2 + 24 = 0$ $a=3$ $b=0$ $c=24$ $0^2 - 4(3)(24)$ $0 - 288$ $= -288$ 2, imaginary	
Choosing the BEST METHOD	Method	Can be used...	When?
	GRAPHING	Sometimes * Not usually practical *	roots are integers
	FACTORING	Sometimes	roots are rational
	SQUARE ROOTS	Sometimes	no b-term (x-term)
	COMPLETING THE SQUARE	always	no x^2 coefficient + even b-term (x-term)
	QUADRATIC FORMULA	always	anytime - use when other situations are inefficient

Choosing the Best Method to Solve a Quadratic Equation

Step 1: Find the discriminant of each equation and give the number and type of roots.

EQUATION A	EQUATION B	EQUATION C	EQUATION D
$x^2 - 9x + 21 = 0$	$-2x^2 + 144 = 0$	$3x^2 - 22x + 7 = 0$	$x^2 + 6x - 15 = 0$
$(-9)^2 - 4(1)(21)$	$0^2 - 4(-2)(144)$	$(-22)^2 - 4(3)(7)$	$6^2 - 4(1)(-15)$
$81 - 84$	$0 + 1152$	$484 - 84$	$36 + 60$
$= -3$	$= 1152$	$= 400$	$= 96$
2, imaginary	2, irrational	2, rational	2, irrational

Step 2: Solve each equation using the methods below. You can only use each method once.

FACTORING

$$3x^2 - 22x + 7 = 0$$

$$x^2 - 22x + 21 = 0$$

$$(x - \frac{21}{3})(x - \frac{1}{3}) = 0$$

$$\begin{array}{l|l} (x-7)(3x-1)=0 & \\ \hline x-7=0 & 3x-1=0 \\ x=7 & x=\frac{1}{3} \end{array}$$

$$x = \left\{ \frac{1}{3}, 7 \right\}$$

C

SQUARE ROOTS

$$-2x^2 + 144 = 0$$

$$-2x^2 = -144$$

$$x^2 = 72$$

$$x = \pm \sqrt{72}$$

$$x = \pm \sqrt{36} \sqrt{2}$$

$$x = \left\{ \pm 6\sqrt{2} \right\}$$

B

COMPLETE THE SQUARE

$$x^2 + 6x - 15 = 0$$

$$x^2 + 6x = 15$$

$$\downarrow$$

$$3^2 = 9$$

$$x^2 + 6x + 9 = 15 + 9$$

$$(x+3)^2 = 24$$

$$x+3 = \pm \sqrt{24} \leftarrow \sqrt{4} \sqrt{6}$$

$$x+3 = \pm 2\sqrt{6}$$

$$x = \left\{ -3 \pm 2\sqrt{6} \right\}$$

D

THE QUADRATIC FORMULA

$$x^2 - 9x + 21 = 0$$

$$x = \frac{9 \pm \sqrt{(-9)^2 - 4(1)(21)}}{2(1)}$$

$$x = \frac{9 \pm \sqrt{81 - 84}}{2}$$

$$x = \frac{9 \pm \sqrt{-3}}{2}$$

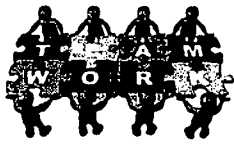
$$x = \left\{ \frac{9 \pm i\sqrt{3}}{2} \right\}$$

A

Be prepared to justify your choices!

Group Members: _____

Per: _____



REVIEW: Solving Quadratic Equations

Directions: Solve using the indicated method. Work together to complete each problem. Do not divide up the work! Each person should be participating.

FACTORING	1. $x^2 - 7 = 6x$ $x^2 - 6x - 7 = 0$ $(x-7)(x+1) = 0$ $x-7=0 \quad x+1=0$ $x=7 \quad x=-1$ $x = \{-1, 7\}$	2. $4x^2 + 35x + 24 = 0$ $x^2 + 35x + 96 = 0$ $(x+32)(x+3) = 0$ $\frac{4}{4} \quad \frac{4}{4}$ $(x+8)(4x+3) = 0$ $x+8=0 \quad 4x+3=0$ $x=-8 \quad x=-\frac{3}{4}$ $x = \{-8, -\frac{3}{4}\}$
SQUARE ROOTS	3. $10x^2 + 2 = 282$ $10x^2 = 280$ $x^2 = 28$ $x = \pm \sqrt{28}$ $x = \pm \sqrt{4} \sqrt{7}$ $x = \{\pm 2\sqrt{7}\}$	4. $2x^2 + 50 = -166$ $2x^2 = -216$ $x^2 = -108$ $x = \pm \sqrt{-108}$ $x = \pm i \sqrt{36} \sqrt{3}$ $x = \{\pm 6i\sqrt{3}\}$
COMPLETING THE SQUARE	5. $x^2 + 12x + 3 = 7$ $x^2 + 12x = 4$ $\downarrow$ $6^2 = 36$ $x^2 + 12x + 36 = 4 + 36$ $(x+6)^2 = 40$ $x+6 = \pm \sqrt{40}$ $x+6 = \pm \sqrt{4} \sqrt{10}$ $x+6 = \pm 2\sqrt{10}$ $x = \{-6 \pm 2\sqrt{10}\}$	6. $6x^2 - 12x - 38 = 10$ $\frac{6x^2}{6} - \frac{12x}{6} = \frac{48}{6}$ $x^2 - 2x = 8$ $\downarrow$ $(-1)^2 = 1$ $x^2 - 2x + 1 = 8 + 1$ $(x-1)^2 = 9$ $x-1 = \pm \sqrt{9}$ $x-1 = \pm 3$ $x = 1 \pm 3$ $x = \{-2, 4\}$

QUADRATIC FORMULA

7. $5x^2 + 4x + 1 = 0$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(5)(1)}}{2(5)}$$

$$x = \frac{-4 \pm \sqrt{16 - 20}}{10}$$

$$x = \frac{-4 \pm \sqrt{-4}}{10}$$

$$x = \frac{-4 \pm 2i}{10}$$

$$x = \left\{ \frac{-2 \pm i}{5} \right\}$$

8. $-2x^2 + 2x = 5$

$$-2x^2 + 2x - 5 = 0$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(-2)(-5)}}{2(-2)}$$

$$x = \frac{-2 \pm \sqrt{4 - 40}}{-4}$$

$$x = \frac{-2 \pm \sqrt{-36}}{-4}$$

$$x = \frac{-2 \pm 6i}{-4}$$

$$x = \left\{ \frac{1 \pm 3i}{2} \right\}$$

CHOOSE EACH METHOD ONCE!

9. $2x^2 = 3x + 10$

$$2x^2 - 3x - 10 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(2)(-10)}}{2(2)}$$

$$x = \frac{3 \pm \sqrt{9 + 80}}{4}$$

$$x = \left\{ \frac{3 \pm \sqrt{89}}{4} \right\}$$

- ☐ F
☐ SR
☐ CS
☒ QF

10. $x^2 + 10x = 2x$

$$x^2 + 8x = 0$$

$$x(x + 8) = 0$$

$$\begin{array}{l|l} x=0 & x+8=0 \\ & x=-8 \end{array}$$

$$x = \{-8, 0\}$$

- ☒ F
☐ SR
☐ CS
☐ QF

11. $4x^2 - 39 = 10$

$$\frac{4x^2}{4} = \frac{49}{4}$$

$$x^2 = \frac{49}{4}$$

$$x = \pm \sqrt{\frac{49}{4}}$$

$$x = \left\{ \pm \frac{7}{2} \right\}$$

- ☐ F
☒ SR
☐ CS
☐ QF

12. $-x^2 + 2x = 7$

$$\frac{-x^2 + 2x}{-1} = \frac{7}{-1}$$

$$x^2 - 2x = -7$$

$$\downarrow$$

$$(x-1)^2 = 1$$

$$x^2 - 2x + 1 = -7 + 1$$

$$(x-1)^2 = -6$$

$$x-1 = \pm \sqrt{-6}$$

$$x-1 = \pm i\sqrt{6}$$

$$x = \{1 \pm i\sqrt{6}\}$$

- ☐ F
☐ SR
☒ CS
☐ QF

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 9: Solving Quadratics Review

**** This is a 2-page document! ******Directions:** Solve each equation using the indicated method.

1. $x^2 = 6x + 91$

(Factoring)

$$x^2 - 6x - 91 = 0$$

$$(x+7)(x-13) = 0$$

$$x+7=0 \quad | \quad x-13=0$$

$$x=-7 \quad | \quad x=13$$

$$x = \{-7, 13\}$$

2. $-2x^2 + 146 = -34$

(Square Roots)

$$\frac{-2x^2}{-2} = \frac{-180}{-2}$$

$$x^2 = 90$$

$$x = \pm \sqrt{90}$$

$$x = \pm \sqrt{9} \sqrt{10}$$

$$x = \{\pm 3\sqrt{10}\}$$

3. $x^2 - 14x + 53 = 9$

(Completing the Square)

$$x^2 - 14x = -44$$

$$\downarrow$$

$$(-7)^2 = 49$$

$$x^2 - 14x + 49 = -44 + 49$$

$$(x-7)^2 = 5$$

$$x-7 = \pm \sqrt{5}$$

$$x = \{7 \pm \sqrt{5}\}$$

4. $3x^2 + 6x - 62 = 7$

(Completing the Square)

$$3x^2 + 6x = 69$$

$$x^2 + 2x = 23$$

$$\downarrow$$

$$1^2 = 1$$

$$x^2 + 2x + 1 = 23 + 1$$

$$(x+1)^2 = 24$$

$$x+1 = \pm \sqrt{24}$$

$$x+1 = \pm \sqrt{4} \sqrt{6}$$

$$x = \{-1 \pm 2\sqrt{6}\}$$

5. $-x^2 + 7x = 1$

(Quadratic Formula)

$$-x^2 + 7x - 1 = 0$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(-1)(-1)}}{2(-1)}$$

$$x = \frac{-7 \pm \sqrt{49-4}}{-2}$$

$$x = \frac{-7 \pm \sqrt{45}}{-2} \leftarrow \sqrt{9} \sqrt{5}$$

$$x = \frac{-7 \pm 3\sqrt{5}}{-2}$$

$$x = \left\{ \frac{7 \pm 3\sqrt{5}}{2} \right\}$$

6. $5x^2 = x - 2$

(Quadratic Formula)

$$5x^2 - x + 2 = 0$$

$$x = \frac{1 \pm \sqrt{(-1)^2 - 4(5)(2)}}{2(5)}$$

$$x = \frac{1 \pm \sqrt{1-40}}{10}$$

$$x = \frac{1 \pm \sqrt{-39}}{10}$$

$$x = \left\{ \frac{1 \pm i\sqrt{39}}{10} \right\}$$

7. Step 1: Find the discriminant of each equation, then determine the number and type of roots.

EQUATION A $-x^2 + 4x - 11 = 0$ $4^2 - 4(-1)(-11)$ $16 - 44$ $= -28$ 2, imaginary	EQUATION B $4x^2 - 20x + 25 = 0$ $(-20)^2 - 4(4)(25)$ $400 - 400$ $= 0$ 1, rational	EQUATION C $-3x^2 - 11x + 2 = 0$ $(-11)^2 - 4(-3)(2)$ $121 + 24$ $= 145$ 2, irrational	EQUATION D $\frac{1}{2}x^2 - 8 = 0$ $0^2 - 4(\frac{1}{2})(-8)$ $0 + 16$ $= 16$ 2, rational
---	---	--	--

Step 2: Solve each equation using the methods below. You can only use each method once.

FACTORING

$$\begin{aligned}
 &4x^2 - 20x + 25 = 0 \\
 &x^2 - 20x + 100 = 0 \\
 &(x - \frac{10}{4})(x - \frac{10}{4}) = 0 \\
 &\frac{(2x - 5)(2x - 5)}{2x - 5 = 0} = 0 \\
 &\quad \quad \quad x = \frac{5}{2} \quad \quad x = \frac{5}{2}
 \end{aligned}$$

$$x = \left\{ \frac{5}{2} \right\}$$

B

SQUARE ROOTS

$$\begin{aligned}
 &\frac{1}{2}x^2 - 8 = 0 \\
 &2 \cdot \frac{1}{2}x^2 = 8 \cdot 2 \\
 &x^2 = 16 \\
 &x = \pm \sqrt{16} \\
 &x = \{ \pm 4 \}
 \end{aligned}$$

D

COMPLETE THE SQUARE

$$\begin{aligned}
 &-x^2 + 4x - 11 = 0 \\
 &x^2 - 4x = -11 \\
 &\quad \downarrow \\
 &(-2)^2 = 4 \\
 &x^2 - 4x + 4 = -11 + 4 \\
 &(x - 2)^2 = -7 \\
 &x - 2 = \pm \sqrt{-7} \\
 &x - 2 = \pm i\sqrt{7}
 \end{aligned}$$

$$x = \{ 2 \pm i\sqrt{7} \}$$

A

THE QUADRATIC FORMULA

$$\begin{aligned}
 &-3x^2 - 11x + 2 = 0 \\
 &x = \frac{11 \pm \sqrt{(-11)^2 - 4(-3)(2)}}{2(-3)} \\
 &x = \frac{11 \pm \sqrt{121 + 24}}{-6} \\
 &x = \frac{11 \pm \sqrt{145}}{-6}
 \end{aligned}$$

$$x = \left\{ \frac{-11 \pm \sqrt{145}}{6} \right\}$$

C

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 4: Solving Quadratic Equations

Quiz 4-3: Solving Quadratic Equations (All Methods)

Solve by factoring:

1. $\frac{3x^2}{3} - \frac{3x}{3} - \frac{60}{3} = 0$

$$x^2 - x - 20 = 0$$

$$(x-5)(x+4) = 0$$

$$x-5=0 \quad x+4=0$$

$$x=5 \quad x=-4$$

2. $4x^2 + 31x = 3x$

$$4x^2 + 28x = 0$$

$$4x(x+7) = 0$$

$$4x=0 \quad x+7=0$$

$$x=0 \quad x=-7$$

1. $x = \{-4, 5\}$

2. $x = \{-7, 0\}$

3. $x = \{\pm \frac{5}{3}\}$

4. $x = \{\pm 5\sqrt{6}\}$

5. $x = \{-2 \pm 3i\sqrt{7}\}$

6. $x = \{\frac{-5 \pm \sqrt{29}}{2}\}$

Solve by square roots:

3. $9x^2 - 18 = 7$

$$9x^2 = 25$$

$$x^2 = \frac{25}{9}$$

$$x = \pm \sqrt{\frac{25}{9}}$$

$$x = \pm \frac{5}{3}$$

4. $\frac{1}{3}x^2 - 7 = 43$

$$\frac{1}{3}x^2 = 50$$

$$x^2 = 150$$

$$x = \pm \sqrt{150}$$

$$x = \pm \sqrt{25 \cdot 6}$$

$$x = \pm 5\sqrt{6}$$

Solve by completing the square:

5. $2x^2 + 8x + 134 = 0$

$$x^2 + 4x = -67$$

$$\downarrow$$

$$2^2 = 4$$

$$x^2 + 4x + 4 = -67 + 4$$

$$(x+2)^2 = -63$$

$$x+2 = \pm \sqrt{-63}$$

$$x+2 = \pm i\sqrt{9 \cdot 7}$$

$$x+2 = \pm 3i\sqrt{7}$$

$$x = -2 \pm 3i\sqrt{7}$$

Solve by the quadratic formula:

6. $x^2 + 12x = 7x + 1$

$$x^2 + 5x - 1 = 0$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1)(-1)}}{2}$$

$$x = \frac{-5 \pm \sqrt{25 + 4}}{2}$$

$$x = \frac{-5 \pm \sqrt{29}}{2}$$

7. **Step 1:** Find the discriminant of each equation, then determine the number and type of solutions.

A. $-2x^2 - 196 = 0$ $0^2 - 4(-2)(-196)$ $0 - 1568$ $= -1568$ 2 imaginary	B. $2x^2 - 9x + 8 = 0$ $(-9)^2 - 4(2)(8)$ $81 - 64$ $= 17$ 2 irrational	C. $4x^2 - 21x - 18 = 0$ $(-21)^2 - 4(4)(-18)$ $441 + 288$ $= 729$ 2 rational	D. $x^2 + 18x + 97 = 0$ $18^2 - 4(1)(97)$ $324 - 388$ $= -64$ 2 imaginary
--	--	--	--

Step 2: Solve each equation using the methods below. Each method may only be used once.

FACTORING $4x^2 - 21x - 18 = 0$ $x^2 - 21x - 72 = 0$ $(x - \frac{24}{4})(x + \frac{3}{4}) = 0$ $\frac{(x-6)(4x+3)}{x-6=0 \quad 4x+3=0}$ $x=6 \quad x=-\frac{3}{4}$ $x = \{-\frac{3}{4}, 6\}$ C	SQUARE ROOTS $-2x^2 - 196 = 0$ $-2x^2 = 196$ $x^2 = -98$ $x = \pm \sqrt{-98}$ $x = \pm i\sqrt{49} \sqrt{2}$ $x = \{\pm 7i\sqrt{2}\}$ A
COMPLETE THE SQUARE $x^2 + 18x + 97 = 0$ $x^2 + 18x = -97$ $\downarrow$ $9^2 = 81$ $x^2 + 18x + 81 = -97 + 81$ $(x+9)^2 = -16$ $x+9 = \pm \sqrt{-16}$ $x+9 = \pm 4i$ $x = \{-9 \pm 4i\}$ D	THE QUADRATIC FORMULA $2x^2 - 9x + 8 = 0$ $x = \frac{9 \pm \sqrt{(-9)^2 - 4(2)(8)}}{2(2)}$ $x = \frac{9 \pm \sqrt{81 - 64}}{4}$ $x = \left\{ \frac{9 \pm \sqrt{17}}{4} \right\}$ B

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																																				
GEOMETRIC Problems	1. The width of a rectangle is six meters less than its length. If the area of the rectangle is 112 m^2, find the dimensions of the rectangle. length = x width = $x - 6$ x $A=112$ $x-6$ $x(x-6) = 112$ $x^2 - 6x - 112 = 0$ $(x-14)(x+8) = 0$ <table style="margin-left: auto; margin-right: auto;"> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x-14=0$</td> <td style="padding: 0 5px;">$x+8=0$</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x=14$</td> <td style="padding: 0 5px;">$x=-8$</td> </tr> </table> length = 14 m width = 8 m 2. The length of a rectangle is one foot more than twice its width. If the area of the rectangle is 300 ft^2, find the dimensions of the rectangle. length = $2x+1$ width = x x $A=300$ $2x+1$ $x(2x+1) = 300$ $2x^2 + x - 300 = 0$ $x^2 + x - 600 = 0$ $(x+25)(x-24) = 0$ <table style="margin-left: auto; margin-right: auto;"> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x+25=0$</td> <td style="padding: 0 5px;">$x-24=0$</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x=-25$</td> <td style="padding: 0 5px;">$x=24$</td> </tr> </table> length = 25 ft width = 12 ft $(2x+25)(x-12) = 0$ $2x^2 + 28x - 300 = 0$ $x^2 + 14x - 150 = 0$ $(x+25)(x-12) = 0$ <table style="margin-left: auto; margin-right: auto;"> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x+25=0$</td> <td style="padding: 0 5px;">$x-12=0$</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 0 5px;">$x=-25$</td> <td style="padding: 0 5px;">$x=12$</td> </tr> </table> <tr> <td style="padding: 10px;"></td><td style="padding: 10px;"> 3. 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Dimensions: 8, 15, 17

7. Ken has a rectangular shaped vegetable garden that is 15 feet by 9 feet. He would like to increase the length and width by the same amount, x , so the new garden is twice the current area. Find the value of x to the nearest tenth of a foot.

9 $A=135$
15

9+x $A=270$
15+x

$$(15+x)(9+x) = 270$$

$$x^2 + 24x + 135 = 270$$

$$x^2 + 24x - 135 = 0$$

$$x = \frac{-24 \pm \sqrt{24^2 - 4(1)(-135)}}{2(1)}$$

$$x = \frac{-24 \pm \sqrt{1116}}{2}$$

$$x = \frac{-24 \pm 33.4066}{2}$$

$$x = 4.7, -28.7$$

$$4.7 \text{ ft}$$

8. Given the diagram to the left, if the area of the shaded region is 78 square inches, what are the dimensions of the outside rectangle?

$$(x+1)(2x-2) - x(x-2) = 78$$

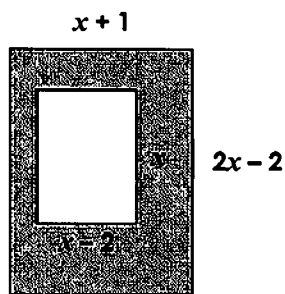
$$2x^2 - 2 - x^2 + 2x = 78$$

$$x^2 + 2x - 80 = 0$$

$$(x+10)(x-8) = 0$$

$x+10=0$	$x-8=0$
$x=-10$	$x=8$

Dimensions:
9in, 14in



CONSECUTIVE INTEGER Problems

9. Find two consecutive positive odd integers such that the square of the smaller integer subtracted from 13 times the larger integer is 68.

1st: x

2nd: $x+2$

$$13(x+2) - x^2 = 68$$

$$13x + 26 - x^2 = 68$$

$$0 = x^2 - 13x + 42$$

$$0 = (x-7)(x-6)$$

$x-7=0$	$x-6=0$
$x=7$	$x=6$

$$7+9$$

10. Find three consecutive positive integers such that the product of the median and largest integer is 8 times the sum of the smallest and largest integer.

1st: x

2nd: $x+1$

3rd: $x+2$

$$(x+1)(x+2) = 8(x+x+2)$$

$$x^2 + 3x + 2 = 16x + 16$$

$$x^2 - 13x - 14 = 0$$

$$(x-14)(x+1) = 0$$

$x-14=0$	$x+1=0$
$x=14$	$x=-1$

$$14, 15, 16$$

Name: _____

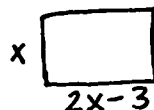
Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 10: Quadratic Equation Applications

**** This is a 2-page document! ****

1. The length of a rectangle is 3 meters less than twice its width. The area of the rectangle is 104 m². Find the dimensions of the rectangle.

length: $2x-3$ width: x 

$$x(2x-3) = 104$$

$$2x^2 - 3x - 104 = 0$$

$$x^2 - 3x - 208 = 0$$

$$(x - \frac{16}{2})(x + \frac{13}{2}) = 0$$

$$(x-8)(2x+13) = 0$$

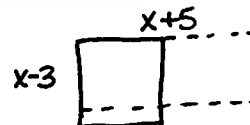
$$x-8=0 \quad 2x+13=0$$

$$x=8 \quad x = -\frac{13}{2}$$

length = 8m

width = 13m

2. The dimensions of a square are altered so that one dimension is increased by 5 feet, while the other is decreased by 3 feet. The area of the resulting rectangle is 84 ft². What was the area of the original square?



$$(x+5)(x-3) = 84$$

$$x^2 + 2x - 15 = 84$$

$$x^2 + 2x - 99 = 0$$

$$(x+11)(x-9) = 0$$

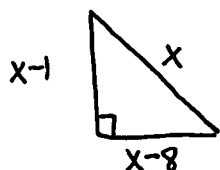
$$x+11=0 \quad x-9=0$$

$$x = -11 \quad x = 9$$

Original area = $9(9)$

$$= 81 \text{ ft}^2$$

3. The shorter leg of a right triangle is 8 less than the hypotenuse. The longer leg is 1 less than the hypotenuse. Find the perimeter of the triangle.



$$(x-1)^2 + (x-8)^2 = x^2$$

$$x^2 - 2x + 1 + x^2 - 16x + 64 = x^2$$

$$x^2 - 18x + 65 = 0$$

$$(x-5)(x-13) = 0$$

$$x-5=0 \quad x-13=0$$

$$x=5 \quad x=13 \rightarrow \text{sides} = 5, 12, 13$$

(Makes the short leg = -3.)

$$\text{Perimeter} = 30$$

4. The height of a trapezoid can be expressed as $x-4$, while the bases can be expressed as $x+4$ and $x+9$. If the area of the trapezoid is 99 cm², find the length of the larger base.

$$99 = \frac{1}{2}(x-4)(x+4+x+9)$$

$$99 = \frac{1}{2}(x-4)(2x+13)$$

$$99 = \frac{1}{2}(2x^2 + 5x - 52)$$

$$198 = 2x^2 + 5x - 52$$

$$0 = 2x^2 + 5x - 250$$

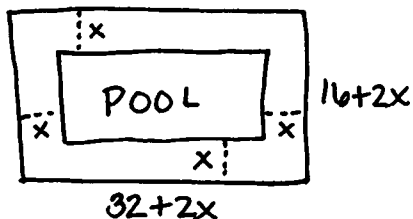
$$0 = (2x+25)(x-10)$$

$$2x+25=0 \quad x-10=0$$

$$x = -\frac{25}{2} \quad x = 10$$

Longer base = 19 cm.

5. An inground rectangular pool has a concrete pathway surrounding the pool. If the pool is 16 feet by 32 feet and the entire area of the pool including the walkway is 924 ft², find the width of the walkway.



$$(32+2x)(16+2x) = 924$$

$$512 + 96x + 4x^2 = 924$$

$$4x^2 + 96x - 412 = 0$$

$$x = \frac{-96 \pm \sqrt{96^2 - 4(4)(-412)}}{2(4)}$$

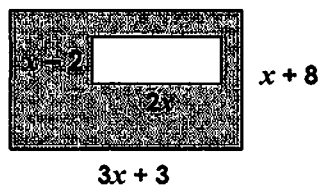
$$x = \frac{-96 \pm \sqrt{15808}}{8}$$

$$x = \frac{-96 \pm 125.730}{8}$$

$$x = 3.72, -27.72$$

3.72 ft

6. Given the diagram below, if the area of the shaded region is 290 ft², what are the dimensions of the inside rectangle?



$$(3x+3)(x+8) - 2x(x-2) = 290$$

$$3x^2 + 27x + 24 - 2x^2 + 4x = 290$$

$$x^2 + 31x - 266 = 0$$

$$(x+38)(x-7) = 0$$

$x+38=0$	$x-7=0$
$x=-38$	$x=7$

Inside Dimensions:
5 ft, 14 ft

7. Find two positive even consecutive integers such that one less than the square of the smaller integer is three more than the four times the larger integer.

1st: x

2nd: $x+2$

$$x^2 - 1 = 4(x+2) + 3$$

$$x^2 - 1 = 4x + 11$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$x-6=0$	$x+2=0$
$x=6$	$x=-2$

6 and 8

8. Find three positive consecutive integers such that the product of the smallest integer and largest integer is 47 more than 8 times the median integer.

1st: x

2nd: $x+1$

3rd: $x+2$

$$x(x+2) = 8(x+1) + 47$$

$$x^2 + 2x = 8x + 55$$

$$x^2 - 6x - 55 = 0$$

$$(x-11)(x+5) = 0$$

$x-11=0$	$x+5=0$
$x=11$	$x=-5$

11, 12, 13

Name:

Date:

Topic:

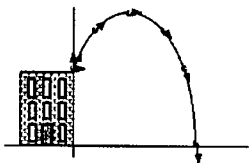
Class:

Main Ideas/Questions

Notes/Examples

PROJECTILE
MOTION

Problems



1. The height h , in feet, of a golf ball t seconds after it was hit from the ground can be modeled by the function $h(t) = -16t^2 + 200t$. Find a) the time it will take the ball to reach its maximum height and b) the maximum height of the ball.

$$a) t = \frac{-200}{2(-16)} = \frac{-200}{-32}$$

$$= 6.25 \text{ seconds}$$

$$b) -16(6.25)^2 + 200(6.25)$$

$$= 625 \text{ ft}$$

2. A rock is thrown off a bridge 40 feet above a river at an initial velocity of 96 feet per second. The height h , in feet, of the rock t seconds after it was thrown can be modeled by the function $h(t) = -16t^2 + 96t + 40$. Find the height of the rock 4 seconds after it was thrown.

$$-16(4)^2 + 96(4) + 40$$

$$= 168 \text{ ft}$$

3. A cannon launched a cannonball from a height of 4 feet with an upward velocity of 125 feet per second. The height h , in feet, of the cannonball t seconds after it was launched can be modeled by the function $h(t) = -16t^2 + 125t + 4$. Find a) the maximum height of the cannonball and b) the time, in seconds, it will take the cannonball to reach the ground.

$$a) t = \frac{-125}{2(-16)} = \frac{-125}{-32} = 3.90625 \text{ sec}$$

$$h = -16(3.90625)^2 + 125(3.90625) + 4$$

$$= 248.14 \text{ ft}$$

$$b) 0 = -16t^2 + 125t + 4$$

$$t = \frac{-125 \pm \sqrt{125^2 - 4(-16)(4)}}{2(-16)}$$

$$= \frac{-125 \pm \sqrt{15881}}{-32}$$

$$= 0.032, 7.84 \text{ sec}$$

4. A dirt bike launches from a 9-foot-tall ramp at an initial velocity of 30 feet per second. The height h , in feet, of the bike t seconds after it launched from the ramp can be modeled by the function $h(t) = -16t^2 + 30t + 9$. After how many seconds will the bike reach the ground?

$$0 = -16t^2 + 30t + 9$$

$$t = \frac{-30 \pm \sqrt{30^2 - 4(-16)(9)}}{2(-16)}$$

$$= \frac{-30 \pm \sqrt{1476}}{-32}$$

$$= -0.26, 2.14 \text{ sec}$$

5. The height h , in feet, of a kangaroo t seconds after it jumps from the ground can be modeled by the function $h(t) = -16t^2 + 28t$. What is the maximum height of the kangaroo?

$$t = \frac{-28}{2(-16)} = \frac{-28}{-32} = \frac{7}{8}$$

6. An arrow is shot from a bow from a height of 4 feet with an initial velocity of 150 feet per second. **a)** Using the formula $h(t) = -16t^2 + vt + s$, if v is the initial velocity, and s is the initial height, write an equation to model the height h of the arrow t seconds after the arrow was shot. Then **b)** find the height of the arrow 7 seconds after it was shot.

$$a) h(t) = -16t^2 + 150t + 4$$

$$b) h(7) = -16(7)^2 + 150(7) + 4$$

$= 270 \text{ ft}$

7. A diver jumps from a diving board 32 feet above the water with an initial velocity of 12 feet per second. height h , in feet, of a the diver t seconds after he jumped can modeled by the function $h(t) = -16t^2 + 12t + 32$. How many seconds will it take the diver to reach the surface of the water?

$$0 = -16t^2 + 12t + 32$$
$$t = \frac{-12 \pm \sqrt{12^2 - 4(-16)(32)}}{2(-16)}$$
$$= \frac{-12 \pm \sqrt{2192}}{-32}$$
$$= -1.69, \boxed{1.84 \text{ sec}}$$

8. The height h , in feet, of a soccer ball t seconds after it was hit by a player with his head can be modeled by the function $h(t) = -16t^2 + 20t + 6.5$. What is the domain and range of this function?

$$t = \frac{-20 \pm \sqrt{20^2 - 4(-16)(6.5)}}{2(-16)}$$

$$t = \frac{-20}{2(-16)} = \frac{-20}{-32} = \frac{5}{8}$$

$$= \frac{-20 \pm \sqrt{816}}{-32}$$

$$h\left(\frac{5}{8}\right) = -16\left(\frac{5}{8}\right)^2 + 20\left(\frac{5}{8}\right) + 6.5$$
$$= 12.75$$

$$= -0.27, 1.52 \text{ sec}$$

Domain: $0 \leq t \leq 1.52$

Range : $0 \leq h \leq 12.75$

9. The height h , in feet, of a baseball t seconds after it is thrown can be modeled by the function $h(t) = -16t^2 + 86t + 5$. If the ball is caught when it is two feet from the ground, how long was the ball in the air?

$$h = 2$$

$$2 = -16t^2 + 86t + 5$$

$$0 = -16t^2 + 86t + 3$$

$$t = \frac{-86 \pm \sqrt{86^2 - 4(-16)(3)}}{2(-16)}$$

$$= \frac{-86 \pm \sqrt{7588}}{-32}$$

$$= -0.03, 5.41 \text{ sec}$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

QUADRATIC

Regression

In a previous unit, you learned how to take data following a linear pattern and write a line of best fit. Sometimes, data follows a curve, and it's more appropriate to write a **curve of best fit** using **quadratic regression**.

Enter your data.

Find the curve of best fit.

- Hit **STAT**, **ENTER**
- Enter **x-values** in **L1**
- Enter **y-values** in **L2**

- Hit **STAT**
- Arrow over to **CALC**
- Choose **#5-QuadReg**
- Hit **ENTER** twice



EXAMPLES

1. An amusement park ride takes riders to the top of a tower and drops them. The table below shows the height, h , of the tower at time t seconds after drop.

t	0	0.5	1	1.5	2
h	200	164	120	68	8

- a. Write an equation for the curve of best fit.

$$h(t) = -16t^2 - 64t + 200$$

- b. How many seconds will it take the tower to reach the ground?

$$\begin{aligned}
 h &= 0 \\
 0 &= -16t^2 - 64t + 200 \\
 t &= \frac{64 \pm \sqrt{(-64)^2 - 4(-16)(200)}}{2(-16)} \\
 t &= \frac{64 \pm \sqrt{16896}}{-32} \\
 t &= -6.66, \boxed{2.06 \text{ sec}}
 \end{aligned}$$

2. Braden jumped off a cliff and into the ocean while vacationing in Hawaii. His height h after time t seconds is given in the table below.

t	0	1	2	3	4
h	480	482	452	390	296

- a. Write an equation for the curve of best fit.

$$h(t) = -16t^2 + 18t + 480$$

- b. How long will it take Braden to reach a height of 150 feet?

$$\begin{aligned}
 -16t^2 + 18t + 480 &= 150 \\
 -16t^2 + 18t + 330 &= 0 \\
 t &= \frac{-18 \pm \sqrt{18^2 - 4(-16)(330)}}{2(-16)} \\
 t &= \frac{-18 \pm \sqrt{21444}}{-32} \\
 t &= -4.64, \boxed{5.139 \text{ sec}}
 \end{aligned}$$

3. Sarah is standing on a bridge that stands 50 feet above the water. She tosses a coin into the water. The height, h , of the coin at time t seconds is given in the table below.

t	0	0.5	1	1.5	2
h	50	64	70	68	58

- a. Write an equation for the curve of best fit.

$$h(t) = -16t^2 + 36t + 50$$

- b. What is the height of the coin after 3 seconds?

$$\begin{aligned} & -16(3)^2 + 36(3) + 50 \\ & = -16(9) + 108 + 50 \\ & = \boxed{14 \text{ ft}} \end{aligned}$$

4. The table below shows how wind affects a runner's performance in the 200 meter dash, where s represents the speed of the wind in meters per second and t represents the change in finishing time. Negative wind speed means the runner is running against the wind while positive wind speed means the runner is running with the wind.

s	-4	-2	0	2	4
t	1.48	0.6	0	-0.5	-1.52

- a. Write an equation for the curve of best fit.

$$t(s) = -0.003s^2 - 0.355s + 0.038$$

- b. Predict the change in finishing time when the wind is -8 meters per second.

$$\begin{aligned} & = -0.003(-8)^2 - 0.355(-8) + 0.038 \\ & = -0.003(64) + 2.84 + 0.038 \\ & = \boxed{2.686 \text{ sec}} \end{aligned}$$

5. A study compared the speed in miles per hour, x , and the average fuel economy in miles per gallon, y , for cars. The results are shown in the table below.

x	20	30	40	50	60	70
y	25.2	28.5	30.1	30.4	27.8	25.0

- a. Write an equation for the curve of best fit.

$$y = -0.008x^2 + 0.752x + 13.553$$

- b. Predict the fuel economy for a car traveling at 35 miles per hour.

$$\begin{aligned} & y = -0.008(35)^2 + 0.752(35) + 13.553 \\ & y = \boxed{30.072 \text{ mpg}} \end{aligned}$$

6. The table below shows the average sale price, p , of a house in Norfolk, Virginia in thousands of dollars during certain years since 2005, t .

	0	1	2	3	4	5
t	2005	2006	2007	2008	2009	2010
p	158	145	132	130	138	142

- a. Write an equation for the curve of best fit.

$$p(t) = 3.018t^2 - 18.032t + 158.25$$

- b. Predict the value of the house in 2015.

$$\begin{aligned} & t = 10 \\ & p(10) = 3.018(10)^2 - 18.032(10) + 158.25 \\ & = \boxed{\$279,730} \end{aligned}$$

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Bell: _____

Homework 11: Projectile Motion &
Quadratic Regression**** This is a 2-page document! ****

1. A pumpkin is launched directly upwards at 72 feet per second from a platform 24 feet high. The height h of the pumpkin (in feet), t seconds after launch can be represented by the function $h(t) = -16t^2 + 72t + 24$. Find the maximum height of the pumpkin and the time it takes to reach this point.

$$t = \frac{-72}{2(-16)} = \frac{-72}{-32} = 2.25$$

Max Height = 105 ft

Time = 2.25 sec

$$h(2.25) = -16(2.25)^2 + 72(2.25) + 24$$

$$= 105$$

2. A squirrel is 24 feet up in a tree and tosses a nut out of the tree with an initial velocity of 8 feet per second. The nut's height, h (in feet) at time t seconds can be represented by the function $h(t) = -16t^2 + 8t + 24$. If the squirrel climbs down the tree in 2 seconds, does it reach the ground before the nut?

$$-16t^2 + 8t + 24 = 0$$

$$t = \frac{-8 \pm \sqrt{8^2 - 4(-16)(24)}}{2(-16)}$$

$$t = \frac{-8 \pm \sqrt{1600}}{-32}$$

$$t = -1, 1.5 \text{ sec}$$

No, the
nut lands
first.

3. A quarterback throws a football from a height of 6 feet with an initial velocity of 20 feet per second. The height, h , of the ball at time t seconds can be represented by the function $h(t) = -16t^2 + 20t + 6$. If the ball is caught at 2 feet, find the range of the function.

$$t = \frac{-20}{2(-16)} = \frac{-20}{-32} = 0.625$$

$$2 \text{ ft} \leq y \leq 12.25 \text{ ft}$$

$$h(0.625) = -16(0.625)^2 + 20(0.625) + 6$$

$$= 12.25$$

(max)

4. A penny is dropped from the top of the Statue of Liberty, which is 305 feet tall. The height of the penny, h , at time t seconds can be represented by the function $h(t) = -16t^2 + 305$. After 4 seconds, find the height of the ball and the time remaining until it reaches the ground.

$$h(4) = -16(4)^2 + 305$$

$$= -16(16) + 305$$

$$= 49 \text{ ft}$$

$$0 = -16t^2 + 305$$

$$-305 = -16t^2$$

$$19.0625 = t^2$$

$$4.366 = t$$

$$4.366 - 4 = 0.366 \text{ sec}$$

5. Ben and Josh went to the roof of their 40-foot-tall high school to throw their math books off the edge. The initial velocity of Ben's book was 60 feet per second while the initial velocity of Josh's book was 48 feet per second. The function $h(t) = -16t^2 + vt + s$ can be used to represent the situation where h is the height of the textbook at time t seconds, v is the initial upward velocity, and s is the starting point. Whose textbook reaches the ground first and by how many seconds?

Ben: $h(t) = -16t^2 + 60t + 40$

$$-16t^2 + 60t + 40 = 0$$

$$t = \frac{-60 \pm \sqrt{60^2 - 4(-16)(40)}}{2(-16)}$$

$$t = \frac{-60 \pm \sqrt{6160}}{-32}$$

$$t = -0.518, 4.328 \text{ sec}$$

Josh: $h(t) = -16t^2 + 48t + 40$

$$-16t^2 + 48t + 40 = 0$$

$$t = \frac{-48 \pm \sqrt{48^2 - 4(-16)(40)}}{2(-16)}$$

$$t = \frac{-48 \pm \sqrt{4864}}{-32}$$

$$t = -0.649, 3.679 \text{ sec}$$

Josh's book landed first,
by 0.649 sec.

6. The table below gives the average cost (to the nearest hundred) of tuition and fees, y , at a four-year college in years since 1991, x .

	0	3	7	9	14	17
x	1991	1994	1998	2000	2005	2008
y	5,500	6,200	7,000	7,200	\$8,800	\$9,600

- a. Write an equation for the curve of best fit. $y = 4.462x^2 + 162.859x + 5562.115$

- b. During what year will tuition and fees reach \$15,000? 2022

$$4.462x^2 + 162.859x + 5562.115 = 15000$$

$$4.462x^2 + 162.859x - 9437.885 = 0$$

$$x = \frac{-162.859 \pm \sqrt{(162.859)^2 - 4(4.462)(-9437.885)}}{2(4.462)}$$

$$x = \frac{-162.859 \pm \sqrt{194970.4254}}{8.924}$$

$$x = -67.729, 31.23 \rightarrow 2022$$

7. The table below shows the hourly temperatures, for a city in Ohio on February 10, 2012, as predicted by the weatherman at the local news station at certain times since 10 a.m..

	0	1	2	3	4	5
Time	10 a.m.	11 a.m.	12 p.m.	1 p.m.	2 p.m.	3 p.m.
Temp ($^{\circ}\text{F}$)	18	20	22	23	25	22

- a. Write an equation for the curve of best fit. $y = -0.446x^2 + 3.261x + 17.607$

- b. Predict the temperature at 7:00 p.m. $y = 10.83$
 $x = 9$

$$y = -0.446(9)^2 + 3.261(9) + 17.607$$

Unit 4 Test Study Guide

(Solving Quadratics & Complex Numbers)

Name: _____

Date: _____ Per: _____

Topic 1: Complex Numbers

Simplify each expression below.

1. $\sqrt{-324}$ $\sqrt{-1 \cdot 324}$ $= \boxed{18i}$	2. $2\sqrt{-147}$ $2\sqrt{-1 \cdot 147}$ $= 2i\sqrt{49}\sqrt{3}$ $= \boxed{14i\sqrt{3}}$	3. $(-4+7i)+(-3-8i)$ $\boxed{-7-i}$	4. $(21+9i)-(13-2i)$ $\boxed{8+11i}$
5. $(-3i^7)^4 \cdot i^{16}$ $81i^{28} \cdot i^{16}$ $= 81i^{44}$ $= 81(i^2)^{22}$ $= 81(-1)^{22} = \boxed{81}$	6. $(1+5i)(4-2i)$ $4-2i+20i-10i^2$ $= 4+18i+10$ $\boxed{14+18i}$	7. $(7-i)^2$ $(7-i)(7-i)$ $= 49-7i-7i+i^2$ $= 49-14i-1$ $= \boxed{48-14i}$	
8. $\frac{8}{-12i} \cdot i$ $\frac{8i}{-12i^2}$ $= \frac{8i}{-12(-1)}$ $= \frac{8i}{12} = \boxed{\frac{2i}{3}}$		9. $\frac{(3-6i)(4+3i)}{(4-3i)(4+3i)}$ $= \frac{12+9i-24i-18i^2}{16+12i-12i-9i^2}$ $= \frac{12-15i+18}{16+9} = \frac{30-15i}{25} = \boxed{\frac{6-3i}{5}}$	

Simplify, then name all sets to which the value belongs.

10. i^{59} $(i^2)^{29} \cdot i$ $(-1)^{29} \cdot i$ $= \boxed{-i}$ $\boxed{i, C}$	11. $(9+5i)(9-5i)$ $81-45i+45i-25i^2$ $81+25$ $= \boxed{106}$ $\boxed{N, W, Z, Q, R, C}$
--	--

For questions 11 and 12, use the complex numbers to write an example of each property.

12. Associative Property of Multiplication $5 \cdot (3i \cdot 2) = (5 \cdot 3i) \cdot 2$	13. Distributive Property $-7(3-5i) = -21 + 35i$
14. Name the additive identity of $(-10+4i)$. 0	15. Name the multiplicative inverse of $7i$. $\frac{1}{7i}$

16. Name all sets that are closed under subtraction.

i, C, R, Z

Topic 2: Standard, Vertex, and Factored Forms of Quadratic Functions

Complete the table below.

Standard Form	Vertex Form (Identify Vertex & Axis of Symmetry)	Factored Form (Identify the Roots)
17. $f(x) = x^2 + 16x + 63$	$f(x) = (x^2 + 16x) + 63$ $\quad \quad \quad \downarrow$ $\quad \quad \quad 8^2 = 64$ $f(x) = (x^2 + 16x + 64) + 63 - 64$ $f(x) = (x + 8)^2 - 1$ Axis: $x = -8$ Vertex: $(-8, -1)$	$f(x) = (x + 9)(x + 7)$ $x = \{-9, -7\}$
18. $f(x) = (x + 2)(x + 2) - 16$ $f(x) = x^2 + 4x + 4 - 16$ $f(x) = x^2 + 4x - 12$	$f(x) = (x + 2)^2 - 16$ Axis: $x = -2$ Vertex: $(-2, -16)$	$f(x) = (x + 6)(x - 2)$ $x = \{-6, 2\}$

Topic 3: Solving Quadratics by Graphing

19. $x^2 + 10x + 24 = 0$ <table border="1" style="display: inline-table; vertical-align: middle;"> <tr><th>x</th><th>y</th></tr> <tr><td>-8</td><td>8</td></tr> <tr><td>-7</td><td>3</td></tr> <tr><td>-6</td><td>0</td></tr> <tr><td>-5</td><td>-1</td></tr> <tr><td>-4</td><td>0</td></tr> <tr><td>-3</td><td>3</td></tr> <tr><td>-2</td><td>8</td></tr> </table> $x = \{-6, -4\}$	x	y	-8	8	-7	3	-6	0	-5	-1	-4	0	-3	3	-2	8	20. $-3x^2 - 6x = 0$ <table border="1" style="display: inline-table; vertical-align: middle;"> <tr><th>x</th><th>y</th></tr> <tr><td>-3</td><td>-9</td></tr> <tr><td>-2</td><td>0</td></tr> <tr><td>-1</td><td>3</td></tr> <tr><td>0</td><td>0</td></tr> <tr><td>1</td><td>-9</td></tr> </table> $x = \{-2, 0\}$	x	y	-3	-9	-2	0	-1	3	0	0	1	-9
x	y																												
-8	8																												
-7	3																												
-6	0																												
-5	-1																												
-4	0																												
-3	3																												
-2	8																												
x	y																												
-3	-9																												
-2	0																												
-1	3																												
0	0																												
1	-9																												

Topic 4: Solving Quadratics by Factoring

21. $x^2 - 11x + 18 = 0$ $(x - 9)(x - 2) = 0$ $\frac{x - 9 = 0}{x = 9} \quad \frac{x - 2 = 0}{x = 2}$ $x = \{2, 9\}$	22. $2x^2 - 32x + 128 = 0$ $2(x^2 - 16x + 64) = 0$ $2(x - 8)(x - 8) = 0$ $\frac{2 \neq 0}{x - 8 = 0} \quad \frac{x - 8 = 0}{x = 8}$ $x = \{8\}$
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23. $8x^2 + 10x = 0$ $\begin{array}{l l} 2x(4x+5)=0 & \\ \hline 2x=0 & 4x+5=0 \\ x=0 & x=-\frac{5}{4} \end{array}$ $x = \left\{ -\frac{5}{4}, 0 \right\}$	24. $7x^2 - 19x - 6 = 0$ $\begin{array}{l l} x^2 - 19x - 42 = 0 & \\ (x-21)(x+2)=0 & \\ \hline (x-3)(7x+2)=0 & \\ x-3=0 & 7x+2=0 \\ x=3 & x=-\frac{2}{7} \end{array}$ $x = \left\{ -\frac{2}{7}, 3 \right\}$
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Topic 5: Solving Quadratics by Square Roots

25. $16x^2 - 1 = 0$ $\begin{aligned} 16x^2 &= 1 \\ x^2 &= \frac{1}{16} \\ x &= \pm \sqrt{\frac{1}{16}} \end{aligned}$ $x = \left\{ \pm \frac{1}{4} \right\}$	26. $-3x^2 + 11 = 95$ $\begin{aligned} -3x^2 &= 84 \\ x^2 &= -28 \\ x &= \pm \sqrt{-28} \\ x &= \pm i\sqrt{4}\sqrt{7} \\ x &= \pm 2i\sqrt{7} \end{aligned}$ $x = \left\{ \pm 2i\sqrt{7} \right\}$
27. $(x+5)^2 = 4$ $\begin{aligned} x+5 &= \pm \sqrt{4} \\ x+5 &= \pm 2 \\ x &= -5 \pm 2 \end{aligned}$ $x = \left\{ -7, -3 \right\}$	28. $(x-8)^2 - 7 = 25$ $\begin{aligned} (x-8)^2 &= 32 \\ x-8 &= \pm \sqrt{32} \\ x-8 &= \pm \sqrt{16}\sqrt{2} \\ x &= 8 \pm 4\sqrt{2} \end{aligned}$ $x = \left\{ 8 \pm 4\sqrt{2} \right\}$

Topic 6: Solving Quadratics by Completing the Square

29. $x^2 - 8x - 10 = 0$ $\begin{aligned} x^2 - 8x &= 10 \\ \downarrow \\ (-4)^2 &= 16 \\ x^2 - 8x + 16 &= 10 + 16 \\ (x-4)^2 &= 26 \\ x-4 &= \pm \sqrt{26} \end{aligned}$ $x = \left\{ 4 \pm \sqrt{26} \right\}$	30. $\frac{-4x^2}{-4} - \frac{48x}{-4} - \frac{20}{-4} = 0$ $\begin{aligned} x^2 + 12x + 5 &= 0 \\ x^2 + 12x &= -5 \\ \downarrow \\ 6^2 &= 36 \\ x^2 + 12x + 36 &= -5 + 36 \\ (x+6)^2 &= 31 \\ x+6 &= \pm \sqrt{31} \end{aligned}$ $x = \left\{ -6 \pm \sqrt{31} \right\}$
--	--

Topic 7: Solving Quadratics by the Quadratic Formula

31. $-x^2 + 3x - 21 = 0$ $x = \frac{-3 \pm \sqrt{3^2 - 4(-1)(-21)}}{2(-1)}$ $x = \frac{-3 \pm \sqrt{9 - 84}}{-2}$ $x = \frac{-3 \pm \sqrt{-75}}{-2} \leftarrow i\sqrt{25}\sqrt{3}$ $x = \frac{-3 \pm 5i\sqrt{3}}{-2}$ $x = \left\{ \frac{3 \pm 5i\sqrt{3}}{2} \right\}$	32. $10x^2 + 8x - 1 = 0$ $x = \frac{-8 \pm \sqrt{8^2 - 4(10)(-1)}}{2(10)}$ $x = \frac{-8 \pm \sqrt{64 + 40}}{20}$ $x = \frac{-8 \pm \sqrt{104}}{20} \leftarrow \sqrt{4}\sqrt{26}$ $x = \frac{-8 \pm 2\sqrt{26}}{20}$ $x = \left\{ \frac{-4 \pm \sqrt{26}}{10} \right\}$
---	---

Topic 8: Discriminant/Choosing the Best Method

Find the discriminant of each equation and determine the number and type of roots. Then solve using the most appropriate method. Use each method once only.		
33. $-x^2 + 2x - 8 = 0$ $x^2 - 2x + 8 = 0$ $x^2 - 2x = -8$ $\downarrow$ $(-1)^2 = 1$ $x^2 - 2x + 1 = -8 + 1$ $(x - 1)^2 = -7$ $x - 1 = \pm \sqrt{-7}$ $x = 1 \pm i\sqrt{7}$	$\begin{aligned} 2^2 - 4(-1)(-8) \\ 4 - 32 \\ = -28 \end{aligned}$	Discriminant: -28 #/Type of Roots: 2, imaginary Method for Solving (Circle One) F / SR / <u>CS</u> / QF Solution(s): $x = \{1 \pm i\sqrt{7}\}$
34. $-2x^2 + 8 = x^2 - 28$ $0 = 3x^2 - 36$ $0 = x^2 - 12$ $12 = x^2$ $\pm \sqrt{12} = x$ $\pm \sqrt{4}\sqrt{3} = x$ $x = \pm 2\sqrt{3}$	$\begin{aligned} 0^2 - 4(3)(-36) \\ 0 + 432 \\ = 432 \end{aligned}$	Discriminant: 432 #/Type of Roots: 2, irrational Method for Solving (Circle One) F / <u>SR</u> / CS / QF Solution(s): $x = \{\pm 2\sqrt{3}\}$

35. $-2x^2 - 5x - 4 = 0$ $X = \frac{5 \pm \sqrt{(-5)^2 - 4(-2)(-4)}}{2(-2)}$ $X = \frac{5 \pm \sqrt{25 - 32}}{-4}$ $X = \frac{5 \pm \sqrt{-7}}{-4}$ $X = \frac{-5 \pm i\sqrt{7}}{4}$	$\begin{aligned} &(-5)^2 - 4(-2)(-4) \\ &25 - 32 \\ &= -7 \end{aligned}$	Discriminant: -7 #/Type of Roots: 2, imaginary Method for Solving (Circle One) F / SR / CS / <u>QF</u> Solution(s): $X = \left\{ \frac{-5 \pm i\sqrt{7}}{4} \right\}$			
36. $4x^2 + 32x - 36 = 0$ $4(x^2 + 8x - 9) = 0$ $4 (x+9)(x-1) = 0$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>$4 \neq 0$</td> <td>$x+9=0$ $x=-9$</td> <td>$x-1=0$ $x=1$</td> </tr> </table>	$4 \neq 0$	$x+9=0$ $x=-9$	$x-1=0$ $x=1$	$\begin{aligned} &32^2 - 4(4)(-36) \\ &1024 + 576 \\ &= 1600 \end{aligned}$	Discriminant: 1600 #/Type of Roots: 2, rational Method for Solving (Circle One) <u>F</u> / SR / CS / QF Solution(s): $X = \{-9, 1\}$
$4 \neq 0$	$x+9=0$ $x=-9$	$x-1=0$ $x=1$			

Topic 9: Applications

37. The chicken coup at the petting farm is 10 feet by 14 feet. The farm would like to double the current area by adding the same amount, x, to the length and width. What are the dimensions of the new enclosure? Round to the nearest hundredth of a foot. length = $14+x$ width = $10+x$ Area = 280 14.85 ft wide, 18.85 ft long	$\begin{aligned} &(14+x)(10+x) = 280 \\ &x^2 + 24x + 140 = 280 \\ &x^2 + 24x - 140 = 0 \\ &x = \frac{-24 \pm \sqrt{24^2 - 4(1)(-140)}}{2(1)} \\ &x = \frac{-24 \pm \sqrt{1136}}{2} \end{aligned}$ $x = -28.85, 4.85$
38. Find three positive consecutive odd integers such that the sum of the squares of the smallest integer and the largest integer is 129 more than the square of the median integer. 1st: x 2nd: $x+2$ 3rd: $x+4$	$\begin{aligned} &x^2 + (x+4)^2 = (x+2)^2 + 129 \\ &x^2 + x^2 + 8x + 16 = x^2 + 4x + 4 + 129 \\ &x^2 + 4x - 117 = 0 \\ &\frac{(x+13)(x-9)}{x+13=0 \quad x-9=0} \\ &\quad \quad \quad x=-13 \quad x=9 \end{aligned}$ 9, 11, 13

39. Natalie found a tennis ball outside a tennis court. She picked up the ball and threw it over the 12-foot fence into the court. The height (in feet) of the ball, h , at time t seconds is given by the function $h(t) = -16t^2 + 18t + 5$. Will she make it over the fence?

$$t = \frac{-18}{2(-16)} = \frac{-18}{-32} = 0.5625$$

$$h(0.5625) = -16(0.5625)^2 + 18(0.5625) + 5$$

$$= 10.0625 \text{ ft}$$

No, She will not make it over the fence.

40. A punter kicked a football with an initial velocity of 84 feet per second. The height (in feet) of the ball, h , at time t seconds is given by the function $h(t) = -16t^2 + 84t + 2$. If the ball is caught by another player down the field at a height of 4 feet, how long was the ball in the air?

$$-16t^2 + 84t + 2 = 4$$

$$-16t^2 + 84t - 2 = 0$$

$$t = \frac{-84 \pm \sqrt{84^2 - 4(-16)(-2)}}{2(-16)}$$

$$t = \frac{-84 \pm \sqrt{6928}}{-32}$$

$$t = 0.024, \boxed{5.23 \text{ sec}}$$

41. Jack was so frustrated with his slow laptop that he threw it out of his second story window. The height, h , of the laptop at time t seconds can be given by the equation $h(t) = -16t^2 + 28t + 17$. Assuming the laptop hits the ground below, find the domain and range of the function.

$$-16t^2 + 28t + 17 = 0$$

$$t = \frac{-28 \pm \sqrt{28^2 - 4(-16)(17)}}{2(-16)}$$

$$t = \frac{-28 \pm \sqrt{784 + 1088}}{-32}$$

Domain:

$$0 \leq x \leq 2.223 \text{ sec}$$

$$t = \frac{-28 \pm \sqrt{1872}}{-32}$$

$$t = -0.477, 2.223 \text{ sec}$$

42. The table below shows the altitude, in feet, of an airplane at certain times during its flight from Baltimore to Orlando since its departure at 11:00 a.m.. Use a quadratic model for the curve of best fit, then predict the altitude of the plane at 1:00 p.m. $x = 2$

		0	0.5	0.75	1	1.5	1.75
x	Time	11:00 a.m.	11:30 a.m.	11:45 a.m.	12:00 pm.	12:30 p.m.	12:45 p.m.
y	Altitude	0	10,500	15,750	21,500	12,450	10,820

$$y = -16930.28x^2 + 35749.53x - 735.7$$

$$y = 3042.22 \text{ ft}$$

Name: _____

Date: _____ Per: _____

Unit 4 Test

Solving Quadratic Equations (& Complex Numbers)

1. Simplify the expression below.

$$\begin{aligned} &\sqrt{-100} - 2\sqrt{16} \\ &10i - 2 \cdot 4i \\ &10i - 8i \\ &2i \end{aligned}$$

- A. $-12i$
B. $-2i$
C. $12i$
D. $2i$

D

2. Simplify the expression below.

$$\begin{aligned} &-9 + \sqrt{-28} \\ &-9 + i\sqrt{4} \sqrt{7} \\ &-9 + 2i\sqrt{7} \end{aligned}$$

- A. $-9 + 2i\sqrt{7}$
B. $-9 - 2i\sqrt{7}$
C. $i\sqrt{7}$
D. $5i\sqrt{3}$

A

3. Simplify the expression below. Give your answer as a complex number in standard form.

$$\begin{aligned} &(7 + i) - (6 - 4i) \\ &7 + i - 6 + 4i \end{aligned}$$

$$1 + 5i$$

4. Simplify the expression below. Give your answer as a complex number in standard form.

$$\begin{aligned} &(4 - 5i)(-1 - 7i) \\ &-4 - 28i + 5i + 35i^2 \\ &-4 - 23i - 35 \end{aligned}$$

$$-39 - 23i$$

5. Simplify the expression below. Give your answer as a complex number in standard form.

$$\begin{aligned} &(-7 + 3i)^2 + (-2 + 8i) \\ &49 - 21i - 21i + 9i^2 + -2 + 8i \\ &49 - 42i - 9 - 2 + 8i \end{aligned}$$

$$38 - 34i$$

6. Simplify the expression below. Give your answer as a complex number in standard form.

$$\begin{aligned} &\frac{(-9 + 5i)(3 + i)}{(3 - i)(3 + i)} \\ &\frac{-27 - 9i + 15i + 5i^2}{9 + 3i - 3i - i^2} = \frac{-32 + 6i}{10} \end{aligned}$$

$$\frac{-16 + 3i}{5}$$

7. Check all values that are equivalent to $-i$.

$$\begin{aligned} (i^2)^7 &= -1 \\ (i^2)^9 \cdot i &= -i \\ (i^2)^{12} \cdot i &= i \\ (i^2)^{18} \cdot i &= i \end{aligned}$$

☐ i^{14}	☐ i^{42}
☒ i^{19}	☐ i^{57}
☐ i^{25}	☐ i^{64}
☐ i^{37}	☒ i^{71}

$$\begin{aligned} (i^2)^{21} &= -1 \\ (i^2)^{28} \cdot i &= i \\ (i^2)^{32} &= 1 \\ (i^2)^{35} \cdot i &= -i \end{aligned}$$

8. Which sets do not contain the value

$(2 + 6i)(2 - 6i) - 50$? Check all that apply.

$$4 - 12i + 12i - 36i^2 - 50 = 4 + 36 - 50 = -10$$

☐ Real	☒ Whole
☒ Irrational	☒ Natural
☐ Rational	☐ Complex
☐ Integers	☒ Pure Imaginary

9. Which statement shows the associative property of multiplication?

- A. $(-4 + 17i) \cdot 1 = (-4 + 17i)$
 B. $3i + (-1 + 9i) = (-1 + 9i) + 3i$
 C. $(-5 + 3i) \cdot 6i = -5 \cdot 6i + 3i \cdot 6i$
 D. $((1 - i) \cdot 8i) \cdot -2i = (1 - i) \cdot (8i \cdot -2i)$

D

10. Which of the following is the additive identity of $5 + 2i$?

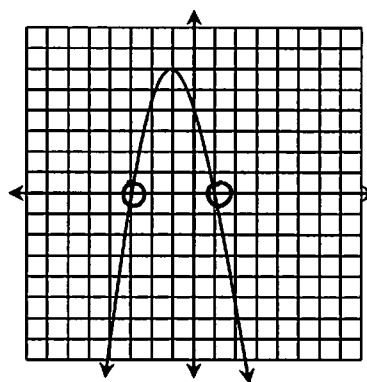
- A. 0
 B. 1
 C. $-(5 + 2i)$
 D. $5 - 2i$

A

11. Which of the following sets are closed under multiplication? Check all that apply.

☒	Real
☐	Irrational
☒	Rational
☒	Integers
☒	Whole
☒	Natural
☒	Complex
☐	Pure Imaginary

12. What is the apparent solution set for the equation associated with the following graph?



- A. $\{-3, 1\}$
 B. $\{-1, 3\}$
 C. $\{-3, 1, 4\}$
 D. $\{-1, 3, 4\}$

A

13. Which of the following gives the factored form and roots of the equation given below?

$$f(x) = 2(x + 4)^2 - 2$$

$$f(x) = 2(x + 4)(x + 4) - 2$$

$$f(x) = 2(x^2 + 8x + 16) - 2$$

$$f(x) = 2x^2 + 16x + 32 - 2$$

$$f(x) = 2x^2 + 16x + 30$$

$$f(x) = 2(x^2 + 8x + 15)$$

$$f(x) = 2(x + 3)(x + 5)$$

- A. $f(x) = 2(x + 3)(x + 5)$; $\{3, 5\}$
 B. $f(x) = 2(x + 3)(x + 5)$; $\{-3, -5\}$
 C. $f(x) = 2(x + 2)(x + 4)$; $\{2, 4\}$
 D. $f(x) = 2(x + 2)(x + 4)$; $\{-4, -2\}$

B

14. Solve by factoring:

$$x^2 + 9x - 83 = 4x + 1$$

$$x^2 + 5x - 84 = 0$$

$$\frac{(x + 12)(x - 7) = 0}{\begin{array}{l|l} x + 12 = 0 & x - 7 = 0 \\ \hline x = -12 & x = 7 \end{array}}$$

$$x = \{-12, 7\}$$

15. Solve by **factoring**:

$$\begin{aligned}
 25x^2 + 1 &= 10 \\
 25x^2 - 9 &= 0 \\
 (5x+3)(5x-3) &= 0 \\
 \hline
 5x+3=0 & \quad 5x-3=0 \\
 x = -\frac{3}{5} & \quad x = \frac{3}{5}
 \end{aligned}$$

$$x = \left\{ \pm \frac{3}{5} \right\}$$

16. Solve by **factoring**:

$$\begin{aligned}
 6x^2 + 7x - 3 &= 0 \\
 x^2 + 7x - 18 &= 0 \\
 (x+\frac{9}{6})(x-\frac{2}{6}) &= 0 \\
 \hline
 (2x+3)(3x-1) &= 0 \\
 \hline
 2x+3=0 & \quad 3x-1=0 \\
 x = -\frac{3}{2} & \quad x = \frac{1}{3}
 \end{aligned}$$

$$x = \left\{ -\frac{3}{2}, \frac{1}{3} \right\}$$

17. Solve by **square roots**:

$$\begin{aligned}
 2x^2 + 11 &= 131 \\
 2x^2 &= 120 \\
 x^2 &= 60 \\
 x &= \pm \sqrt{60} \\
 x &= \pm \sqrt{4} \sqrt{15} \\
 x &= \pm 2\sqrt{15}
 \end{aligned}$$

$$x = \{ \pm 2\sqrt{15} \}$$

18. Solve by **square roots**:

$$\begin{aligned}
 \frac{1}{3}x^2 - 10 &= -37 \\
 \frac{1}{3}x^2 &= -27 \\
 x^2 &= -81 \\
 x &= \pm \sqrt{-81} \\
 x &= \pm 9i
 \end{aligned}$$

$$x = \{ \pm 9i \}$$

19. Solve by **square roots**:

$$\begin{aligned}
 (2x+3)^2 &= 36 \\
 2x+3 &= \pm \sqrt{36} \\
 2x+3 &= \pm 6 \\
 2x &= -3 \pm 6 \\
 x &= \frac{3}{2}, -\frac{9}{2}
 \end{aligned}$$

$$x = \left\{ -\frac{9}{2}, \frac{3}{2} \right\}$$

20. Solve by **completing the square**:

$$\begin{aligned}
 x^2 + 10x + 7 &= 0 \\
 x^2 + 10x &= -7 \\
 \downarrow \\
 5^2 &= 25 \\
 x^2 + 10x + 25 &= -7 + 25 \\
 (x+5)^2 &= 18 \\
 x+5 &= \pm \sqrt{18} \leftarrow \sqrt{9} \sqrt{2} \\
 x+5 &= \pm 3\sqrt{2} \\
 x &= -5 \pm 3\sqrt{2}
 \end{aligned}$$

$$x = \{ -5 \pm 3\sqrt{2} \}$$

21. Solve by completing the square:

$$\begin{array}{r} -2x^2 + 6x + 7 = 2x - 3 \\ \frac{-2x^2}{-2} + \frac{4x}{-2} + \frac{10}{-2} = \frac{0}{-2} \end{array}$$

$$x^2 - 2x = 5$$

$$\downarrow$$

$$(-1)^2 = 1$$

$$x^2 - 2x + 1 = 5 + 1$$

$$(x-1)^2 = 6$$

$$x-1 = \pm \sqrt{6}$$

$$x = 1 \pm \sqrt{6}$$

$$x = \{1 \pm \sqrt{6}\}$$

22. Solve by the quadratic formula:

$$\begin{array}{r} -x^2 + 16 = 6x + 1 \\ -x^2 - 6x + 15 = 0 \end{array}$$

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4(-1)(15)}}{2(-1)}$$

$$x = \frac{6 \pm \sqrt{36 + 60}}{-2}$$

$$x = \frac{6 \pm \sqrt{96}}{-2} \quad \leftarrow \sqrt{16} \sqrt{6}$$

$$x = \frac{6 \pm 4\sqrt{6}}{-2}$$

$$x = -3 \pm 2\sqrt{6}$$

$$x = \{-3 \pm 2\sqrt{6}\}$$

23. Which functions have at least one real root? Check all that apply.

☒ $f(x) = -x^2 + 10x - 14$	☒ $f(x) = -x^2 + 4$
☒ $f(x) = 2x^2 - 12x + 18$	☐ $f(x) = -3x^2 + x - 2$
☐ $f(x) = x^2 - 3x + 25$	☒ $f(x) = 10x^2 - 4x$

For questions 24-27, find the discriminant of the equation and determine the number and type of roots, then solve the equation. Each method may only be used once!

24. $3x^2 + 15x + 19 = 0$

$$15^2 - 4(3)(19)$$

$$225 - 228$$

$$= -3$$

$$x = \frac{-15 \pm \sqrt{15^2 - 4(3)(19)}}{2(3)}$$

$$x = \frac{-15 \pm \sqrt{-3}}{6}$$

$$x = \frac{-15 \pm i\sqrt{3}}{6}$$

Discriminant:

-3

/ Type of Roots:

2, imaginary

Method (Circle one):

F

SR

CS

☒ QF

Solution(s):

$$x = \left\{ \frac{-15 \pm i\sqrt{3}}{6} \right\}$$

25. $9x^2 + 24x = 0$ $24^2 - 4(9)(0)$ $= 576 - 0$ $= 576$ $\begin{array}{l l} 3x & (3x+8) = 0 \\ \hline 3x=0 & 3x+8=0 \\ x=0 & x=-\frac{8}{3} \end{array}$	Discriminant: 576 # / Type of Roots: 2, rational Method (Circle one): ☒ F SR CS QF Solution(s): $x = \{-\frac{8}{3}, 0\}$
26. $-x^2 - 4x + 16 = 0$ $(-4)^2 - 4(-1)(16)$ $16 + 64$ $= 80$ $\begin{array}{l} -x^2 - 4x = -16 \\ x^2 + 4x = 16 \\ \downarrow \\ (2)^2 = 4 \\ x^2 + 4x + 4 = 16 + 4 \\ (x+2)^2 = 20 \\ x+2 = \pm \sqrt{20} \leftarrow \sqrt{4}\sqrt{5} \\ x+2 = \pm 2\sqrt{5} \\ x = -2 \pm 2\sqrt{5} \end{array}$	Discriminant: 80 # / Type of Roots: 2, irrational Method (Circle one): F SR ☒ CS QF Solution(s): $x = \{-2 \pm 2\sqrt{5}\}$
27. $4x^2 + 28 = 0$ $0^2 - 4(4)(28)$ $0 - 448$ -448 $\begin{array}{l} 4x^2 = -28 \\ x^2 = -7 \\ x = \pm \sqrt{-7} \\ x = \pm i\sqrt{7} \end{array}$	Discriminant: -448 # / Type of Roots: 2, imaginary Method (Circle one): F ☒ SR CS QF Solution(s): $x = \{\pm i\sqrt{7}\}$

28. The dimensions of a square are altered so that 8 inches is added to one side while 3 inches is subtracted from the other. The area of the resulting rectangle is 126 in^2 . What was the original side length of the square?

length: $x+8$

width: $x-3$

$$(x+8)(x-3) = 126$$

$$x^2 + 5x - 24 = 126$$

$$x^2 + 5x - 150 = 0$$

$$\frac{(x+15)(x-10) = 0}{\begin{array}{l|l} x+15=0 & x-10=0 \\ x=-15 & x=10 \end{array}}$$

10 inches

29. A juggler tosses a ball into the air. The ball's height, h , and time t seconds can be represented by the equation $h(t) = -16t^2 + 40t + 4$. Suppose the juggler missed and the ball hit the ground. Find the maximum height of the ball and the time it took to reach the ground. Round all answers to the nearest hundredth.

$$t = \frac{-40}{2(-16)} = \frac{-40}{-32} = \frac{5}{4}$$

$$h\left(\frac{5}{4}\right) = -16\left(\frac{5}{4}\right)^2 + 40\left(\frac{5}{4}\right) + 4$$

$$= 29$$

$$-16t^2 + 40t + 4 = 0$$

$$t = \frac{-40 \pm \sqrt{40^2 - 4(-16)(4)}}{2(-16)}$$

$$t = \frac{-40 \pm \sqrt{1856}}{-32}$$

$$t = \frac{-40 \pm 43.0813}{-32}$$

$$t = -0.09629, 2.59629$$

Maximum Height: 29 ft

Time to reach ground: 2.60 sec.

30. The table below gives the average temperature of a lake in months since April in a certain year. Write an equation for the curve of best fit using a quadratic model, then predict the average temperature of the lake during December of the same year. Round all answers to the nearest hundredth.

$x = 8$

		0	1	2	3	4
x	Month	April	May	June	July	August
y	Temp (°F)	59	68	71	78	74

Equation: $y = -1.57x^2 + 10.29x + 58.86$

Temperature: 40.7°

Name:

Date:

Topic:

Class:

Main Ideas/Questions

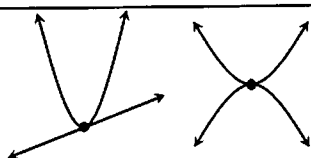
Notes/Examples

NONLINEAR SYSTEM of Equations

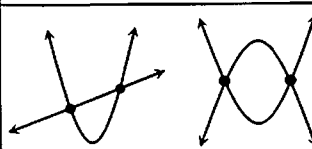
A system of equations in which one or more equations are not linear.

- The **solution set** to a system of equations is the set of points in which the equations intersect.
- There can be **one solution**, **multiple solutions**, or **no solution**. See the examples below.

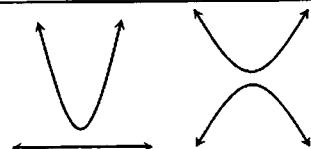
ONE SOLUTION



MULTIPLE SOLUTIONS



NO SOLUTION



SOLVE by Graphing

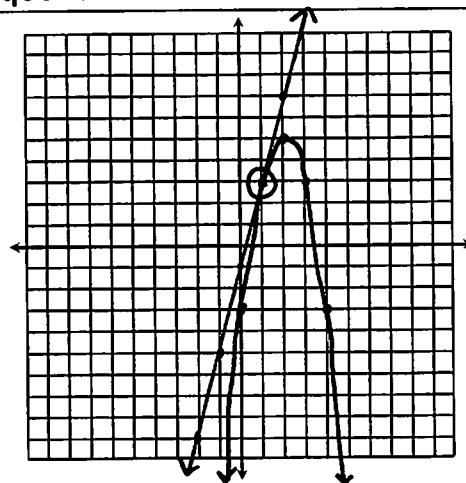
SET 1:
Linear/Quadratic

Solve the system by graphing each equation.

$$1. \begin{cases} 4x - y = 1 \\ y = -2x^2 + 8x - 3 \end{cases} \rightarrow \begin{cases} -y = -4x + 1 \\ y = 4x - 1 \end{cases}$$

x	y
-1	-13
0	-3
1	3
2	5
3	3
4	-3
5	-13

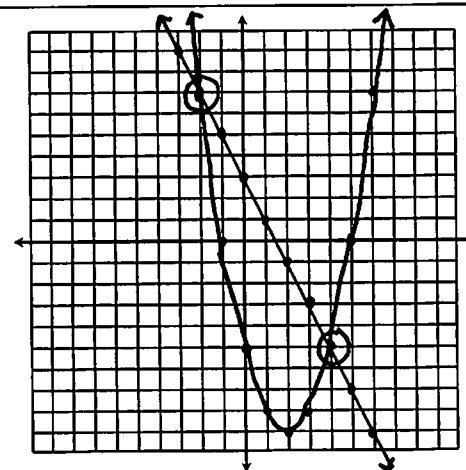
Solution(s): (1, 3)



$$2. \begin{cases} y = x^2 - 4x - 5 \\ 2x + y = 3 \end{cases} \rightarrow y = -2x + 3$$

x	y
-2	7
-1	0
0	-5
1	-8
2	-9
3	-8
4	-5
5	0
6	7

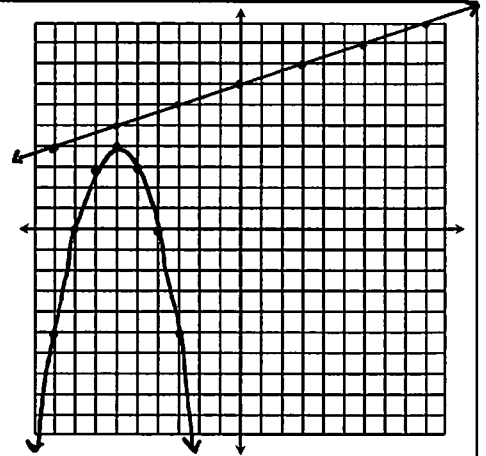
Solution(s): (-2, 7), (4, -5)



$$3. \begin{cases} y = -x^2 - 12x - 32 \\ x - 3y = -21 \rightarrow -3y = -x - 21 \\ y = \frac{1}{3}x + 7 \end{cases}$$

X	y
-9	-5
-8	0
-7	3
-6	4
-5	3
-4	0
-3	-5

Solution(s): No Solution



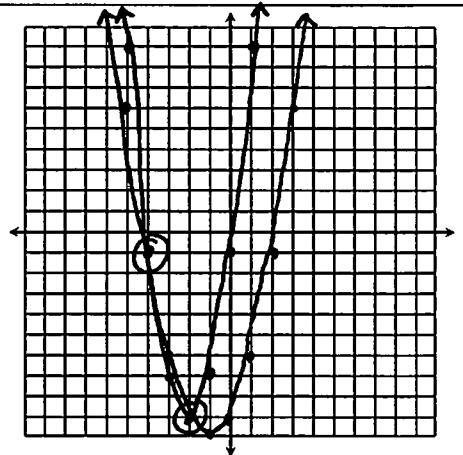
SET 2:
Quadratic/
Quadratic

$$4. \begin{cases} y = 2x^2 + 8x - 1 \\ y = x^2 + 2x - 9 \end{cases}$$

X	y
-5	9
-4	-1
-3	-7
-2	-9
-1	-7
0	-1
1	9

X	y
-5	-1
-4	-6
-3	-9
-2	-10
-1	-9
0	-6
1	-1
2	6
3	13

Solution(s): $(-4, -1), (-2, -9)$

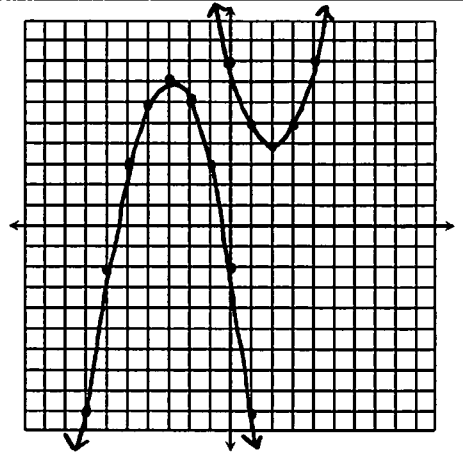


$$5. \begin{cases} y = -x^2 - 6x - 2 \\ y = x^2 - 4x + 8 \end{cases}$$

X	y
-7	-9
-6	-2
-5	3
-4	6
-3	7
-2	6
-1	3
0	-2
1	-9

X	y
-1	13
0	8
1	5
2	4
3	5
4	8
5	13

Solution(s): No Solution

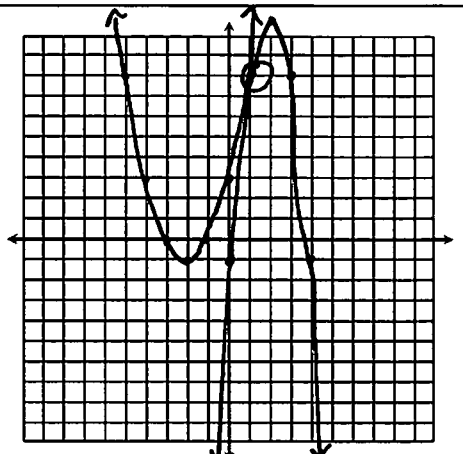


$$6. \begin{cases} y = -3x^2 + 12x - 1 \\ y = x^2 + 4x + 3 \end{cases}$$

X	y
-1	-16
0	-1
1	8
2	11
3	8
4	-1
5	-16

X	y
-5	8
-4	3
-3	0
-2	-1
-1	0
0	3
1	8

Solution(s): $(1, 8)$



Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

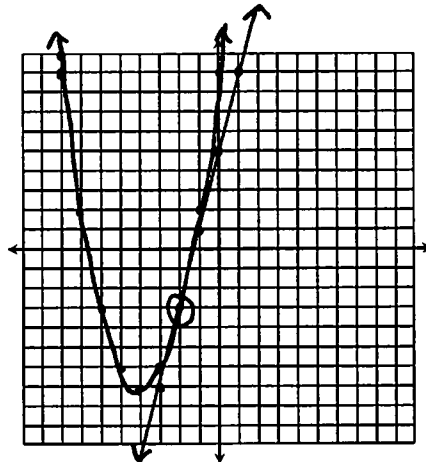
Homework 12: Solving Nonlinear Systems Graphically

**** This is a 2-page document! ****

Solve each system by graphing. Be sure to identify the solution(s).

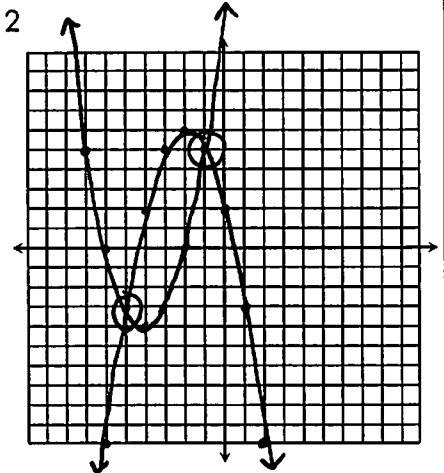
1. $\begin{cases} y = 4x + 5 \\ y = x^2 + 8x + 9 \end{cases}$

$(-2, -3)$



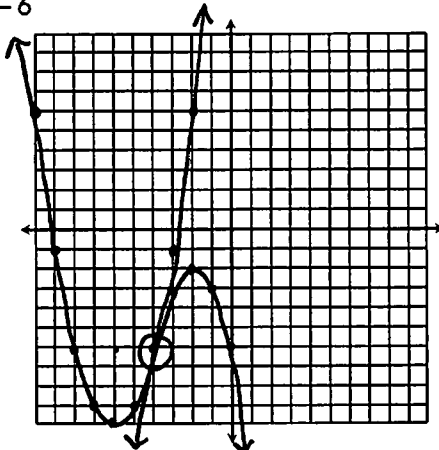
2. $\begin{cases} y = -x^2 - 4x + 2 \\ y = x^2 + 8x + 12 \end{cases}$

$(-5, -3),$
 $(-1, 5)$



3. $\begin{cases} y = x^2 + 12x + 26 \\ y = -x^2 - 4x - 6 \end{cases}$

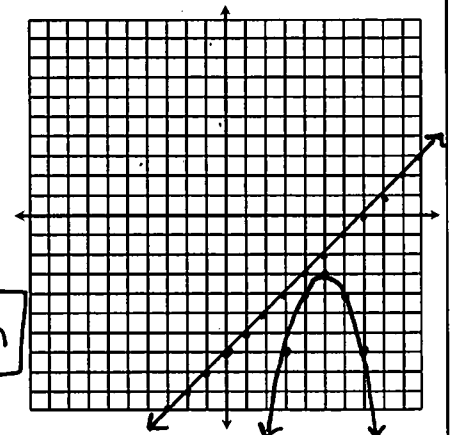
$(-4, -6)$



4. $\begin{cases} y = -x^2 + 10x - 28 \\ x - y = 7 \end{cases}$

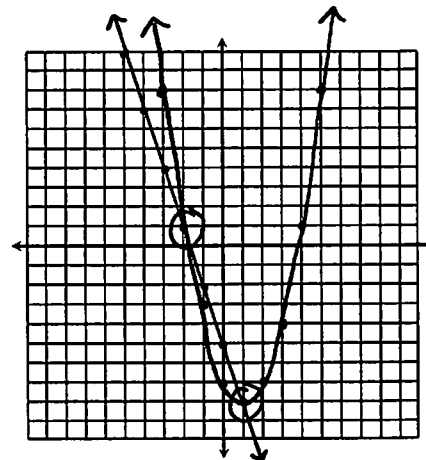
$\downarrow$
 $-y = -x + 7$
 $y = x - 7$

No Solution



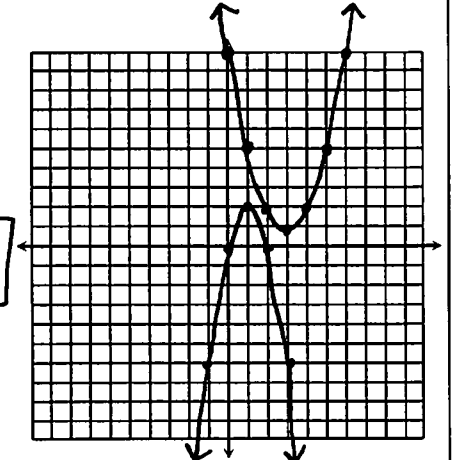
5. $\begin{cases} y = x^2 - 2x - 7 \\ y = -3x - 5 \end{cases}$

$(-2, 1),$
 $(1, -8)$



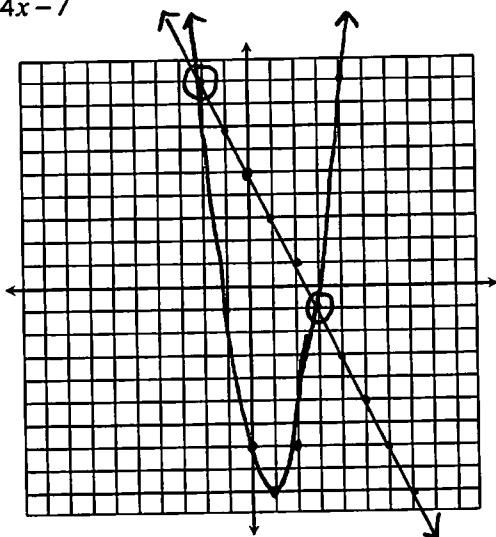
6. $\begin{cases} y = x^2 - 6x + 10 \\ y = -2x^2 + 4x \end{cases}$

No Solution



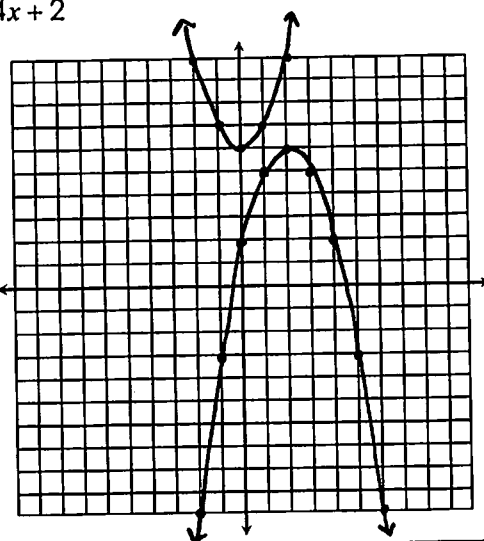
7. $\begin{cases} 2x + y = 5 \rightarrow y = -2x + 5 \\ y = 2x^2 - 4x - 7 \end{cases}$

$(-2, 9),$
 $(3, -1)$



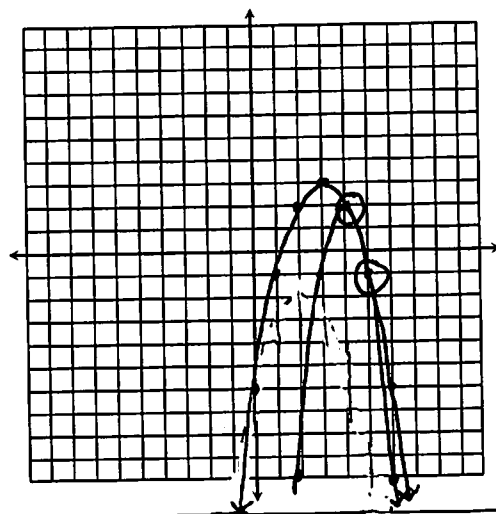
8. $\begin{cases} y = x^2 + 6 \\ y = -x^2 + 4x + 2 \end{cases}$

No
Solution



9. $\begin{cases} y = -x^2 + 6x - 6 \\ y = -3x^2 + 24x - 46 \end{cases}$

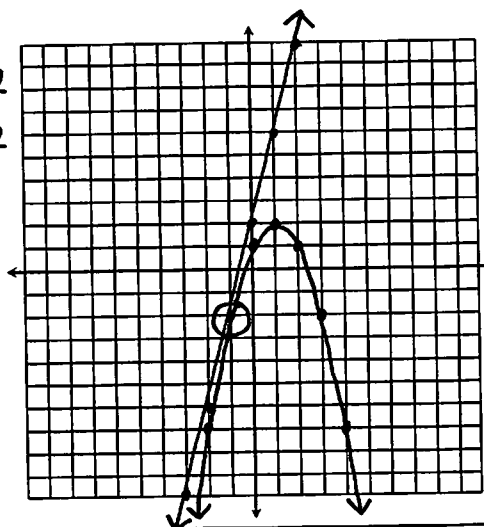
$(4, 2),$
 $(5, -1)$



10. $\begin{cases} y = -x^2 + 2x + 1 \\ 4x - y = -2 \end{cases}$

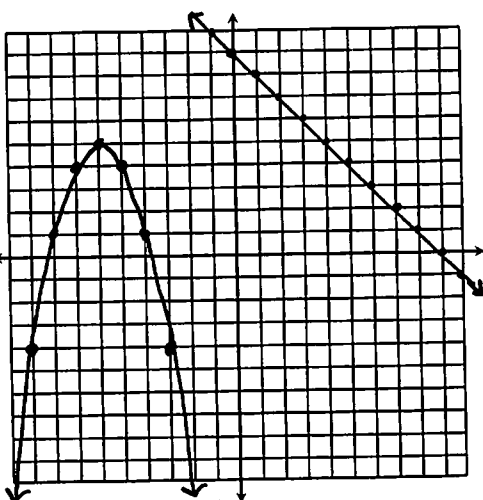
$\downarrow$
 $-y = -4x - 2$
 $y = 4x + 2$

$(-1, -2)$



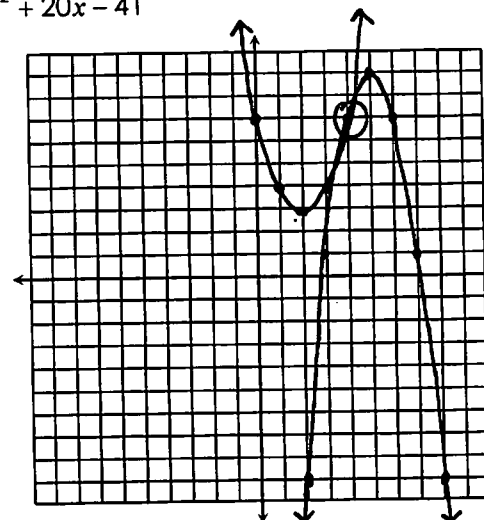
11. $\begin{cases} y = -x^2 - 12x - 31 \\ x + y = 9 \end{cases}$
 $\downarrow$
 $y = -x + 9$

No
Solution



12. $\begin{cases} y = x^2 - 4x + 7 \\ y = -2x^2 + 20x - 41 \end{cases}$

$(4, 7)$



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples							
NONLINEAR SYSTEMS: Solving Algebraically *Remember: A nonlinear system can have one solution, multiple solutions, or no solution.	Follow the steps below to solve the system algebraically by substitution.							
	Steps	Example						
	STEP 1: Isolate a variable in one of the equations.	$\begin{cases} 2x + y = 1 \\ y = x^2 - 8x + 6 \end{cases}$						
	STEP 2: Substitute the expression equal to your isolated variable into the other equation and solve.	$2x + (x^2 - 8x + 6) = 1$ $x^2 - 6x + 6 = 1$ $x^2 - 6x + 5 = 0$ $(x - 5)(x - 1) = 0$ <table><tr><td style="border-right: 1px solid black;">$x - 5 = 0$</td><td>$x - 1 = 0$</td></tr><tr><td style="border-right: 1px solid black;">$x = 5$</td><td>$x = 1$</td></tr></table>	$x - 5 = 0$	$x - 1 = 0$	$x = 5$	$x = 1$		
	$x - 5 = 0$	$x - 1 = 0$						
$x = 5$	$x = 1$							
STEP 3: Substitute your solution(s) into either original equation to complete the ordered pairs.	<table><tr><td>$2(5) + y = 1$ $10 + y = 1$ $y = -9$</td><td>$2(1) + y = 1$ $2 + y = 1$ $y = -1$</td></tr></table>	$2(5) + y = 1$ $10 + y = 1$ $y = -9$	$2(1) + y = 1$ $2 + y = 1$ $y = -1$					
$2(5) + y = 1$ $10 + y = 1$ $y = -9$	$2(1) + y = 1$ $2 + y = 1$ $y = -1$							
STEP 4: Identify the solution(s).	<table><tr><td>$(5, -9)$</td><td>$(1, -1)$</td></tr></table>	$(5, -9)$	$(1, -1)$					
$(5, -9)$	$(1, -1)$							
MORE EXAMPLES	Solve each system.							
	1. $\begin{cases} y = -2x^2 - 12x - 10 \\ 4x + y = -2 \end{cases}$ $4(-2) + y = -2$ $-8 + y = -2$ $y = 6$ $(-2, 6)$	$4x + (-2x^2 - 12x - 10) = -2$ $-2x^2 - 8x - 10 = -2$ $-2x^2 - 8x - 8 = 0$ $-2(x^2 + 4x + 4) = 0$ $-2(x + 2)(x + 2) = 0$ <table><tr><td style="border-right: 1px solid black;">$-2 \neq 0$</td><td style="border-right: 1px solid black;">$x + 2 = 0$</td><td>$x + 2 = 0$</td></tr><tr><td style="border-right: 1px solid black;"></td><td style="border-right: 1px solid black;">$x = -2$</td><td>$x = -2$</td></tr></table>	$-2 \neq 0$	$x + 2 = 0$	$x + 2 = 0$		$x = -2$	$x = -2$
	$-2 \neq 0$	$x + 2 = 0$	$x + 2 = 0$					
	$x = -2$	$x = -2$						
2. $\begin{cases} x - y = 8 \\ y = x^2 + 6x + 2 \end{cases}$	$x - (x^2 + 6x + 2) = 8$ $x - x^2 - 6x - 2 = 8$ $-x^2 - 5x - 10 = 0$ $x^2 + 5x + 10 = 0$ * cannot be factored; imaginary roots * No Solution							

Gina Wilson (All Things Algebra®, LLC).

$$3. \begin{cases} 2x - y = -4 \\ y = -x^2 + 4x + 7 \end{cases}$$

$$(3, 10), (-1, 2)$$

$$2x - (-x^2 + 4x + 7) = -4$$

$$2x + x^2 - 4x - 7 = -4$$

$$x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\begin{array}{c|c} x-3=0 & x+1=0 \\ \hline x=3 & x=-1 \end{array}$$

$$x=3 \quad x=-1$$

$$2(3) - y = -4$$

$$-y = -10$$

$$y = 10$$

$$2(-1) - y = -4$$

$$-y = -2$$

$$y = 2$$

$$4. \begin{cases} y = x^2 + 12x + 24 \\ y = -x^2 - 2x + 4 \end{cases}$$

$$x^2 + 12x + 24 = -x^2 - 2x + 4$$

$$2x^2 + 14x + 20 = 0$$

$$2(x^2 + 7x + 10) = 0$$

$$y = (-5)^2 + 12(-5) + 24$$

$$= 25 - 60 + 24$$

$$= -11$$

$$y = (-2)^2 + 12(-2) + 24$$

$$= 4 - 24 + 24$$

$$= 4$$

$$2(x+5)(x+2) = 0$$

$$\begin{array}{c|c|c} 2 \neq 0 & x+5=0 & x+2=0 \\ \hline & x=-5 & x=-2 \end{array}$$

$$x=-5 \quad x=-2$$

$$(-5, -11), (-2, 4)$$

$$5. \begin{cases} y = 2x^2 - 8x - 1 \\ y = -2x^2 + 16x - 37 \end{cases}$$

$$2x^2 - 8x - 1 = -2x^2 + 16x - 37$$

$$4x^2 - 24x + 36 = 0$$

$$4(x^2 - 6x + 9) = 0$$

$$4(x-3)(x-3) = 0$$

$$\begin{array}{c|c|c} 4 \neq 0 & x-3=0 & x-3=0 \\ \hline & x=3 & x=3 \end{array}$$

$$x=3 \quad x=3$$

$$y = 2(3)^2 - 8(3) - 1$$

$$y = 18 - 24 - 1$$

$$y = -7$$

$$(3, -7)$$

$$6. \begin{cases} y = x^2 - 4x - 11 \\ y = -x^2 - 12x - 30 \end{cases}$$

$$x^2 - 4x - 11 = -x^2 - 12x - 30$$

$$2x^2 + 8x + 19 = 0$$

* Cannot be factored;
imaginary roots *

No Solution

$$7. \begin{cases} y = 3x^2 - 12x + 1 \\ y = -x^2 + 8x - 15 \end{cases}$$

$$3x^2 - 12x + 1 = -x^2 + 8x - 15$$

$$4x^2 - 20x + 16 = 0$$

$$4(x^2 - 5x + 4) = 0$$

$$4(x-4)(x-1) = 0$$

$$\begin{array}{c|c|c} 4 \neq 0 & x-4=0 & x-1=0 \\ \hline & x=4 & x=1 \end{array}$$

$$x=4 \quad x=1$$

$$y = -(4)^2 + 8(4) - 15$$

$$= -16 + 32 - 15$$

$$= 1$$

$$y = -(1)^2 + 8(1) - 15$$

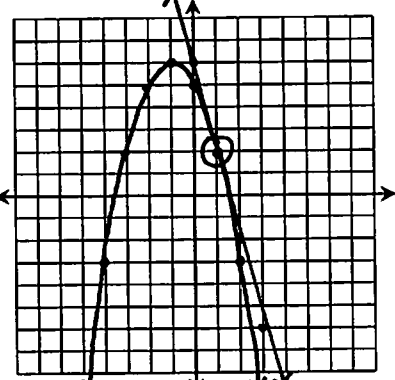
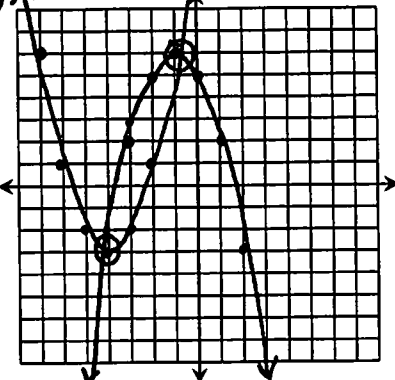
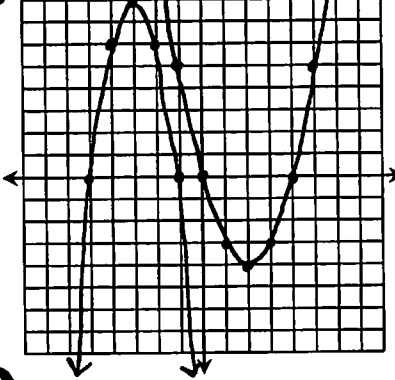
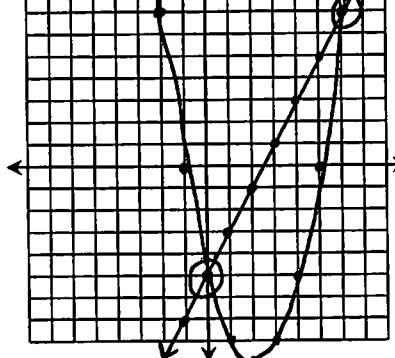
$$= -1 + 8 - 15$$

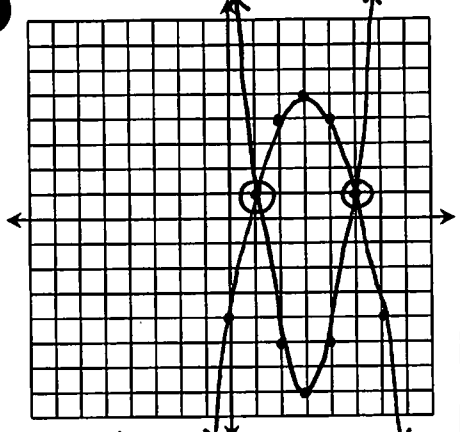
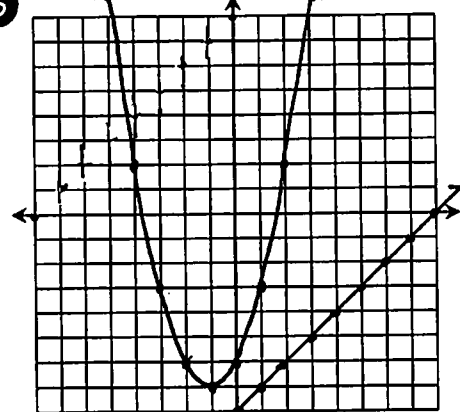
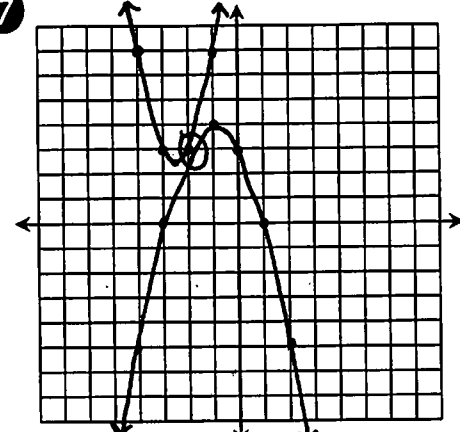
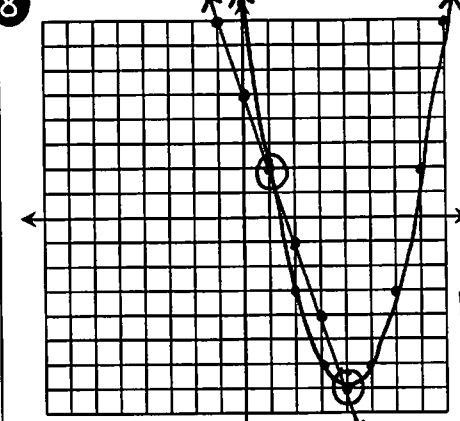
$$= -8$$

$$(4, 1), (1, -8)$$

GRAPHING vs. SUBSTITUTION

Directions: Solve each given system of equations by graphing and substitution.
If done correctly, your answers should be the same!

Graphing Method	Substitution Method
1  $\begin{cases} y = -x^2 - 2x + 5 \\ y = -4x + 6 \end{cases}$ $(1, 2)$	$\begin{aligned} -4x + 6 &= -x^2 - 2x + 5 \\ x^2 - 2x + 1 &= 0 \\ (x-1)(x-1) &= 0 \\ \begin{array}{c c} x-1=0 & x-1=0 \\ \hline x=1 & x=1 \end{array} \end{aligned}$ $\begin{aligned} y &= -4(1) + 6 \\ y &= -4 + 6 \\ y &= 2 \end{aligned}$ $(1, 2)$
2  $\begin{cases} y = x^2 + 8x + 13 \\ y = -x^2 - 2x + 5 \end{cases}$ $(-4, -3), (-1, 6)$	$\begin{aligned} x^2 + 8x + 13 &= -x^2 - 2x + 5 \\ 2x^2 + 10x + 8 &= 0 \\ 2(x^2 + 5x + 4) &= 0 \\ 2(x+4)(x+1) &= 0 \\ \begin{array}{c c c} 2 \neq 0 & x+4=0 & x+1=0 \\ \hline & x=-4 & x=-1 \end{array} \end{aligned}$ $\begin{aligned} y &= (-4)^2 + 8(-4) + 13 & y &= (-1)^2 + 8(-1) + 13 \\ y &= 16 - 32 + 13 & & = 1 - 8 + 13 \\ y &= -3 & & = 6 \end{aligned}$ $(-4, -3), (-1, 6)$
3  $\begin{cases} y = x^2 - 4x \\ y = -2x^2 - 12x - 10 \end{cases}$ No Solution	$\begin{aligned} x^2 - 4x &= -2x^2 - 12x - 10 \\ 3x^2 + 8x + 10 &= 0 \end{aligned}$ * cannot be factored; imaginary roots * No Solution
4  $\begin{cases} 2x - y = 5 \\ y = x^2 - 4x - 5 \end{cases}$ $(0, -5), (6, 7)$	$\begin{aligned} 2x - (x^2 - 4x - 5) &= 5 \\ -x^2 + 6x + 5 &= 5 \\ 0 &= x^2 - 6x \\ 0 &= x(x-6) \\ \begin{array}{c c} x=0 & x-6=0 \\ \hline & x=6 \end{array} \end{aligned}$ $\begin{aligned} 2(0) - y &= 5 & 2(6) - y &= 5 \\ -y &= 5 & -y &= -7 \\ y &= -5 & y &= 7 \end{aligned}$ $(0, -5), (6, 7)$

Graphing Method	Substitution Method				
5  $\begin{cases} y = 2x^2 - 12x + 11 \\ y = -x^2 + 6x - 4 \end{cases}$ $(1, 1), (5, 1)$	$2x^2 - 12x + 11 = -x^2 + 6x - 4$ $3x^2 - 18x + 15 = 0$ $3(x^2 - 6x + 5) = 0$ $3(x-5)(x-1) = 0$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>$x-5=0$</td><td>$x-1=0$</td></tr> <tr> <td>$x=5$</td><td>$x=1$</td></tr> </table> $y = 2(5)^2 - 12(5) + 11$ $y = 50 - 60 + 11$ $y = 1$ $y = 2(1)^2 - 12(1) + 11$ $y = 2 - 12 + 11$ $y = 1$ $(5, 1), (1, 1)$	$x-5=0$	$x-1=0$	$x=5$	$x=1$
$x-5=0$	$x-1=0$				
$x=5$	$x=1$				
6  $\begin{cases} x - y = 8 \\ -y = -x + 8 \\ y = x - 8 \end{cases}$ $\begin{cases} x - y = 8 \\ y = x^2 + 2x - 6 \end{cases}$ No Solution	$x - (x^2 + 2x - 6) = 8$ $x - x^2 - 2x + 6 = 8$ $0 = x^2 + x + 2$ * cannot be factored; imaginary roots * No Solution				
7  $\begin{cases} y = 2x^2 + 10x + 15 \\ y = -x^2 - 2x + 3 \end{cases}$ $(-2, 3)$	$2x^2 + 10x + 15 = -x^2 - 2x + 3$ $3x^2 + 12x + 12 = 0$ $3(x^2 + 4x + 4) = 0$ $3(x+2)(x+2) = 0$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>$x+2=0$</td><td>$x+2=0$</td></tr> <tr> <td>$x=-2$</td><td>$x=-2$</td></tr> </table> $y = 2(-2)^2 + 10(-2) + 15$ $y = 8 - 20 + 15$ $y = 3$ $y = 2(-2)^2 - 12(-2) + 11$ $y = 8 + 24 + 11$ $y = 43$ $(-2, 3)$	$x+2=0$	$x+2=0$	$x=-2$	$x=-2$
$x+2=0$	$x+2=0$				
$x=-2$	$x=-2$				
8  $\begin{cases} 3x + y = 5 \\ y = -3x + 5 \end{cases}$ $\begin{cases} y = x^2 - 8x + 9 \\ 3x + y = 5 \end{cases}$ $(1, 2), (4, -7)$	$3x + x^2 - 8x + 9 = 5$ $x^2 - 5x + 4 = 0$ $(x-4)(x-1) = 0$ <table border="1" style="margin-left: auto; margin-right: auto;"> <tr> <td>$x-4=0$</td><td>$x-1=0$</td></tr> <tr> <td>$x=4$</td><td>$x=1$</td></tr> </table> $3(4) + y = 5$ $12 + y = 5$ $y = -7$ $3(1) + y = 5$ $3 + y = 5$ $y = 2$ $(4, -7), (1, 2)$	$x-4=0$	$x-1=0$	$x=4$	$x=1$
$x-4=0$	$x-1=0$				
$x=4$	$x=1$				

Name: _____

Unit 4: Solving Quadratic Equations

Date: _____ Per: _____

Homework 12: Solving Nonlinear Systems Algebraically

**** This is a 2-page document! ****

Solve each system algebraically. Then verify your solution with a graph.

1.
$$\begin{cases} y = x^2 - 6x + 2 \\ 2x - y = 5 \end{cases}$$

$$2x - (x^2 - 6x + 2) = 5$$

$$2x - x^2 + 6x - 2 = 5$$

$$-x^2 + 8x - 7 = 0$$

$$x^2 - 8x + 7 = 0$$

$$(x-7)(x-1) = 0$$

$$\begin{array}{c|c} x=7 & x=1 \end{array}$$

$$2(7) - y = 5$$

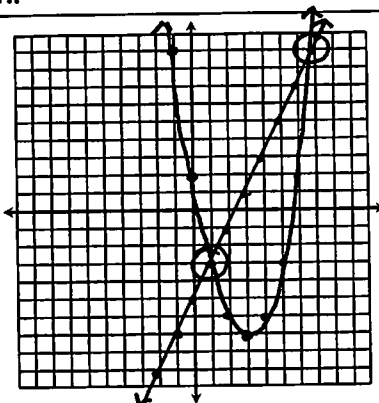
$$14 - y = 5$$

$$y = 9$$

$$2(1) - y = 5$$

$$2 - y = 5$$

$$y = -3$$

 $(7, 9),$
 $(1, -3)$ 

2.
$$\begin{cases} y = 3x^2 + 30x + 70 \\ y = -x^2 - 2x + 6 \end{cases}$$

$$3x^2 + 30x + 70 = -x^2 - 2x + 6$$

$$4x^2 + 32x + 64 = 0$$

$$4(x^2 + 8x + 16) = 0$$

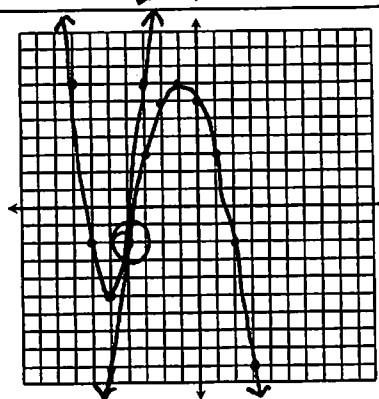
$$4(x+4)(x+4) = 0$$

$$\begin{array}{c|c} 4 \neq 0 & x = -4 \quad x = -4 \end{array}$$

$$y = -(-4)^2 - 2(-4) + 6$$

$$y = -16 + 8 + 6$$

$$y = -2$$

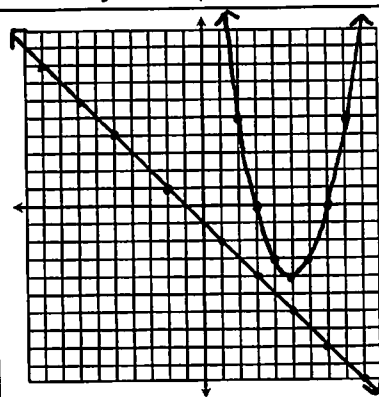
 $(-4, -2)$ 

3.
$$\begin{cases} x + y = -1 \\ y = x^2 - 10x + 21 \end{cases}$$

$$x + x^2 - 10x + 21 = -1$$

$$x^2 - 9x + 22 = 0$$

* cannot be factored -
imaginary roots *

No Solution

4.
$$\begin{cases} y = 2x^2 - 8x - 1 \\ y = -x^2 - 2x - 1 \end{cases}$$

$$2x^2 - 8x - 1 = -x^2 - 2x - 1$$

$$3x^2 - 6x = 0$$

$$3x(x-2) = 0$$

$$\begin{array}{c|c} x=0 & x=2 \end{array}$$

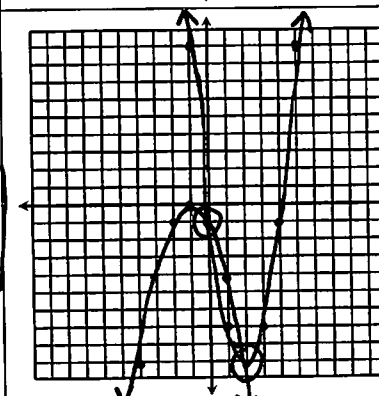
$$y = -(0)^2 - 2(0) - 1$$

$$y = -1$$

$$y = -(2)^2 - 2(2) - 1$$

$$y = -4 - 4 - 1$$

$$y = -9$$

 $(0, -1),$
 $(2, -9)$ 

Solve each system algebraically.

5. $\begin{cases} y = x^2 + 12x + 40 \\ y = -x^2 + 8x - 9 \end{cases}$
 $x^2 + 12x + 40 = -x^2 + 8x - 9$
 $2x^2 + 4x + 49 = 0$
 * cannot be factored;
 imaginary roots *

No Solution

6. $\begin{cases} y = -x^2 - 4x + 3 \\ 2x - y = -8 \end{cases}$
 $2x - (-x^2 - 4x + 3) = -8$
 $2x + x^2 + 4x - 3 = -8$
 $x^2 + 6x + 5 = 0$
 $(x+5)(x+1) = 0$
 $x = -5 \quad x = -1$
 $2(-5) - y = -8$
 $-10 - y = -8$
 $-y = 2$
 $y = -2$
 $2(-1) - y = -8$
 $-2 - y = -8$
 $-y = -6$
 $y = 6$
 $(-5, -2), (-1, 6)$

7. $\begin{cases} y = 2x^2 - 4x + 2 \\ y = 8x - 16 \end{cases}$
 $2x^2 - 4x + 2 = 8x - 16$
 $2x^2 - 12x + 18 = 0$
 $2(x^2 - 6x + 9) = 0$
 $2(x-3)(x-3) = 0$
 $2 \neq 0 \quad x = 3 \quad x = 3$

$y = 8(3) - 16$
 $y = 8$

(3, 8)

8. $\begin{cases} y = -x^2 - 10x - 23 \\ y = (x+3)^2 \end{cases}$
 $-x^2 - 10x - 23 = (x+3)^2$
 $-x^2 - 10x - 23 = x^2 + 6x + 9$
 $0 = 2x^2 + 16x + 32$
 $0 = 2(x^2 + 8x + 16)$
 $0 = 2(x+4)(x+4)$
 $2 \neq 0 \quad x = -4 \quad x = -4$

$y = (-4+3)^2$
 $y = (-1)^2$
 $y = 1$

(-4, 1)

9. $\begin{cases} y = 3x^2 - 18x + 17 \\ y = -x^2 + 2x + 1 \end{cases}$
 $3x^2 - 18x + 17 = -x^2 + 2x + 1$
 $4x^2 - 20x + 16 = 0$
 $4(x^2 - 5x + 4) = 0$
 $4(x-4)(x-1) = 0$
 $4 \neq 0 \quad x = 4 \quad x = 1$

$y = -(4)^2 + 2(4) + 1$
 $y = -16 + 8 + 1$
 $y = -7$

$y = -(1)^2 + 2(1) + 1$
 $y = -1 + 2 + 1$
 $y = 2$

(4, -7), (1, 2)

10. $\begin{cases} y = x^2 + 14x + 35 \\ 3x - y = 1 \end{cases}$
 $3x - (x^2 + 14x + 35) = 1$
 $3x - x^2 - 14x - 35 = 1$
 $-x^2 - 11x - 36 = 0$
 $x^2 + 11x + 36 = 0$

* Cannot be factored;
 imaginary roots *

No Solution

Name: _____

Algebra 2

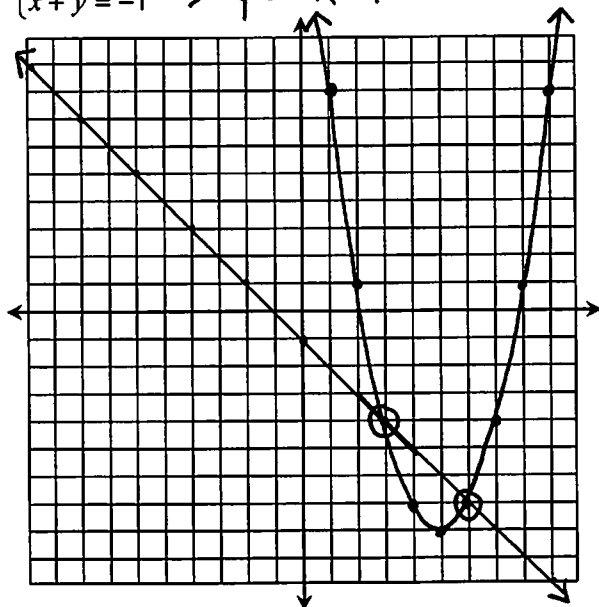
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Unit 4: Solving Quadratic Equations

Quiz 4-4: Linear-Quadratic and Quadratic-Quadratic-Systems

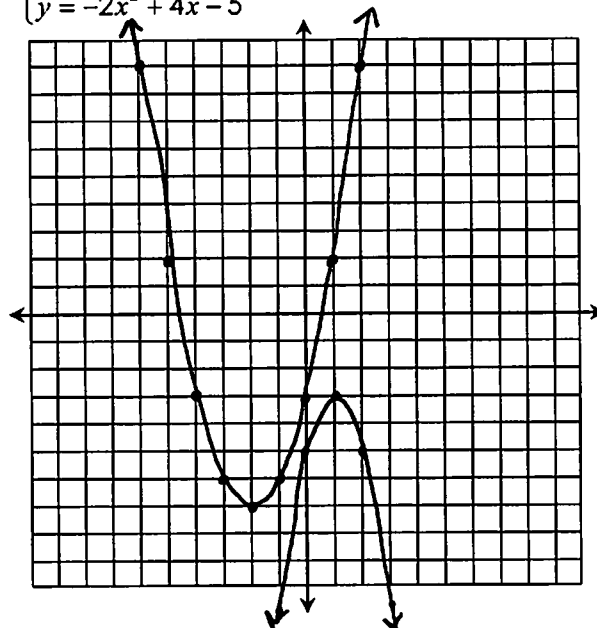
Solve by graphing:

1. $\begin{cases} y = x^2 - 10x + 17 \\ x + y = -1 \end{cases} \rightarrow y = -x - 1$



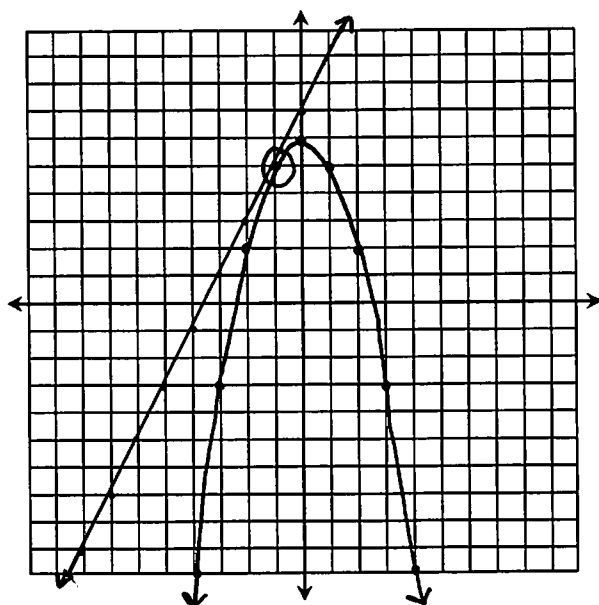
Solution(s): $(3, -4), (6, -7)$

2. $\begin{cases} y = x^2 + 4x - 3 \\ y = -2x^2 + 4x - 5 \end{cases}$



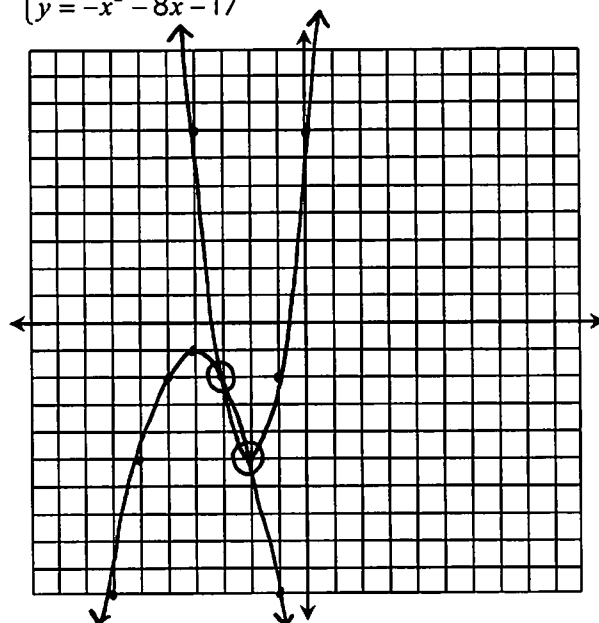
Solution(s): No Solution

3. $\begin{cases} y = -x^2 + 6 \\ 2x - y = -7 \end{cases}$



Solution(s): $(-1, 5)$

4. $\begin{cases} y = 3x^2 + 12x + 7 \\ y = -x^2 - 8x - 17 \end{cases}$



Solution(s): $(-3, -2), (-2, -5)$

Solve by algebraically:

$$5. \begin{cases} y = x^2 + 4x + 2 \\ x - y = 2 \end{cases}$$

$$x - (x^2 + 4x + 2) = 2$$

$$x - x^2 - 4x - 2 = 2$$

$$-x^2 - 3x - 4 = 0$$

$$x^2 + 3x + 4 = 0$$

* Cannot be factored;
imaginary roots*

Solution(s): No Solution

$$7. \begin{cases} y = -3x^2 - 10x - 5 \\ y = x^2 + 6x + 11 \end{cases}$$

$$-3x^2 - 10x - 5 = x^2 + 6x + 11$$

$$0 = 4x^2 + 16x + 16$$

$$0 = 4(x^2 + 4x + 4)$$

$$0 = 4(x+2)(x+2)$$

$$4 \neq 0 \quad | \quad x = -2 \quad | \quad x = -2$$

$$y = (-2)^2 + 6(-2) + 11$$

$$y = 4 - 12 + 11$$

$$y = 3$$

Solution(s): $(-2, 3)$

$$6. \begin{cases} y = x^2 - 8x + 8 \\ y = -x^2 + 10x - 20 \end{cases}$$

$$x^2 - 8x + 8 = -x^2 + 10x - 20$$

$$2x^2 - 18x + 28 = 0$$

$$2(x^2 - 9x + 14) = 0$$

$$\frac{2(x-7)(x-2)}{2 \neq 0} = 0 \quad | \quad x = 7 \quad | \quad x = 2$$

$$y = 7^2 - 8(7) + 8$$

$$y = 49 - 56 + 8$$

$$y = 1$$

$$y = 2^2 - 8(2) + 8$$

$$y = 4 - 16 + 8$$

$$y = -4$$

Solution(s): $(7, 1), (2, -4)$

$$8. \begin{cases} y = 2x^2 + 11x - 4 \\ 3x - y = -6 \end{cases}$$

$$3x - (2x^2 + 11x - 4) = -6$$

$$3x - 2x^2 - 11x + 4 = -6$$

$$-2x^2 - 8x + 10 = 0$$

$$-2(x^2 + 4x - 5) = 0$$

$$\frac{-2(x+5)(x-1)}{-2 \neq 0} = 0 \quad | \quad x = -5 \quad | \quad x = 1$$

$$3(-5) - y = -6$$

$$-15 - y = -6$$

$$-y = 9$$

$$y = -9$$

$$3(1) - y = -6$$

$$-y = -9$$

$$y = 9$$

Solution(s): $(-5, -9), (1, 9)$

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!